ELLIPTIC CURVE COUNTS AND GENERALIZED WEINSTEIN CONJECTURES

Abstract. We use elliptic curve Gromov–Witten theory to probe various generalizations of the Weinstein conjecture. For instance, we prove new cases of Mazzucchelli's generalization of the Weinstein conjecture, and versions of Sandon's translated points conjecture. We also state an analogue of the Weinstein conjecture in lcs geometry, directly generalizing the classical Weinstein conjecture, and discuss various partial verifications. A counterexample for a stronger form of this conjecture is given. Along the way we give an intriguing formula for certain $\mathbb{Q}$-valued elliptic Gromov–Witten curve counts in terms of the dynamical Fuller index, which might be of independent interest. The paper aims to be self-contained with no knowledge beyond basic Gromov–Witten theory required.

Contents

1. Introduction 2
1.1. Fixed Reeb strings 2
1.2. Liouville-lcs geometry 4
1.3. Conformal symplectic Weinstein conjecture 5
1.4. Organization 8
2. Preliminaries on lcs manifolds 8
3. On Reeb curves 11
4. J-holomorphic curves in certain lcs manifolds 12
4.1. Charged elliptic curves in an lcs manifold 14
4.2. Reeb holomorphic tori in $(C \times S^1, J^\lambda)$ 16
5. Elements of Gromov–Witten theory of an almost complex manifold 17
5.1. Implicit atlases 18
5.2. The Gromov–Witten invariant and its dependence 20
5.3. The space of smooth stable maps 21
6. Elliptic curves in the lcs-fication of a contact manifold and the Fuller index 22
6.1. Preliminaries on Reeb tori 22
7. Proofs of main theorems 30
7.1. Mapping tori and Reeb 2-curves 30
7.2. Proofs of the remaining theorems 34
Acknowledgements 36
A. Fuller index 36
The Morse–Bott case 37

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B. Remark on multiplicity 39

References 40

1. Introduction

The Weinstein conjecture asserts that the Reeb vector field of any contact form on a closed manifold has a closed orbit [42]. It has been very influential in contact geometry. Attacks on it have produced a range of techniques, most notably the theory of pseudo-holomorphic curves in symplectizations, initiated by Hofer [16], which in turn led to the embedded contact homology of Hutchings [19]. In dimension 3 the conjecture was proved in full by Taubes [40], using a connection to Seiberg–Witten theory.

We study some natural generalizations of the Weinstein conjecture. We argue that one natural home for such a generalization is locally conformally symplectic (lcs) geometry, and more specifically what we call Liouville-lcs geometry, Section 1.2. Our methods are based on the Fuller index and on Gromov–Witten counts of elliptic curves in lcs manifolds. For the lcs-fication of a contact manifold, Example 2.4, these counts are computed by the Fuller index of its Reeb vector field, Theorem 6.14.

The Gromov–Witten theory studied here is still in its infancy, and is partly local in nature. The paper also has global results, for example Theorems 1.8, 1.14 and 6.14, and Corollary 1.19. Moreover, there is strong evidence that a global theory exists. Most notably, [38] proves topological invariance of certain rational closed geodesic counts on surfaces. These geodesic counts directly correspond to rational Gromov–Witten counts of elliptic curves, of the type studied here, in the lcs-fications $S^*X \times S^1$ of the unit cotangent bundles S^*X of closed oriented Riemannian surfaces X, Example 2.4. The topological invariance then translates to invariance of these Gromov–Witten counts under the Liouville-lcs equivalences, Definition 1.11, given by families of metrics on X, cf. Theorem 6.14. We leave exploration of this to future study.

1.1. Fixed Reeb strings.

We will start with the more basic story of fixed Reeb strings of strict contactomorphisms, Definition 1.1. This will turn out to be closely tied to counts of elliptic curves in certain associated lcs manifolds. From fixed Reeb strings, we are then led to cases of Mazzucchelli's generalization of the Weinstein conjecture, Sandon's translated points conjecture, and then to the formulation of the "conformal symplectic Weinstein conjecture".

Let (C^{2n+1}, λ) be a closed contact manifold, Terminology 2.2, with λ a contact form, that is a 1-form with $\lambda \wedge (d\lambda)^n$ nowhere vanishing. Denote by R^λ the Reeb vector field satisfying:

$$d\lambda(R^\lambda, \cdot) = 0, \quad \lambda(R^\lambda) = 1.$$

Recall that a ***closed λ-Reeb orbit*** (or just closed Reeb orbit when λ is implicit) is a smooth map

$$o : (S^1 = \mathbb{R}/\mathbb{Z}) \to C$$

such that

$$o'(t) = cR^\lambda(o(t)),$$

with $o'(t)$ denoting the time derivative, for some $c > 0$, called the ***period*** of o. Let $S(R^\lambda, \beta)$ denote the space of all closed Reeb orbits in free homotopy class β, with its compact open topology. Set

$$\mathcal{O}(R^\lambda, \beta) := S(R^\lambda, \beta)/S^1,$$

where S^1 is acting naturally by reparametrization $(\theta \cdot o)(t) = o(t + \theta)$. Its elements are called ***Reeb strings***. They are the orbit strings of Appendix A for $X = R^\lambda$.

Definition 1.1. Let ϕ be a strict contactomorphism of a contact manifold (C,λ), Terminology 2.2. As $\phi^*\lambda=\lambda$, $\phi_*R^\lambda=R^\lambda$. So $o\mapsto\phi\circ o$ takes closed Reeb orbits to closed Reeb orbits of the same period, commutes with reparametrization, and induces a map $\mathcal{O}(R^\lambda,\beta)\to\mathcal{O}(R^\lambda,\phi_*\beta)$. A ***fixed Reeb string of ϕ in class*** β is an $o\in\mathcal{O}(R^\lambda,\beta)$ with $\phi\circ o=o$. Then $\phi_*\beta=\beta$.

Here and below, "homotopic to the id via strict contactomorphisms" means that $\phi=\phi_1$ for a smooth family $\{\phi_t\}_{t\in[0,1]}$ of strict contactomorphisms with $\phi_0=\mathrm{id}$.

Theorem 1.2 (Special case of Theorem 1.7, proved in Section 7)**.** *For any contact form λ on S^{2n+1}, $n\geq 1$, sufficiently C^1-close to λ_{st} (the standard contact form), any strict contactomorphism ϕ of (S^{2n+1},λ) homotopic to the* id *via strict contactomorphisms has a fixed Reeb string.*

Let ϕ be a strict contactomorphism of (C,λ). Following Mazzucchelli [23], a ***ϕ-invariant Reeb orbit*** is a trajectory γ of R^λ s.t. for some $T\neq 0$ and all $t\in\mathbb{R}$, $\phi(\gamma(t))=\gamma(t+T)$. Hence id-invariant Reeb orbits are precisely the trajectories $t\mapsto o(t/c)$ of closed Reeb orbits o of period c. Note also that given a fixed Reeb string of ϕ, represented by o of period c, the trajectory $t\mapsto o(t/c)$ is a ϕ-invariant Reeb orbit.

Conjecture 1.3 (Strict form of Mazzucchelli [23, Conjecture 1.2])**.** *For any (C,λ) with C closed, every strict contactomorphism ϕ homotopic to the identity via strict contactomorphisms admits a ϕ-invariant Reeb orbit.*

Mazzucchelli [23, Conjecture 1.2] only requires ϕ to preserve λ and to be contact isotopic to the identity, not necessarily through strict contactomorphisms. The above is the special case that we can address. Theorem 1.2, and more generally Theorem 1.8 below, give:

Corollary 1.4. *Conjecture 1.3 holds for (S^{2n+1},λ), $n\geq 1$, for λ sufficiently C^1-close to λ_{st}. Furthermore, it holds for every (C,λ) satisfying the hypothesis of Theorem 1.8 below, in particular for the contact forms of Theorem 1.7.*

There is a partial connection of Theorem 1.2 with the theory of translated points.

Definition 1.5 (Sandon [35])**.** Given a general contactomorphism ϕ of (C,λ), a point $p\in C$ is called a ***translated point*** provided that $\phi^*\lambda(p)=\lambda(p)$ and $\phi(p)$ lies on the λ-Reeb flow line passing through p.

Let ϕ be a strict contactomorphism of (C,λ), o a fixed Reeb string of ϕ, and $p\in\operatorname{image} o$. Then p is a translated point of ϕ. Indeed, $\phi^*\lambda=\lambda$, and $\phi(p)\in\operatorname{image} o$, which is the Reeb flow line through p. Hence we get:

Corollary 1.6. *Let λ be a contact form on S^{2n+1}, $n\geq 1$, sufficiently C^1-close to λ_{st}, and ϕ a strict contactomorphism of (S^{2n+1},λ) homotopic to the* id *via strict contactomorphisms. Then ϕ has a non-empty continuum of translated points.*

Note that the general form of Sandon's conjecture on existence of translated points [35] has counterexamples on S^{2n+1} for the standard contact form λ_{st}, see Cant [6]. However, it likely holds for any strict contactomorphism of (S^{2n+1},λ) contact isotopic to the identity, for λ any contact form of the standard contact structure.

Theorem 1.7 (Proof in Section 7)**.** *Suppose that (C,λ) is closed and Morse–Bott, and some connected component $N\subset\mathcal{O}(R^\lambda,\beta)$, with the union of the images of its orbits orientable, has non-vanishing orbifold Euler characteristic $\chi^{orb}(N)$, defined in Appendix A. Then for any contact form λ' on C, sufficiently C^1-close to λ, any strict contactomorphism ϕ of (C,λ'), homotopic to the* id *via strict contactomorphisms, has a class β fixed Reeb string.*

Both of the theorems above are consequences of Theorem 1.8 below. Let λ be a contact form on a closed manifold C, $N \subset \mathcal{O}(R^\lambda, \beta)$ open and compact, and let $i(N, R^\lambda, \beta) \in \mathbb{Q}$ denote the Fuller index, Appendix A. For example, if λ is Morse–Bott (see [4]) and N is a connected component of $\mathcal{O}(R^\lambda, \beta)$, such that the union of the images of the orbits in N is an orientable submanifold of C, then N is open and compact, and naturally a closed orbifold. Moreover

$$i(N, R^\lambda, \beta) = \pm\chi^{orb}(N),$$

for $\chi^{orb}(N)$ the orbifold Euler characteristic of N, see (A.4) in Appendix A, cf. [37, Section 2.1.1, proof of Lemma 2.13]. If all the orbits in N have the same multiplicity m, Appendix A, then N is a closed manifold and $\chi^{orb}(N) = \chi(N)/m$, for $\chi(N)$ the Euler characteristic. Examples of (C, λ) with such a component N with $\chi^{orb}(N) \neq 0$ are (S^{2n+1}, λ_{st}) and the unit cotangent bundle of a round sphere, with its canonical contact form. See Bourgeois [4] for more.

Theorem 1.8 (Proof in Section 7)**.** *Let (C, λ) be a closed contact manifold, and let $N \subset \mathcal{O}(R^\lambda, \beta)$ be open and compact with $i(N, R^\lambda, \beta) \neq 0$. Then any strict contactomorphism $\phi : (C, \lambda) \to (C, \lambda)$, homotopic to the* id *via strict contactomorphisms, has a class β fixed Reeb string $o \in N$.*

The index assumption of the theorem thus holds for the Morse–Bott examples above. It persists for every λ' sufficiently C^1-close to λ, with N replaced by an open compact $N' \subset \mathcal{O}(R^{\lambda'}, \beta)$, by Lemma A.1. This gives Theorem 1.7. In this Morse–Bott case, if the underlying topological space of N has $\chi(N) \neq 0$, the theorem may be verified by elementary considerations. Indeed, ϕ as above induces a homeomorphism ϕ_N of N, homotopic to the identity, hence with non-zero Lefschetz number. So in this case the result follows by the Lefschetz fixed point theorem.

In general an open compact $N \subset \mathcal{O}(R^\lambda, \beta)$ may not be a finite simplicial complex, or indeed any kind of topological space to which the classical Lefschetz fixed point theorem may apply. Also the relationship of $i(N, R^\lambda, \beta)$ with $\chi(N)$ breaks down in general, as $i(N, R^\lambda, \beta)$ is partly sensitive to the dynamics of R^λ.

1.2. **Liouville-lcs geometry.** For preliminaries on lcs geometry see Section 2. In particular lcs pairs (λ, α) and their lcs forms $\omega = d_\alpha\lambda$ are defined in Definition 2.5.

Definition 1.9. Let (λ, α) be an lcs pair on M and $\omega = d_\alpha\lambda$, for d_α the Lichnerowicz differential (2.3). Define X_λ by $\omega(X_\lambda, \cdot) = \lambda$ and X_α by $\omega(X_\alpha, \cdot) = \alpha$, and generally X_η by $\omega(X_\eta, \cdot) = \eta$ for a 1-form η. Let $\mathcal{D}$ denote the (generalized) distribution spanned by X_α, X_λ, meaning $\mathcal{D}(p) := \text{span}(X_\alpha(p), X_\lambda(p))$. This will be called the ***canonical distribution***.

Definition 1.10. An lcs pair (λ, α) on M satisfies the ***Reeb condition*** if

$$\lambda(X_\alpha) > 0.$$

We then call X_α the ***generalized Reeb vector field*** of the pair. An lcs pair satisfying the Reeb condition is called a ***Reeb pair***, and a manifold with a Reeb pair a ***Liouville-lcs manifold***.

Definition 1.11. Two Reeb pairs (λ_0, α_0) on M_0 and (λ_1, α_1) on M_1 are ***Liouville-lcs equivalent*** if there is a diffeomorphism $\phi : M_0 \to M_1$ such that $\phi^*(\lambda_1, \alpha_1)$ and (λ_0, α_0) are joined by a smooth family of Reeb pairs on M_0. Such ϕ is called a ***Liouville-lcs equivalence***.

Liouville-lcs equivalences compose and are invertible, by pulling back, concatenating and reversing the families, reparametrized to be constant near their endpoints. So Reeb pairs with Liouville-lcs equivalences form a groupoid, the ***Liouville-lcs category***. Liouville isomorphisms, Definition 2.7, are the morphisms for which the family may be taken constant.

Conceptually this paper is about Liouville-lcs geometry, not lcs geometry in general. An lcs pair (λ, α) is the lcs analogue of a Liouville structure, with ***Liouville field*** X_λ, $\iota_{X_\lambda}\omega = \lambda$. The lcs-fication of

(C,λ), Example 2.4, is the symplectization $(\mathbb{R}\times C, d(e^r\lambda))$ divided by $r\mapsto r+1$, with $\theta=-r$. Its lcs form is the invariant form $e^{-r}d(e^r\lambda)=d\lambda+dr\wedge\lambda$. The Liouville field ∂_r descends to $X_\lambda=-\partial_\theta$. The almost complex structure J^λ of Example 4.6 is the standard cylindrical one, $J^\lambda\partial_r=R^\lambda$.

We call the foliation $\ker\alpha$ the ***Lee foliation***. It is a foliation where $\alpha\neq 0$, hence everywhere under the Reeb condition, as then $X_\alpha\neq 0$. Its leaves L satisfy $\omega|_L=d(\lambda|_L)$. Their characteristic line field is spanned by X_α, as $X_\alpha\in\ker\alpha$ is ω-orthogonal to $\ker\alpha$. As $\alpha(X_\lambda)=-\lambda(X_\alpha)$, the Reeb condition implies that the Liouville field is transverse to the Lee foliation, that is $\lambda(X_\alpha)$ vanishes nowhere. On a closed connected M^{2n}, oriented by ω^n, the Reeb condition and this transversality are equivalent. Indeed, $d(\lambda\wedge\omega^{n-1})=(1-\lambda(X_\alpha))\omega^n$, so $\int_M\lambda(X_\alpha)\omega^n=\int_M\omega^n>0$ by Stokes. Where $\alpha\neq 0$, transversality is equivalent to $\lambda|_L$ being a contact form on each leaf L, with Reeb vector field $X_\alpha/\lambda(X_\alpha)$, by the computation in the proof of Proposition 1.18.

1.3. **Conformal symplectic Weinstein conjecture.** The goal is now to describe one analogue of the Weinstein conjecture in Liouville-lcs geometry.

1.3.1. *Reeb 2-curves.* The canonical distribution $\mathcal{D}$ of an lcs pair is one analogue of the Reeb distribution on contact manifolds. A Reeb 2-curve, defined below, is a certain kind of singular surface tangent to $\mathcal{D}$. So it is a 2-dimensional analogue of a Reeb orbit.

Definition 1.12. Let Σ be a closed nodal Riemann surface (the set of nodes can be empty). Let $u:\Sigma\to M$ be a smooth map and let $\widetilde{u}:\widetilde{\Sigma}\to M$ be its normalization (see Definition 4.2). We say that u is a ***Reeb 2-curve*** of an lcs pair (λ,α) on M, if the following is satisfied:

(1) For each $z\in\widetilde{\Sigma}$ at which $d\widetilde{u}(z):T_z\widetilde{\Sigma}\to T_{\widetilde{u}(z)}M$ is non-zero, $\dim\mathcal{D}(\widetilde{u}(z))=2$ and $d\widetilde{u}(z)(T_z\widetilde{\Sigma})=\mathcal{D}(\widetilde{u}(z))$.

(2) $0\neq u^*[\alpha]\in H^1(\Sigma,\mathbb{R})$.

Example 1.13. The simplest example of a Reeb 2-curve in a manifold M with an lcs pair (λ,α), in the case $\mathcal{D}$ is a true 2-dimensional distribution (for example if (λ,α) is first kind, Definition 2.6, or a Reeb pair, Definition 1.10), is a closed oriented immersed surface $u:\Sigma\to M$ tangent to $\mathcal{D}$ with $u^*[\alpha]\neq 0$, for any complex structure on Σ. The last condition is automatic when $\lambda(X_\alpha)\neq 0$ along u, in particular in the first kind case. Indeed, ω is then non-degenerate on $\mathcal{D}$, so $(u^*\lambda,u^*\alpha)$ is an lcs pair on the closed surface Σ, and Lemma 3.1 applies. However, it will be necessary to consider more general curves, such as the nodal J-curves of Lemma 6.1.

It is tempting to conjecture that every closed Liouville-lcs manifold has a Reeb 2-curve, in analogy to the Weinstein conjecture. This is false, even in dimension 4 and for first kind pairs, where $\mathcal{D}$ is integrable. Moreover the generalized Reeb vector field can simultaneously have no closed orbits:

Theorem 1.14 (Proof in Section 7)**.** *Let (T^2,g_{st}) be the 2-torus with its standard flat metric, x_1,x_2 the coordinates of $T^2=\mathbb{R}^2/\mathbb{Z}^2$, and (C,λ) its unit cotangent bundle with the canonical contact form. Let ϕ be the strict contactomorphism of C induced by the translation of T^2 by (a,b), with $1,a,b$ linearly independent over $\mathbb{Q}$. Let $M_{\phi,1}$ be the mapping torus, Example 2.9, and for $\epsilon=(\epsilon_1,\epsilon_2)\in\mathbb{R}^2$ set*

$$\alpha_\epsilon:=\pi^*d\theta+\epsilon_1dx_1+\epsilon_2dx_2,$$

for dx_1,dx_2 pulled back to $C\times\mathbb{R}$, which descend to $M_{\phi,1}$ as ϕ preserves them. Then $(\lambda_\phi,\alpha_\epsilon)$ is a first kind Reeb pair on $M_{\phi,1}$ with no Reeb 2-curves. If $1,\epsilon_1,\epsilon_2$ are linearly independent over the field $\mathbb{Q}(a,b)$, then moreover α_ϵ is scale irrational, Definition 2.6, and X_{α_ϵ} has no closed orbits.

1.3.2. *Reeb 1-curves.* We can repair the tempting conjecture above by working with a more flexible notion, which is as follows.

Definition 1.15. A smooth map $o : S^1 \to M$ is a ***Reeb 1-curve*** of an lcs pair (λ, α) on M if

$$\forall t \in S^1 : (\lambda(o'(t)) > 0) \wedge (o'(t) \in \mathcal{D}(o(t))).$$

If (λ, α) is a Reeb pair then every closed orbit of X_α, parametrized by S^1, is a Reeb 1-curve, as $X_\alpha \in \mathcal{D}$ and $\lambda(X_\alpha) > 0$. We call such Reeb 1-curves ***strict***. By Theorem 1.14 a closed Liouville-lcs 4-manifold need not have a strict Reeb 1-curve. The a, b, ϵ required there exist, as $\mathbb{Q}$ and $\mathbb{Q}(a, b)$ are countable. By Corollary 1.19 below, a Reeb pair on a closed 4-manifold with scale integral α has a strict Reeb 1-curve. So in Theorem 1.14 the absence of closed orbits of X_{α_ϵ} forces α_ϵ to be scale irrational, in particular $\epsilon \neq 0$. By Theorem 2.10, the pairs there with α_ϵ scale irrational are not Liouville isomorphic to mapping tori.

The following is elementary. Its proof is in Section 3.

Proposition 1.16. *Every closed connected Liouville-lcs surface is a torus, and has a Reeb 1-curve.*

We will refer to the following as the ***conformal symplectic Weinstein conjecture***.

Conjecture 1.17. *Every closed Liouville-lcs manifold of dimension* 4 *has a Reeb 1-curve.*

Let (λ, α) be a Reeb pair on a closed connected M. For scale integral α, Definition 2.6, the leaves of $\ker \alpha$ are the components of the fibers of a fibration $M \to S^1$, which are closed contact manifolds. So in this case Conjecture 1.17 follows from the Weinstein conjecture for these fibers, by Proposition 1.18. This is Corollary 1.19 below. For scale irrational α every leaf is dense, so no leaf is compact. Indeed, a codimension-one foliation without holonomy on a closed manifold, such as $\ker \alpha$, is either a fibration over S^1 or has every leaf dense, Imanishi [20], see also [15, Chapter VIII]. In the first case let $f : M \to S^1$ be the fibration. As α is closed and vanishes on the connected fibers of f, $\alpha = f^*\eta$ for a nowhere vanishing 1-form η on S^1. So $[\alpha] = (\int_{S^1} \eta)\, f^*[d\theta]$, with $f^*[d\theta]$ integral. Hence α is scale integral. A Reeb 1-curve replaces a closed characteristic: a loop tangent to $\mathcal{D} = \operatorname{span}(X_\lambda, X_\alpha)$ and positive on λ, which may drift along the Liouville field.

The scale integral case is the following.

Proposition 1.18. *Let (λ, α) be an lcs pair on a closed manifold M^{2n}, with α scale integral: $c\alpha$ has integral periods for some $0 \neq c \in \mathbb{R}$, see Definition 2.6. Suppose that $\alpha(X_\lambda)$ vanishes nowhere, that is α vanishes nowhere and X_λ is transverse to $\ker \alpha$. This holds under the Reeb condition, Section 1.2. Then there is a map $p : M \to S^1 = \mathbb{R}/\mathbb{Z}$ with $p^*d\theta = c\alpha$, for $d\theta$ as in Example 2.4, and it is a fibration. Each fiber $C_t = p^{-1}(t)$ is a closed contact manifold with contact form $\lambda|_{C_t}$. Moreover, $X_\alpha|_{C_t} = \lambda(X_\alpha)R_t$ for R_t the Reeb vector field of $\lambda|_{C_t}$. In particular every closed Reeb orbit of $(C_t, \lambda|_{C_t})$ is a Reeb 1-curve of (λ, α).*

Proof. Integrating $c\alpha$ along paths from a base point in each component of M defines p modulo $\mathbb{Z}$, as $c\alpha$ has integral periods. As α vanishes nowhere, p is a submersion, and a fibration as M is closed. On $TC_t = \ker \alpha$ the form $\alpha \wedge \lambda$ vanishes, so $\omega = d\lambda$ there. Hence

$$\lambda \wedge (d\lambda)^{n-1}|_{C_t} = \lambda \wedge \omega^{n-1}|_{C_t} = \frac{1}{n} \iota_{X_\lambda} \omega^n|_{C_t},$$

which vanishes nowhere, as ω^n is a volume form and X_λ is transverse to C_t. So $\lambda|_{C_t}$ is contact.

Next

$$\alpha(X_\alpha) = \omega(X_\alpha, X_\alpha) = 0,$$

so X_α is tangent to C_t. Also

$$\iota_{X_\alpha} d\lambda = \iota_{X_\alpha}\omega + \iota_{X_\alpha}(\alpha \wedge \lambda) = (1 - \lambda(X_\alpha))\alpha$$

vanishes on TC_t. Thus $X_\alpha|_{C_t}$ lies in the line field $\ker d\lambda|_{C_t}$. Also $\lambda(X_\alpha) = -\alpha(X_\lambda) \neq 0$. So $X_\alpha = \lambda(X_\alpha)R_t$.

A closed orbit $o : \mathbb{R}/\mathbb{Z} \to C_t$ of R_t of period $T > 0$ has $o' = TR_t \in \mathbb{R}X_\alpha \subset \mathcal{D}$ and $\lambda(o') = T > 0$, so it is a Reeb 1-curve. □

Corollary 1.19. *Let (λ, α) be a Reeb pair on a closed 4-manifold M, with α scale integral. Then (λ, α) has a strict Reeb 1-curve. In particular Conjecture 1.17 holds for (λ, α).*

Proof. By Proposition 1.18 the fibers C_t are closed contact 3-manifolds, and $X_\alpha|_{C_t} = \lambda(X_\alpha)R_t$ with $\lambda(X_\alpha) > 0$. Each C_t has a closed Reeb orbit by Taubes [40]. Reparametrized, it is a closed orbit of X_α, so a strict Reeb 1-curve. □

By Corollary 1.19, Conjecture 1.17 is open only for scale irrational α. Scale integrality holds for every closed 1-form α when $b_1(M) = 1$, for instance on $S^3 \times S^1$. So the conjecture is open only when $b_1(M) \geq 2$.

The following is also elementary. Unlike Corollary 1.19, it allows scale irrational α.

Theorem 1.20 (Proof in Section 3)**.** *Let (C, λ) be a closed contact manifold with a non-degenerate closed Reeb orbit, or as in Theorem 1.7. Then every lcs pair (λ', α') on $C \times S^1$ sufficiently C^1-close to the lcs-fication of λ, Example 2.4, is a Reeb pair and has a Reeb 1-curve, contained in a fiber $C \times \{\theta_0\}$.*

For example, let C be the unit cotangent bundle of a closed hyperbolic surface, with its canonical contact form. Its closed Reeb orbits are the lifts of the closed geodesics. They are non-degenerate, as the geodesic flow is Anosov. Here $b_1(C \times S^1) \geq 5$. So the theorem applies to pairs (λ', α') with α' scale irrational, which exist by the remarks at the end of Section 2.

The Reeb 1-curve constructed in the proof is a closed orbit, in the fiber, of the line field cut out on $C \times \{\theta_0\}$ by the canonical distribution $\mathcal{D}'$ of (λ', α'). This line field is C^0-close to the Reeb line field of λ. For scale irrational α' and connected C every leaf of the Lee foliation is dense, as noted after Conjecture 1.17. So the Reeb 1-curve meets every leaf, unless it is tangent to the Lee foliation. In the latter case it is a strict Reeb 1-curve up to reparametrization, as $\mathcal{D}' \cap \ker \alpha' = \mathbb{R}X_{\alpha'}$, by $\alpha'(X_{\lambda'}) = -\lambda'(X_{\alpha'}) \neq 0$. By contrast the generalized Reeb vector field is tangent to the leaves, and may in general have no closed orbits, Theorem 1.14.

Proposition 1.21. *The analogue of Conjecture 1.17, in all dimensions, implies the Weinstein conjecture: every closed contact manifold (C, λ) has a closed Reeb orbit.*

Proof. Let (λ, α) be the lcs-fication on $M = C \times S^1$ of a closed contact manifold (C, λ), see Example 2.4. Then it satisfies the Reeb condition, as $X_\alpha = (R^\lambda, 0)$ and $\lambda(R^\lambda) = 1$, see Example 4.6. By the assumed analogue of Conjecture 1.17, there is a Reeb 1-curve $o : S^1 \to M$. By Example 4.6, $\mathcal{D} = \operatorname{span}((R^\lambda, 0), (0, \partial_\theta))$, so $o' = a(R^\lambda, 0) + b(0, \partial_\theta)$ with $a = \lambda(o') > 0$. Hence $(pr_C \circ o)' = aR^\lambda$, for $pr_C : C \times S^1 \to C$ the projection. So $pr_C \circ o$ is a closed Reeb orbit of (C, λ) after reparametrization to constant speed. □

1.4. **Organization.** Section 2 presents preliminaries on lcs geometry, which should make this paper self-contained and accessible to a general reader. Section 3 proves Theorem 1.20, Proposition 1.16 and Lemma 3.1. Section 4 introduces charged elliptic curves and the admissible almost complex structures. Section 5 sets up the Gromov–Witten counts of such curves and their invariance in families. Section 6 identifies these counts, for lcs-fications, with the Fuller index. Theorems 1.2, 1.7, 1.8 and 1.14 are proved in Section 7. Appendix A recalls the Fuller index, and Appendix B the multiplicities of multiply covered curves.

2. Preliminaries on lcs manifolds

Definition 2.1. A ***locally conformally symplectic manifold***, or just an ***lcs manifold***, is a smooth $2n$-fold M with an ***lcs structure***, or ***lcs form***: a non-degenerate 2-form ω, with the property that for every $p \in M$ there is an open $U \ni p$ such that $\omega|_U = f_U \cdot \omega_U$, for some symplectic form ω_U defined on U and some smooth positive function f_U on U.

These kinds of structures were originally considered by Lee in [22], arising naturally as part of an abstract study of "a kind of even dimensional differential geometry". They were then further studied by a number of authors, see for instance [2] and [41]. An lcs manifold admits all the interesting classical notions of a symplectic manifold, like Lagrangian submanifolds and Hamiltonian dynamics. At the same time lcs manifolds form a much more flexible class. For example Eliashberg and Murphy show that if a closed almost complex $2n$-fold M has $H^1(M,\mathbb{R}) \neq 0$ then it admits an lcs form, [8]. Apostolov–Dloussky [1] show that any compact complex surface with an odd first Betti number admits an lcs form taming the complex structure.

To see the connection with the first cohomology group $H^1(M,\mathbb{R})$, mentioned above, let us point out right away the most basic invariant of an lcs form ω, when M has dimension at least 4. We take a cover of M by open sets U_a in which $\omega = e^{f_a} \cdot \omega_a$ for ω_a symplectic. Then we have 1-forms $d(f_a)$ on each U_a, which glue to a well-defined closed 1-form α on M, as shown by Lee. This form is called the ***Lee form***. It is characterized by

$$d\omega = \alpha \wedge \omega.$$

Its class $[\alpha] \in H^1(M,\mathbb{R})$ is the ***Lee class***. We may denote this 1-form and its cohomology class both by α.

An lcs form ω on M is ***globally conformally symplectic*** if it is of the form $e^f \cdot \omega'$ with ω' symplectic. For $\dim M \geq 4$ this holds iff its Lee class vanishes. Indeed, if $\alpha = df$ then $d(e^{-f}\omega) = 0$. Conversely, if $\omega = e^f \omega'$ then $\alpha = df$ by the characterization above. Let $\widetilde{M}$ be the α-covering space of M, the covering of the proof of Lemma 6.1, taken componentwise if M is not connected. On it α pulls back to an exact form. So, by this criterion on $\widetilde{M}$, the lift $\widetilde{\omega}$ of ω is globally conformally symplectic.

There is a class of explicit examples of lcs manifolds. Start with a symplectic cobordism (see [8]) from (C,λ_-) to (C,λ_+), for C closed and $\lambda_+ = c\lambda_-$, with $c > 0$ a constant. Rescale its symplectic form by a positive function, equal to 1 near the negative end and to $1/c$ near the positive end. The rescaled form has matching ends, and we glue them.[1]

Terminology 2.2. For us a contact manifold is a pair (C,λ) where C is a manifold and λ a contact form on the $(2n+1)$-fold C: $\forall p \in C : \lambda \wedge (d\lambda)^n(p) \neq 0$. This is not a completely common terminology as usually it is the equivalence class of (C,λ) that is called a contact manifold. Here $(C,\lambda) \sim (C,\lambda')$ if $\lambda = f\lambda'$ for f a positive function. (So the contact structure $\ker\lambda$, in the classical sense, is co-oriented.) A ***contactomorphism*** between (C_1,λ_1), (C_2,λ_2) is a diffeomorphism $\phi : C_1 \to C_2$ s.t. $\phi^*\lambda_2 = f\lambda_1$ for some $f > 0$. It is called ***strict*** if $\phi^*\lambda_2 = \lambda_1$.

[1] This basic picture is how this author got interested in the subject after listening to ideas of Penrose on conformal cyclic cosmology.

For a closed 1-form α on M, the operator

$$d_\alpha : \Omega^k(M) \to \Omega^{k+1}(M), \quad d_\alpha \eta := d\eta - \alpha \wedge \eta, \tag{2.3}$$

is called the ***Lichnerowicz differential***. It satisfies $d_\alpha \circ d_\alpha = 0$, as $d\alpha = 0$. So we have an associated Lichnerowicz cochain complex.

The following example shows how lcs manifolds generalize contact manifolds. It is a variant of the above cobordism construction: the trivial cobordism $[0,1] \times C$ with $d(e^r\lambda)$, rescaled by e^{-r}, Section 1.2.

Example 2.4 (Banyaga). Let (C, λ) be a contact manifold, $S^1 = \mathbb{R}/\mathbb{Z}$, $d\theta$ the standard nowhere vanishing 1-form on S^1 with $\int_{S^1} d\theta = 1$. Take $M = C \times S^1$ with the 2-form

$$\omega_\lambda = d_\alpha \lambda = d\lambda - \alpha \wedge \lambda,$$

for d_α the Lichnerowicz differential (2.3). Here $\alpha := pr_{S^1}^* d\theta$, for $pr_{S^1} : C \times S^1 \to S^1$ the projection. Likewise λ denotes the pull-back of λ by the projection $C \times S^1 \to C$. We call the pair (λ, α), an lcs pair in the sense of Definition 2.5 below, and also (M, ω_λ), the ***lcs-fication*** of (C, λ). This is also a basic example of a first kind lcs pair, Definition 2.6 ahead, as $\ker d\lambda = \operatorname{span}(R^\lambda, \partial_\theta)$.

Definition 2.5. An ***lcs pair*** on M is a pair of 1-forms (λ, α), with α closed, such that $\omega = d_\alpha \lambda$ is non-degenerate. Then ω is an lcs form, as locally $\alpha = dg$ and $\omega = e^g d(e^{-g}\lambda)$. It is called the lcs form of the pair. An lcs form is called ***exact*** if it is the lcs form of some lcs pair.

An lcs pair determines a generalized distribution $\mathcal{V}_\lambda$ on M:

$$\mathcal{V}_\lambda(p) = \{v \in T_pM \mid d\lambda(v, \cdot) = 0\},$$

which we call the ***vanishing distribution***. For each $p \in M$, $\mathcal{V}_\lambda(p)$ has dimension at most 2 since $d\lambda - \alpha \wedge \lambda$ is non-degenerate, see the computation following Definition 2.6. If M^{2n} is closed, $\mathcal{V}_\lambda$ cannot identically vanish, since $(d\lambda)^n$ cannot be nowhere vanishing by Stokes theorem.

Definition 2.6. Let α be a closed 1-form on M. We call α ***integral*** if its periods are integral, ***scale integral*** if $c\alpha$ is integral for some $0 \neq c \in \mathbb{R}$, and ***scale irrational*** otherwise. These are properties of $[\alpha] \in H^1(M, \mathbb{R})$. An lcs pair (λ, α) is called integral, scale integral or scale irrational if α is. If $\mathcal{V}_\lambda$ is non-zero at each point of M, then the lcs pair (λ, α) is called ***first kind***. Then $\mathcal{V}_\lambda$ is a smooth 2-distribution, as it has dimension at most 2 and $d\lambda$ has even rank. An exact lcs form ω is called first kind if $\omega = d_\alpha \lambda$ for some first kind lcs pair (λ, α). When $\dim M \geq 4$, α is the Lee form of $d_\alpha \lambda$, as $d(d_\alpha \lambda) = \alpha \wedge d_\alpha \lambda$. For $\dim M \geq 4$ we call any lcs form integral, scale integral or scale irrational if its Lee form is.

For any lcs pair, $\mathcal{V}_\lambda \subset \mathcal{D}$, for $\mathcal{D}$ the canonical distribution of Definition 1.9. Indeed, if $d\lambda(v, \cdot) = 0$ then $\iota_v \omega = \lambda(v)\alpha - \alpha(v)\lambda$. So $v = \lambda(v)X_\alpha - \alpha(v)X_\lambda$. If the lcs pair is first kind then $\mathcal{V}_\lambda$ is 2-dimensional and $\dim \mathcal{D} \leq 2$, so $\mathcal{V}_\lambda = \mathcal{D}$. Then $X_\alpha \in \ker d\lambda$. So $0 = d\lambda(X_\alpha, \cdot) = (1 - \lambda(X_\alpha))\alpha$, using $\alpha(X_\alpha) = \omega(X_\alpha, X_\alpha) = 0$. Moreover α vanishes nowhere, as $\dim \mathcal{D} = 2$ forces $X_\alpha \neq 0$. Hence $\lambda(X_\alpha) = 1$ everywhere for a first kind lcs pair.

Definition 2.7. A ***conformal symplectomorphism*** of lcs manifolds $\phi : (M_1, \omega_1) \to (M_2, \omega_2)$ is a diffeomorphism ϕ s.t. $\phi^* \omega_2 = e^f \omega_1$, for some f. Note that in this case we have an induced relation (when M_1, M_2 have dimension at least 4):

$$\phi^* \alpha_2 = \alpha_1 + df,$$

where α_i is the Lee form of ω_i, $i = 1, 2$. If $f = 0$ we call ϕ a ***symplectomorphism***. A (conformal) symplectomorphism of lcs pairs (λ_1, α_1), (λ_2, α_2) on M_1 respectively M_2 is a (conformal) symplectomorphism of the corresponding lcs 2-forms. If a diffeomorphism $\phi : M_1 \to M_2$ satisfies $\phi^* \lambda_2 = \lambda_1$

and $\phi^*\alpha_2 = \alpha_1$ we call it a ***Liouville isomorphism*** of the lcs pairs. This is analogous to a strict contactomorphism of contact manifolds.

With the above notions we have the following basic points. Except for the third, their proofs are left to the reader.

(1) A Liouville isomorphism of lcs pairs (λ_1, α_1), (λ_2, α_2) preserves the first kind condition, and moreover preserves the corresponding vanishing distributions.

(2) A symplectomorphism of lcs forms preserves the first kind condition for lcs forms, Definition 2.6.

(3) A symplectomorphism, conformal or not, of lcs pairs generally does not preserve the first kind condition for lcs pairs, as it only relates the two lcs forms. For instance, let (λ, α) be the lcs-fication of (S^3, λ_{st}), and $g : S^3 \to \mathbb{R}$ with $dg(R^{\lambda_{st}}) \neq 0$ somewhere, pulled back to $S^3 \times S^1$. As $d_\alpha(dg - g\alpha) = 0$, the pair $(\lambda + dg - g\alpha, \alpha)$ has the same lcs form. So the identity is a symplectomorphism of these lcs pairs. The second pair is not first kind. Indeed, $X_\alpha = (R^{\lambda_{st}}, 0)$, Example 4.6, is the same for both pairs. So the second pair has $(\lambda+dg-g\alpha)(X_\alpha) = 1+dg(R^{\lambda_{st}})$, which is not 1 everywhere. But a first kind pair has $\lambda(X_\alpha) = 1$ everywhere, by the computation following Definition 2.6.

(4) A (conformal) symplectomorphism of first kind lcs pairs generally does not preserve the vanishing distributions. (Similar to a contactomorphism not preserving Reeb distributions.)

(5) A conformal symplectomorphism of lcs forms and of lcs pairs preserves the integrality, scale integrality and scale irrationality conditions, in dimension at least 4, where the Lee form is determined by the lcs form.

Remark 2.8. We say that ω_0 is ***conformally equivalent*** to ω_1 if $\omega_1 = e^f\omega_0$ for some f, i.e. the identity map is a conformal symplectomorphism $\mathrm{id} : (M, \omega_0) \to (M, \omega_1)$. For us the structure is the form ω. For some authors it is the conformal equivalence class of ω. So for us conformally equivalent lcs forms on a given manifold are isomorphic, but in general distinct, objects of the category whose objects are lcs manifolds and whose morphisms are conformal symplectomorphisms.

Example 2.9. One example of a first kind lcs pair is carried by the mapping torus of a strict contactomorphism, see Banyaga [2]. Let ϕ be a strict contactomorphism of (C, λ) and $0 \neq c \in \mathbb{R}$. The mapping torus is $M_{\phi,c} := C \times \mathbb{R}/\sim$, $(x, \theta) \sim (\phi(x), \theta + 1)$. As a manifold it does not depend on c. It fibers over S^1 with fiber C,

$$\pi : M_{\phi,c} \to S^1.$$

It carries the lcs pair (λ_ϕ, α). Here λ_ϕ is induced by λ, well defined since $\phi^*\lambda = \lambda$, and $\alpha = c\pi^*(d\theta)$. This is an lcs pair, as $(d_\alpha\lambda_\phi)^{n+1} = -(n+1)\,\alpha \wedge \lambda_\phi \wedge (d\lambda_\phi)^n$ vanishes nowhere, for $\dim C = 2n+1$. It is scale integral, as $c^{-1}\alpha$ is integral, and first kind, as $\ker d\lambda_\phi = \mathrm{span}(R^\lambda, \partial_\theta)$. For $\phi = \mathrm{id}$ and $c = 1$ it is the lcs-fication, Example 2.4.

Moreover we have:

Theorem 2.10 (Only reformulating Bazzoni–Marrero [3, Theorem 4.7])**.** *A first kind lcs pair (λ, α) on a closed connected manifold M of dimension at least 4 is Liouville isomorphic to some mapping torus $(M_{\phi,c}, \lambda_\phi, c\pi^*d\theta)$ of Example 2.9 if and only if it is scale integral.*

Bazzoni–Marrero assume M connected and that the Lee foliation $\ker\alpha$ has a compact leaf L. For a first kind pair on closed M the latter holds when α is scale integral. Indeed, the leaves are then the components of the fibers of the map p of Proposition 1.18, so they are compact. That proposition applies as $\alpha(X_\lambda) = -\lambda(X_\alpha) = -1$. They also assume their anti-Lee field U complete. Here $U = -X_\lambda$,

as $\alpha(-X_\lambda) = 1$ and $-\iota_U\omega = \lambda$ is their η, [3, Section 2.1]. It is complete as M is closed. Their isomorphisms preserve the Lee form and η, [3, Definition 4.4]. So they are Liouville isomorphisms. Their theorem also allows an isomorphism $L \times \mathbb{R} \to M$. This is excluded as M is compact.

The scale integrality condition is necessary. Indeed, a Liouville isomorphism pulls back the form $c\pi^* d\theta$ of a mapping torus to α, and $c\pi^* d\theta$ is scale integral. Thus we may understand scale irrational first kind lcs pairs as first (and rather dramatic) departures from the world of contact manifolds into a brave new lcs world. Basic examples are the pairs of Theorem 1.14 with $1, \epsilon_1, \epsilon_2$ linearly independent over $\mathbb{Q}(a, b)$.

In general, on a closed manifold M with $b_1(M) \geq 2$ we may perturb a scale integral lcs pair (λ, α) to a scale irrational one (λ, α'), with α' closed and C^∞-close to α. Indeed, the scale integral classes form a countable union of lines through 0 in $H^1(M, \mathbb{R})$, whose complement is dense for $b_1(M) \geq 2$. This keeps $\mathcal{V}_\lambda$, hence the first kind condition, and $d_{\alpha'}\lambda$ stays non-degenerate. When $b_1(M) = 1$ every closed 1-form is scale integral, see the remarks following Corollary 1.19. Theorem 1.14, Theorem 1.20 and Proposition 7.3 concern deformations of this sort.

3. On Reeb curves

We now develop some basic theory to be used in the analysis of Reeb curves.

Lemma 3.1. *Let (λ, α) be an lcs pair on a closed manifold M. Then $0 \neq [\alpha] \in H^1(M, \mathbb{R})$.*

Proof. Suppose by contradiction that α is exact and let g be a primitive. Then computing we get: $d_\alpha\lambda = \frac{1}{f} d(f\lambda)$ with $f = e^{-g}$. Consequently, $d(f\lambda)$ is non-degenerate on M. This contradicts Stokes theorem. □

Lemma 3.2. *Let η be a nowhere vanishing 1-form on a closed connected oriented surface S. Then there is an embedded loop $\sigma : S^1 \to S$ with $\eta(\sigma') > 0$.*

Proof. The foliation $\ker \eta$ is transversely oriented by η, so it is tangent to a nowhere vanishing vector field V on the oriented surface S. If some orbit of V is not closed, take a point q in its limit set, in the dynamical sense, and a short arc τ through q transverse to V. The orbit crosses τ at infinitely many distinct points. An orbit segment between two consecutive crossings, closed up by the sub-arc of τ between them, can be perturbed to a closed transversal σ, see for instance [5, Proposition 1.17]. If every orbit of V is closed, the first return map to a transverse arc at any orbit is an orientation preserving homeomorphism of an interval all of whose points are periodic, hence the identity. So the orbits are the fibers of a circle bundle over a circle, and a section σ of this bundle is a closed transversal. In both cases $\eta(\sigma')$ vanishes nowhere, as σ is transverse to $\ker \eta$. So $\eta(\sigma') > 0$ after possibly reversing the orientation of σ. □

Proof of Proposition 1.16. Let (λ, α) be a Reeb pair on a closed connected surface M. Then ω is an area form, so M is oriented. As $\lambda(X_\alpha) > 0$, the form $\alpha = \iota_{X_\alpha}\omega$ vanishes nowhere, so $\chi(M) = 0$ and M is a torus. Also $\mathcal{D} = TM$, as $\omega(X_\lambda, X_\alpha) = \lambda(X_\alpha) \neq 0$, and λ vanishes nowhere, as $\lambda(X_\alpha) > 0$. By Lemma 3.2 there is a loop σ with $\lambda(\sigma') > 0$. Its velocity lies in $\mathcal{D} = TM$. So σ is a Reeb 1-curve. □

Proof of Theorem 1.20. In both cases there is a class β and an open compact $N \subset \mathcal{O}(R^\lambda, \beta)$ with non-zero Fuller index $i(N, R^\lambda, \beta)$, Appendix A. The rest of the proof uses only this. A non-degenerate closed Reeb orbit o_1 of class β is isolated in $\mathcal{O}(R^\lambda, \beta)$, Appendix A. So $N := \{o_1\}$ is open and compact, and $i(N, R^\lambda, \beta) = i(o_1)/m(o_1) = \pm 1/m(o_1) \neq 0$ by (A.2). In the setting of Theorem 1.7, the component N there is open and compact with $i(N, R^\lambda, \beta) = \pm\chi^{orb}(N) \neq 0$, by (A.4).

Let (λ_0, α_0) be the lcs-fication of (C, λ) on $M = C \times S^1$. Then $X_{\alpha_0} = (R^\lambda, 0)$ and $X_{\lambda_0} = (0, -\partial_\theta)$, Example 4.6.

Let (λ', α') be an lcs pair on M, sufficiently C^1-close to (λ_0, α_0). The fields $X_{\lambda'}$ and $X_{\alpha'}$ are obtained from λ', α' and $d\lambda'$ by inverting $d_{\alpha'}\lambda'$. So they are C^0-close to X_{λ_0} and X_{α_0}. In particular $\lambda'(X_{\alpha'}) > 0$, so (λ', α') is a Reeb pair, and $d\theta(X_{\lambda'}) \neq 0$. Hence $d\theta$ does not vanish on the canonical distribution $\mathcal{D}' = \mathrm{span}(X_{\lambda'}, X_{\alpha'})$. Here $\mathcal{D}'$ is 2-dimensional, as $d_{\alpha'}\lambda'(X_{\lambda'}, X_{\alpha'}) = \lambda'(X_{\alpha'}) \neq 0$. So $\ell := \mathcal{D}' \cap \ker d\theta$ is a line field on M, tangent to the fibers $C \times \{\theta\}$ and C^0-close to $\mathbb{R}(R^\lambda, 0)$. So λ' does not vanish on ℓ, as $\lambda_0(R^\lambda, 0) = 1$. Let W be the section of ℓ with $\lambda'(W) = 1$. Fix θ_0. The restriction W_0 of W to $C \times \{\theta_0\} = C$ is a vector field on C, C^0-close to R^λ.

By Lemma A.1, W_0 has a closed orbit γ of class β. Set $o(t) := (\gamma(t), \theta_0)$. Then $o' = p(\gamma)\, W \in \mathcal{D}'$, for $p(\gamma) > 0$ the period of γ, Appendix A. So $\lambda'(o') = p(\gamma) > 0$. So o is a Reeb 1-curve of (λ', α'). □

4. J-holomorphic curves in certain lcs manifolds

Let (M, J) be an almost complex manifold and (Σ, j) a Riemann surface. Recall that a map $u : \Sigma \to M$ is said to be ***J-holomorphic*** if $du \circ j = J \circ du$.

Notation 4.1. We will often say ***J-curve*** in place of J-holomorphic curve.

To set notation and terminology we review the basic definition of a nodal Riemann surface.

Definition 4.2. A ***nodal Riemann surface*** (without boundary) is a pair $\Sigma = (\widetilde{\Sigma}, \mathcal{N})$ where $\widetilde{\Sigma}$ is a possibly disconnected Riemann surface, and $\mathcal{N}$ a set of pairs of points of $\widetilde{\Sigma}$: $\mathcal{N} = \{(z_1^0, z_1^1), \dots, (z_n^0, z_n^1)\}$, $n \geq 0$, with $z_i^j \neq z_k^l$ for $(i,j) \neq (k,l)$. By slight abuse, we may also denote by Σ the quotient space $\widetilde{\Sigma}/\sim$, where the equivalence relation is generated by $z_i^0 \sim z_i^1$. Let $q_\Sigma : \widetilde{\Sigma} \to (\widetilde{\Sigma}/\sim)$ denote the quotient map. The points $q_\Sigma(z_i^0) = q_\Sigma(z_i^1) \in \widetilde{\Sigma}/\sim$ are called ***nodes***. Let M be a smooth manifold. By a map $u : \Sigma \to M$ of a nodal Riemann surface Σ, we mean a set map $u : (\widetilde{\Sigma}/\sim) \to M$. The map u is called smooth, an immersion or J-holomorphic (when M is almost complex) if the map $\widetilde{u} = u \circ q_\Sigma$ is smooth, an immersion or J-holomorphic, respectively. We call $\widetilde{u}$ the ***normalization of*** u. The map u is called an embedding if u is a topological embedding and its normalization is an immersion. The cohomology groups of Σ are defined as $H^\bullet(\Sigma) := H^\bullet(\widetilde{\Sigma}/\sim)$, likewise with homology. The genus of Σ, for Σ connected, is its arithmetic genus $\sum_i g(\Sigma_i) + |\mathcal{N}| - c + 1$, for Σ_i the components of $\widetilde{\Sigma}$ and c their number. The nodal surfaces of genus one are described in Section 4.1.

We shall say that $(\widetilde{\Sigma}, \mathcal{N})$ is ***normal*** if $\mathcal{N} = \emptyset$. Similarly, $u : \Sigma \to M$, $\Sigma = (\widetilde{\Sigma}, \mathcal{N})$ is called ***normal*** if $\mathcal{N} = \emptyset$. Note that if u is a Reeb 2-curve, its normalization $\widetilde{u}$ need not be a Reeb 2-curve (the second condition of Definition 1.12 may fail).

Tamed first kind lcs pairs, Example 4.4 below, give immediate examples of almost complex manifolds where the L^2 energy functional is unbounded on the moduli spaces of fixed homology class J-curves, as well as where null-homologous J-curves can be non-constant. We give such an example at the beginning of Section 4.1.

Definition 4.3. Let (λ, α) be a Reeb pair on M, Definition 1.10, that is with $\omega(X_\lambda, X_\alpha) = \lambda(X_\alpha) > 0$, where $\omega = d_\alpha\lambda$. In this case $\mathcal{D}$ is a 2-dimensional distribution, as $\omega(X_\lambda, X_\alpha) \neq 0$. We say that an ω-compatible J, that is one with $\omega(\cdot, J\cdot)$ a Riemannian metric, is ***(λ, α)-admissible*** if J preserves the canonical distribution: $J(\mathcal{D}) \subset \mathcal{D}$. Then J also preserves the ω-orthogonal complement $\mathcal{D}^\perp$ of $\mathcal{D}$. Indeed, for $v \in \mathcal{D}$ and $w \in \mathcal{D}^\perp$ we have $\omega(v, Jw) = \omega(Jv, J^2w) = -\omega(Jv, w) = 0$, as $Jv \in \mathcal{D}$. Moreover, $d\lambda$ tames J on $\mathcal{D}^\perp$, that is $d\lambda(v, Jv) > 0$ for $0 \neq v \in \mathcal{D}^\perp$. Indeed, $\mathcal{D}^\perp = \ker\alpha \cap \ker\lambda$, as $\iota_{X_\alpha}\omega = \alpha$ and $\iota_{X_\lambda}\omega = \lambda$. On $\ker\alpha \cap \ker\lambda$ we have $\omega = d\lambda$. So $d\lambda(v, Jv) = \omega(v, Jv) > 0$, as $Jv \in \mathcal{D}^\perp$. We call (λ, α, J) as above a ***tamed lcs pair***.

Admissible J exist, and the space of such J is contractible. Indeed, they are the compatible structures on the symplectic subbundles $\mathcal{D}$ and $\mathcal{D}^\perp$ taken separately, and each such space is contractible, cf. [27, 2nd ed., Proposition 2.50].

Example 4.4. If (λ, α) is a first kind lcs pair on M then by the computation following Definition 2.6, $\omega(X_\lambda, X_\alpha) = 1$ everywhere. In particular, we may find a J such that (λ, α, J) is a tamed lcs pair, and the space of such J is contractible. We call such (λ, α, J) a ***tamed first kind lcs pair***.

The following lemma is used in the proof of Lemma 6.1, Section 6.

Lemma 4.5. *Let (λ, α, J) be a tamed first kind lcs pair on M. Then given a smooth $u : \Sigma \to M$, where Σ is a closed (nodal) Riemann surface, u is J-holomorphic only if*

$$\operatorname{image} d\widetilde{u}(z) \subset \mathcal{V}_\lambda(\widetilde{u}(z))$$

for all $z \in \widetilde{\Sigma}$, where $\widetilde{u} : \widetilde{\Sigma} \to M$ is the normalization of u (see Definition 4.2). In particular $\widetilde{u}^* d\lambda = 0$.

Proof. By the first kind condition (see the computation following Definition 2.6) we have

$$V := \mathcal{V}_\lambda = \mathcal{D}.$$

Let u be J-holomorphic, so that $\widetilde{u}$ is J-holomorphic (by definition of a J-holomorphic nodal map). We have

$$\int_{\widetilde{\Sigma}} \widetilde{u}^* d\lambda = 0$$

by Stokes theorem. Let $V^\perp := \mathcal{D}^\perp$ be the ω-orthogonal complement, as in the definition of admissibility, and $\operatorname{proj}(p) : T_pM \to V^\perp(p)$ the projection induced by the splitting $TM = V \oplus V^\perp$.

Write $d\widetilde{u}^\perp := \operatorname{proj} \circ d\widetilde{u}$. As J preserves the splitting $TM = V \oplus V^\perp$, $d\widetilde{u}^\perp \circ j = J \circ d\widetilde{u}^\perp$. As $\iota_v d\lambda = 0$ for $v \in V$, by definition of $\mathcal{V}_\lambda$, for $a \in T_z\widetilde{\Sigma}$ we get

$$\widetilde{u}^* d\lambda(a, ja) = d\lambda(d\widetilde{u}^\perp(a), Jd\widetilde{u}^\perp(a)) \geq 0,$$

with equality iff $d\widetilde{u}^\perp(a) = 0$, as $d\lambda$ tames J on $V^\perp$. So $\widetilde{u}^* d\lambda$ is a non-negative multiple of a positive 2-form on $\widetilde{\Sigma}$, positive on the open set $\{d\widetilde{u}^\perp \neq 0\}$. If this set is non-empty, $\int_{\widetilde{\Sigma}} \widetilde{u}^* d\lambda > 0$, a contradiction. Thus,

$$\forall z \in \widetilde{\Sigma} : d\widetilde{u}^\perp(z) = 0,$$

so

$$\forall z \in \widetilde{\Sigma} : \operatorname{image} d\widetilde{u}(z) \subset \mathcal{V}_\lambda(\widetilde{u}(z)).$$

□

Example 4.6. Let (λ, α) on $C \times S^1$ be the lcs-fication of a contact manifold (C, λ). In this case

$$X_\alpha = (R^\lambda, 0),$$

where R^λ is the Reeb vector field and

$$X_\lambda = (0, -\partial_\theta),$$

minus the vector field generating the natural action of S^1 on $C \times S^1$. The sign is forced by

$$\omega(\partial_\theta, \cdot) = -(\alpha \wedge \lambda)(\partial_\theta, \cdot) = -\lambda.$$

Let $\xi := \ker \lambda \cap \ker \alpha = \{(v, 0) \mid v \in \ker \lambda\} \subset T(C \times S^1)$, the horizontal lift of the contact distribution of C. Then $\xi = \mathcal{D}^\perp$, the ω-orthogonal complement of the canonical distribution, by Definition 4.3. We then take J to be an almost complex structure on ξ, which is S^1-invariant, and compatible with $d\lambda$. The latter means that

$$g_J(\cdot, \cdot) := d\lambda|_\xi(\cdot, J\cdot)$$

is a J-invariant Riemannian metric on the distribution ξ.

There is an induced almost complex structure J^λ on $C \times S^1$, which is S^1-invariant, coincides with J on ξ, and satisfies:

$$J^\lambda(X_\alpha) = -X_\lambda, \quad \text{that is} \quad J^\lambda(R^\lambda, 0) = (0, \partial_\theta).$$

It preserves ξ and $\mathcal{D}$. It is ω-compatible. Indeed, $\xi = \mathcal{D}^\perp$ and $\omega = d\lambda$ on ξ, while $\omega(X_\alpha, J^\lambda X_\alpha) = \omega((R^\lambda, 0), (0, \partial_\theta)) = \lambda(R^\lambda) = 1$. This suffices on $\mathcal{D}$, as $\mathcal{D}$ is 2-dimensional and J^λ-invariant. Then $(\lambda, \alpha, J^\lambda)$ is a tamed integral first kind lcs pair on $C \times S^1$.

4.1. **Charged elliptic curves in an lcs manifold.** We now study moduli spaces of elliptic curves in an lcs manifold, constrained to have a certain charge.[2] In the present context, one reason for the introduction of "charge" is that it is now possible for non-constant holomorphic curves to be null-homologous, so the homology class does not distinguish them from constant curves.

Here is a simple example: take $S^3 \times S^1$ with $J = J^\lambda$, for λ the standard contact form. Then all the Reeb tori (Section 4.2) are null-homologous. Moreover, let o be a Reeb orbit of period c, and u_o its Reeb torus. Then $\int_{T^2} u_o^* \omega = c$, as $u_o^* d\lambda = 0$ and $u_o^*(\alpha \wedge \lambda) = -c\, ds \wedge dt$. This is half its L^2 energy for the metric $\omega(\cdot, J^\lambda \cdot)$, as $|du|^2 \operatorname{dvol} = 2u^*\omega$ pointwise for a J^λ-holomorphic u, cf. [28, 2nd ed., Lemma 2.2.1]. So the energy is unbounded on the Reeb tori of the iterates of a Hopf fiber, all of class 0. They all have charge 1, as $u_o^* \alpha = dt$, and trivial π-class, as S^3 is simply connected, see Definition 4.7 below. So fixing the charge class, Definition 4.8 below, does not bound the energy either. Compactness of the sets of curves we count is a hypothesis below, Section 5.

By an ***elliptic curve*** we mean a closed connected Riemann surface of genus one, that is a complex torus. By a ***nodal elliptic curve*** we mean a closed connected nodal Riemann surface of genus one, see Definition 4.2. Such a curve is a complex torus with trees of spheres attached, or a cycle of spheres, of length one (a torus with a pinched cycle) or more, with trees of spheres attached. Indeed, $|\mathcal{N}| - c + 1$ is the first Betti number of the dual graph, which has a vertex for each component of $\widetilde{\Sigma}$ and an edge for each node. So genus one means one of two things. Either there is one torus component, the others are spheres, and the graph is a tree. Or all components are spheres and the graph has exactly one cycle. We use no marked points.

The notion of charge can be defined in a general setting. Let M be a manifold endowed with a closed integral 1-form α. Its class $[\alpha]$ lies in $H^1(M, \mathbb{Z}) = \operatorname{Hom}(H_1(M, \mathbb{Z}), \mathbb{Z})$, which we use below. Let $u : T^2 \to M$ be a continuous map with $0 \neq u^*[\alpha] \in H^1(T^2, \mathbb{Z})$. Let $g : T^2 \to S^1$ be a fibration with $g^*[d\theta] = u^*[\alpha]$, for instance a linear map $T^2 = \mathbb{R}^2/\mathbb{Z}^2 \to \mathbb{R}/\mathbb{Z}$ in that class. Take a component of a fiber of g, an embedded circle. Let $[\gamma] \in H_1(T^2, \mathbb{Z})$ be its class for one of its two orientations, represented by $\gamma : S^1 \to T^2$. Let $[\rho] \in H_1(T^2, \mathbb{Z})$, likewise represented by $\rho : S^1 \to T^2$, be a class with $[\gamma] \cdot [\rho] = 1$, where $\cdot$ is the intersection pairing for the orientation of T^2, the complex orientation when T^2 is a Riemann surface. Such $[\rho]$ exists, as $[\gamma]$ is primitive: γ is an embedded circle bounding no disc, since g has no critical points. Then

$$\langle [\gamma], u^*[\alpha] \rangle = 0, \quad \langle [\rho], u^*[\alpha] \rangle \neq 0,$$

where $\langle , \rangle$ is the natural pairing of homology and cohomology. The first holds as $g \circ \gamma$ is constant, the second since $[\gamma], [\rho]$ generate $H_1(T^2, \mathbb{Z})$. Reversing the orientation of γ replaces $[\rho]$ by $-[\rho]$ modulo $[\gamma]$ and changes the sign of $\langle [\rho], u^*[\alpha] \rangle$. We orient γ so that $\langle [\rho], u^*[\alpha] \rangle > 0$. We call $[\gamma]$, so oriented, the ***fiber class*** of u.

Definition 4.7. For u, γ, ρ as above we call

$$k = |\langle [\rho], u^*[\alpha] \rangle| \in \mathbb{N}_{>0}$$

the α***-charge of*** u, or just the charge of u when α is implicit. The class $[u \circ \gamma] \in \pi_1^{\mathrm{free}}(M)$ will be called the π***-class of*** u. Here and below $\pi_1^{\mathrm{free}}(X)$ denotes the set of free homotopy classes of loops in a space X, the set of conjugacy classes in $\pi_1(X)$ when X is connected.

[2] The name charge is inspired by the notion of charge in Oh–Wang [30], in the context of contact instantons. However, the respective notions are not obviously related.

The α-charge and the π-class do not depend on the choices above. Indeed, the kernel of $u^*[\alpha]$ on $H_1(T^2,\mathbb{Z})$ has rank one, and $[\gamma]$ is a primitive class in it. So the fiber class is the primitive class f in this kernel with $\langle[\rho],u^*[\alpha]\rangle > 0$ whenever $f\cdot[\rho]=1$. Such $[\rho]$ is unique modulo f, so the charge depends only on f. As $\pi_1(T^2)$ is abelian, loops representing f are freely homotopic. So the π-class also depends only on f. The π-class depends on the orientation of T^2: reversing it replaces the π-class by that of the inverse loop. When $u^*[\alpha]=0$ we set the charge to be 0.

We extend the charge and the π-class to continuous maps $u:\Sigma\to M$ with Σ a nodal elliptic curve. There are two cases, see the description of nodal elliptic curves above. If $\widetilde{\Sigma}$ has a component Σ^T of genus one, a torus component, the fiber class, the charge and the π-class of u are those of $\widetilde{u}|_{\Sigma^T}$, with the fiber class pushed forward to $H_1(\Sigma,\mathbb{Z})$ by q_Σ. Otherwise Σ is a cycle of spheres with trees of spheres attached, and $H_1(\Sigma,\mathbb{Z})\simeq\mathbb{Z}$. In this case the charge of u is $|\langle[\rho],u^*[\alpha]\rangle|$ for $[\rho]$ a generator of $H_1(\Sigma,\mathbb{Z})$, and the π-class of u is defined to be the trivial class, that of a constant loop in $u(\Sigma)$. With this definition the charge is locally constant in the C^0-topology of Lemma 4.9 below. So is the π-class where the charge is non-zero. Gromov convergence implies convergence in this topology, Theorem 5.2 below. So the charge is preserved under Gromov convergence of stable maps. But it is not determined by the homology class. For instance, the Reeb tori above have class 0 and charge 1, while constant maps have class 0 and charge 0.

Definition 4.8. By the above, a continuous map $u:\Sigma\to M$, with Σ a nodal elliptic curve and non-zero α-charge k, determines a triple $(A,\beta,k)\in H_2(M,\mathbb{Z})\times\pi_1^{\mathrm{free}}(M)\times\mathbb{N}_{>0}$. Here $A=u_*[\Sigma]$ is its homology class, for $[\Sigma]:=q_{\Sigma*}\sum_S[S]\in H_2(\Sigma,\mathbb{Z})$, the sum over the components S of $\widetilde{\Sigma}$. And β is its π-class. This triple will be called the ***charge class*** of u.

Let (M,J) be an almost complex manifold and α a closed integral 1-form on M with $[\alpha]\neq 0$. We call (M,J,α) a ***Lee manifold***. For $k\geq 1$, $A\in H_2(M,\mathbb{Z})$ and $\beta\in\pi_1^{\mathrm{free}}(M)$ we define

$$\overline{\mathcal{M}}_1^k(J,A,\beta)$$

as the set of equivalence classes of pairs (u,Σ), for Σ a possibly nodal elliptic curve and $u:\Sigma\to M$ a J-holomorphic map of charge class (A,β,k), understood as a Kontsevich stable map when $\widetilde{\Sigma}$ has spherical components. Stability is spelled out in Lemma 4.9 below. The bar indicates, as usual, that nodal domains are allowed. A priori the space need not be compact. The equivalence relation is $(u_1,\Sigma')\sim(u_2,\Sigma'')$ if there is a biholomorphism $\phi:\Sigma'\to\Sigma''$ s.t. $u_2\circ\phi=u_1$. Here a biholomorphism of nodal Riemann surfaces $\Sigma'=(\widetilde{\Sigma}',\mathcal{N}')$, $\Sigma''=(\widetilde{\Sigma}'',\mathcal{N}'')$ is a biholomorphism $\widetilde{\phi}:\widetilde{\Sigma}'\to\widetilde{\Sigma}''$ taking $\mathcal{N}'$ onto $\mathcal{N}''$, as sets of unordered pairs. It descends to a homeomorphism ϕ of the quotients, and $u_2\circ\phi=u_1$ means $\widetilde{u}_2\circ\widetilde{\phi}=\widetilde{u}_1$. Such a ϕ preserves the charge class, so that $\overline{\mathcal{M}}_1^k(J,A,\beta)$ is well defined.

There are no marked points, so the automorphism group of an element is its stabilizer in the automorphism group of the domain. This stabilizer is finite. Indeed, an element is non-constant, as $k\geq 1$. For a torus domain, the stabilizer is a closed subgroup of the automorphism group of the torus. That group is a compact Lie group whose identity component is the translation group. So the stabilizer is finite unless it contains a one-parameter group of translations. A non-constant J-holomorphic torus is invariant under no such group, as a J-holomorphic map whose differential has rank at most one everywhere is constant. A stable map whose domain has spherical components has finitely many automorphisms by stability.

By slight abuse we denote such an equivalence class simply by u, so we may write $u\in\overline{\mathcal{M}}_1^k(J,A,\beta)$, with Σ implicit. The expected dimension of $\overline{\mathcal{M}}_1^k(J,A,\beta)$ is $2c_1(A)$. Indeed, the linearized Cauchy–Riemann operator at u, coupled with the variations of the complex structure of the domain, cf. (6.8), has Fredholm index $2c_1(A)+2$ by the Riemann–Roch theorem. From it we subtract 2, the dimension of the translation group of the domain torus.

Let (λ,α) on $M=C\times S^1$ be the lcs-fication of a closed contact manifold (C,λ). For $\beta\in\pi_1^{\mathrm{free}}(C)$ set $A_\beta:=\beta\otimes[S^1]\in H_2(M,\mathbb{Z})$, for β also denoting its Hurewicz image in $H_1(C,\mathbb{Z})$. Then $c_1(A_\beta)=0$ for every J homotopic to J^λ. This includes every (λ,α)-admissible J, as these form a contractible space containing J^λ. For J^λ, $TM=\mathcal{D}\oplus\xi$ as complex vector bundles, with $\mathcal{D}$ trivial and ξ the pullback of $\ker\lambda$ from C. So $c_1(TM)=pr_C^*c_1(\ker\lambda)$. It pairs to zero with A_β, as $pr_{C*}A_\beta=0$. So the expected dimension is 0 in the classes A_β.

Lemma 4.9. *Let M be a manifold with a closed integral 1-form α, and let $u_i:\Sigma_i\to M$ be continuous stable maps of nodal elliptic curves (stable meaning that each component S of $\widetilde{\Sigma}_i$ on which $\widetilde{u}_i$ is constant is a stable curve, with the points of S over the nodes of Σ_i as special points) converging in the C^0-topology of Siebert* [39, Definition 3.6] *to $u:\Sigma\to M$. Then for i large $u_i^*[\alpha]\neq 0$ iff $u^*[\alpha]\neq 0$, and in this case u_i and u have the same charge class. In particular, for M compact and J an almost complex structure on M, $\overline{\mathcal{M}}_1^k(J,A,\beta)$ is open and closed in the space of J-holomorphic stable maps of nodal elliptic curves in class A, with the Gromov topology of Section 5.*

Proof. Siebert's C^0-topology is defined for all continuous stable maps. On J-holomorphic stable maps it is the Gromov topology of Section 5, [39, Remark 3.12]. By its definition, [39, Definitions 2.13, 3.6], for i large there is a continuous surjection $\varphi_i:\Sigma_i\to\Sigma$ with u_i uniformly close to $u\circ\varphi_i$, hence homotopic to it. Moreover, φ_i collapses finitely many disjoint embedded circles in Σ_i to nodes of Σ and is an orientation preserving diffeomorphism elsewhere. Write H_1 for integral first homology. Collapsing a circle is, up to homotopy equivalence, attaching a disc along it, so $\varphi_{i*}:H_1(\Sigma_i)\to H_1(\Sigma)$ is onto. Also $\varphi_{i*}[\Sigma_i]=[\Sigma]$, so $u_{i*}[\Sigma_i]=u_*[\Sigma]$: the homology classes agree.

The charge of a continuous map $v:\Sigma'\to M$ of a nodal elliptic curve is the non-negative generator of the subgroup $\langle v_*H_1(\Sigma'),[\alpha]\rangle$ of $\mathbb{Z}$, zero iff $v^*[\alpha]=0$. When $\widetilde{\Sigma}'$ has a torus component Σ'^T this uses $H_1(\Sigma')\cong H_1(\Sigma'^T)$, the other components forming trees of spheres. As $u_{i*}=u_*\varphi_{i*}$ on H_1 and φ_{i*} is onto, u_i and u have the same charge.

Suppose the charge k is positive. If $\widetilde{\Sigma}_i$ has no torus component then neither has $\widetilde{\Sigma}$, as φ_{i*} maps onto $H_1(\Sigma)$. So in this case the π-classes of u_i and u are both trivial by definition. If $\widetilde{\Sigma}_i$ has a torus component, let $\gamma_i:S^1\to\Sigma_i$ be a loop in the torus component representing the fiber class of u_i. The π-class of u_i is $[u_i\circ\gamma_i]=[u\circ\varphi_i\circ\gamma_i]$, and $\langle\varphi_{i*}[\gamma_i],u^*[\alpha]\rangle=\langle[\gamma_i],u_i^*[\alpha]\rangle=0$. If $\widetilde{\Sigma}$ has no torus component then $\pi_1(\Sigma)=H_1(\Sigma)=\mathbb{Z}$, on which $\langle\cdot,u^*[\alpha]\rangle$ is injective. So in this case $\varphi_i\circ\gamma_i$ is null-homotopic, and the π-class of u_i is trivial, as is that of u by definition. If $\widetilde{\Sigma}$ has a torus component Σ^T, then φ_{i*}, a surjection $H_1(\Sigma_i)\simeq\mathbb{Z}^2\to H_1(\Sigma)\simeq\mathbb{Z}^2$, is an isomorphism. It preserves the intersection pairing, which on $H_1(\Sigma)$ is that of Σ^T. Indeed, composed with the collapse of the sphere trees of Σ to their attaching points, φ_i restricts to a degree one map between the torus components, being an orientation preserving homeomorphism over the complement of finitely many points. A degree one map between tori induces an element of $SL_2(\mathbb{Z})$ on H_1. So $\varphi_{i*}[\gamma_i]$ is primitive, pairs to zero with $u^*[\alpha]$, and has intersection number one with $\varphi_{i*}[\rho_i]$, which pairs to $k>0$ with $u^*[\alpha]$, for ρ_i as before Definition 4.7. Hence $\varphi_{i*}[\gamma_i]$ is the fiber class $[\gamma]$ of u. As $\pi_1(\Sigma)$ is abelian, $\varphi_i\circ\gamma_i$ is freely homotopic to γ, and the π-classes agree.

Finally, by [39, Definition 3.6] the above applies to every u' in a basic neighborhood of u in place of the u_i. So the set of positive charge is open and closed, and the charge class is locally constant on it. Hence $\overline{\mathcal{M}}_1^k(J,A,\beta)$, the preimage of (A,β,k), is open and closed. □

4.2. **Reeb holomorphic tori in $(C\times S^1,J^\lambda)$.** Let (C,λ) be a contact manifold and let α and J^λ be as in Example 4.6. In particular $(C\times S^1,J^\lambda,\alpha)$ is a Lee manifold.

There is a natural family of charge 1 J^λ-holomorphic tori in $M=C\times S^1$. Let o be a period c closed Reeb orbit of R^λ, and let β be its class in $\pi_1^{\mathrm{free}}(C)\subset\pi_1^{\mathrm{free}}(M)$. The ***Reeb torus*** u_o of o is the

map

$$u_o : T^2 = S^1 \times S^1 \to C \times S^1, \quad u_o(s,t) = (o(s), t).$$

A Reeb torus is J^λ-holomorphic for a uniquely determined complex structure j on T^2, defined by

$$j(\partial_s) = c\,\partial_t.$$

Indeed, $J^\lambda\, du_o(\partial_s) = c\, J^\lambda(R^\lambda, 0) = c\,(0, \partial_\theta) = du_o(c\,\partial_t)$, and du_o is injective. We have $u_o^*\alpha = dt$, so u_o has charge 1. Its fiber class is that of $S^1 \times \{t\}$ oriented by ∂_s, as (∂_s, ∂_t) is the complex orientation and $\langle [\{s\} \times S^1], [dt] \rangle = 1 > 0$, for $\{s\} \times S^1$ oriented by ∂_t. So its π-class is $[o] = \beta$. Moreover, $u_{o*}[T^2] = A_\beta$, the class of Section 4.1.

5. Elements of Gromov–Witten theory of an almost complex manifold

Throughout, (M, J) is a compact almost complex manifold. In this paper almost complex structures are C^∞.

We first recall the moduli spaces and the energy quantization theorem. Let $\mathcal{M}_{h,n}(J, A)$ denote the moduli space of isomorphism classes of stable, class A, J-holomorphic curves in M, with domain a connected genus h closed Riemann surface, with n labeled marked points. Stable means non-constant if $2h - 2 + n \leq 0$. Here an isomorphism between $u_1 : \Sigma_1 \to M$ and $u_2 : \Sigma_2 \to M$ is a biholomorphism of marked Riemann surfaces $\phi : \Sigma_1 \to \Sigma_2$ s.t. $u_2 \circ \phi = u_1$.

Theorem 5.1 ([28, 2nd ed., Proposition 4.1.4])**.** *Let (M, J) be a compact almost complex manifold, and g a Riemannian metric on M. There is an $\hbar = \hbar(g, J) > 0$ s.t. every J-holomorphic map $u : (S^2, j) \to M$ with $e_g(u) < \hbar$ is constant, where e_g is the L^2 energy functional,*

$$e_g(u) = \int_{S^2} |du|^2 \,\mathrm{dvol}\,.$$

For a smooth map $u : \Sigma \to M$ of a closed, possibly nodal, Riemann surface Σ we likewise set $e_g(u) := \int_{\widetilde{\Sigma}} |d\widetilde{u}|^2 \,\mathrm{dvol}$.

Next we state the basic precompactness theorem.

Theorem 5.2. *Let (M, J) be a compact almost complex manifold. Then $\mathcal{M}_{h,n}(J, A)$ is an open subset of the space $\overline{\mathcal{M}}_{h,n}(J, A)$ of Kontsevich stable maps, with the natural Gromov topology, see for instance* [28, 2nd ed., Section 5.6] *for the genus 0 case, and* [31, Section 9.1] *for the general case. Moreover, given $E > 0$, the subspace $\overline{\mathcal{M}}_{h,n}(J, A)_E \subset \overline{\mathcal{M}}_{h,n}(J, A)$ consisting of elements u with $e(u) \leq E$ is compact, where e is the L^2 energy with respect to an auxiliary J-invariant metric.* **Gromov convergence** *of stable maps, also for varying almost complex structures J_j, is the sequential notion of* [28, 2nd ed., Definition 5.2.1] *and* [18]*: the domains converge in a Deligne–Mumford space after adding marked points, the maps converge in C^∞ away from the nodes after reparametrization, and the energies converge. In particular the maps converge uniformly after reparametrization, in the C^0-topology of Siebert used in Lemma 4.9. Sequential compactness also holds for varying J. That is, if $J_j \to J$ in C^∞ and u^j are J_j-holomorphic stable maps in class A with $e(u^j) \leq E$, then a subsequence Gromov converges to a J-holomorphic stable map.*

Proof of Theorem 5.2. (Outline, as the argument is standard.) Suppose that we have a sequence u^j of J-holomorphic maps with L^2 energy $\leq E$. By [28, 2nd ed., Theorem 4.1.1], a sequence u^j of J-holomorphic curves has a convergent subsequence if $\sup_j ||du^j||_{L^\infty} < \infty$. On the other hand, when this condition does not hold, a rescaling argument tells us that a holomorphic sphere bubbles off. The quantization Theorem 5.1 tells us that the energy of a non-constant J-holomorphic map of $\mathbb{CP}^1$ is at least $\hbar > 0$. So if the energy of the maps u^j is bounded from above by E, only finitely many bubbles may appear, so that a subsequence of u^j must Gromov converge to a Kontsevich stable map. When the complex structures of the domains degenerate, the same bubbling analysis applies on the long necks,

whose thin parts carry little energy and become the nodes of the limit. The details, for any genus and for varying J_j, are in [18]. This is sequential compactness. It is compactness, as the Gromov topology is metrizable. In genus 0 metrizability is [28, 2nd ed., Theorem 5.6.6]. For the spaces $\overline{\mathcal{M}}_1^k(J,A,\beta)$, the only case used in this paper, it follows from property (c) of Section 5.3 and the paragraph after it. That paragraph uses only the sequential compactness proved here. □

We now specialize to genus 1 and no marked points, the only case used in this paper. Let (M,J,α) be a Lee manifold, and let $\mathcal{M} := \overline{\mathcal{M}}_1^k(J,A,\beta)$ be the moduli space of Section 4.1. It is an open and closed subset of $\overline{\mathcal{M}}_{1,0}(J,A)$ by Lemma 4.9, so Theorem 5.2 applies to it. It contains no constant maps, as $k \geq 1$. Its Gromov topology is the topology of [31, Section 9.1], there described by neighborhood bases of maps whose graphs over domains, stabilized by extra marked points, are Hausdorff-close. This topology is first countable. By the following lemma, which uses no taming form, its convergent sequences are Gromov convergent. Conversely, Gromov convergence implies convergence in this topology, see Theorem 5.2.

Lemma 5.3. *Let $u_i : \Sigma_i \to M$ be J-holomorphic stable maps converging to a J-holomorphic stable map $u_0 : \Sigma_0 \to M$ in the topology of* [31, Section 9.1]*: the domains converge in a Deligne–Mumford space after adding marked points, and u_i is uniformly δ_i-close to $u_0 \circ \varphi_i$, $\delta_i \to 0$, for collapse maps $\varphi_i : \Sigma_i \to \Sigma_0$, as in the proof of Lemma 4.9. Then the energies $e(u_i)$ are bounded, and u_i Gromov converges to u_0.*

Proof. First replace each φ_i by a collapse map that is C^1-close to conformal outside thin annuli around the collapsed circles, and maps these annuli into small discs around the nodes, with $\int |d\varphi_i|^2$ over these annuli tending to 0 for a fixed metric on $\widetilde{\Sigma}_0$. For instance, a radial collapse in plumbing coordinates has these properties. This changes $u_0 \circ \varphi_i$ little. So, after enlarging the δ_i, still with $\delta_i \to 0$, u_i is uniformly δ_i-close to $u_0 \circ \varphi_i$. Also $\int |d(u_0 \circ \varphi_i)|^2 \leq e(u_0) + \delta_i$. Indeed, outside the annuli the integral is at most $e(u_0)$ plus a term tending to 0, as φ_i is C^1-close to conformal. On the annuli it tends to 0, as $|du_0|$ is bounded. Let $\omega_g := g(J\cdot,\cdot)$ for the auxiliary J-invariant metric g. For a J-holomorphic map u we have $|du|^2 \operatorname{dvol} = 2u^*\omega_g$ pointwise, cf. [28, 2nd ed., Lemma 2.2.1], so $e(u_i) = 2\int u_i^*\omega_g$ and $e(u_0) = 2\int u_0^*\omega_g$. Let $H_i : \Sigma_i \times [0,1] \to M$ be the geodesic homotopy from $u_0 \circ \varphi_i$ to u_i. By Stokes, $\int u_i^*\omega_g - \int (u_0\circ\varphi_i)^*\omega_g = \int H_i^* d\omega_g$. Pointwise $|H_i^* d\omega_g| \leq C\delta_i(|du_i|^2 + |d(u_0\circ\varphi_i)|^2)$, for C depending on g and $d\omega_g$. Now $\int (u_0\circ\varphi_i)^*\omega_g = \int u_0^*\omega_g$, as φ_i has degree one. So $(1-2C\delta_i)e(u_i) \leq (1+2C\delta_i)e(u_0) + 2C\delta_i^2$, and the energies are bounded.

By Theorem 5.2 a subsequence Gromov converges. Its limit is u_0, as Gromov convergence implies convergence in the topology of [31, Section 9.1], which is the C^0-topology of Siebert on J-holomorphic maps and is Hausdorff, [39, Proposition 3.8, Remark 3.12]. This applies to every subsequence, so u_i Gromov converges to u_0. Cf. [39, Remark 3.12], stated there for any almost complex manifold with a Riemannian metric. □

Let $N \subset \mathcal{M}$ be an open compact subset, not necessarily connected. In our applications such N arise from concrete geometry. For example, they may correspond to open compact sets of Reeb strings, Theorem 6.14. Compactness of N is a hypothesis. We know of no general theory, even within lcs geometry, from which it emerges.

We follow mainly Pardon [31], whose setup is more familiar to me. However, the original setup of Fukaya–Ono [12], later expanded with Oh–Ohta [11], should apply here in the same fashion. We do not check this.

5.1. **Implicit atlases.** The construction in [31] of an implicit atlas on a moduli space of J-curves asks for J to be tamed by a symplectic form, but uses this only for compactness. We sketch it, following [31, Section 9.2], to locate that use, and to see that it applies to an open compact $N \subset \mathcal{M}$ as above.

Pardon's implicit atlas is indexed by the set $\mathfrak{A}$ of thickening datums, [31, Definition 9.2.1]. A datum $a \in \mathfrak{A}$ is a tuple $(Z_a, r_a, \Gamma_a, \overline{M}_a, E_a, \Theta_a)$, consisting of: a compact codimension two submanifold with boundary $Z_a \subset M$; an integer $r_a \geq 1$; a finite group Γ_a; a smooth Γ_a-manifold $\overline{M}_a$, Γ_a acting possibly non-effectively, with an isomorphism of orbifolds of $\overline{M}_a/\Gamma_a$ onto an open subset of $\overline{M}_{1,r_a}/S_{r_a}$, for $\overline{M}_{1,r_a}$ the Deligne–Mumford space of stable genus one curves with r_a marked points and S_{r_a} the symmetric group, with the universal curve $\overline{\mathcal{C}}_a \to \overline{M}_a$ pulled back as an orbifold fiber product; a finite dimensional Γ_a-representation E_a; and a Γ_a-equivariant linear map $\Theta_a : E_a \to C^\infty(\overline{\mathcal{C}}_a \times M, \Omega^{0,1}_{\overline{\mathcal{C}}_a/\overline{M}_a} \otimes_{\mathbb{C}} TM)$, supported away from the nodes and marked points of the fibers of $\overline{\mathcal{C}}_a$.

For a finite $I \subset \mathfrak{A}$ the thickened moduli space $\mathcal{M}_I$ is the set of isomorphism classes of tuples $(u, \{\psi_a\}_{a\in I}, \{\varepsilon_a\}_{a\in I})$ as follows. First, $u : \Sigma \to M$ is a smooth stable map of charge class (A, β, k), transverse to each Z_a, $a \in I$, with exactly r_a intersection points, away from the nodes and from ∂Z_a. Here stable means as in Lemma 4.9. Second, $\psi_a : \Sigma \to \overline{\mathcal{C}}_a$ is an isomorphism of Σ, with the points $u^{-1}(Z_a)$ added as marked points, onto a fiber of $\overline{\mathcal{C}}_a \to \overline{M}_a$. Third, $\varepsilon_a \in E_a$, and

$$\overline{\partial}_J u + \sum_{a\in I} \Theta_a(\varepsilon_a)(\psi_a, u) = 0, \tag{5.4}$$

where $\Theta_a(\varepsilon_a)(\psi_a, u)$, the pullback of $\Theta_a(\varepsilon_a)$ by $(\psi_a, u) : \Sigma \to \overline{\mathcal{C}}_a \times M$, is a $(0,1)$-form on Σ with values in u^*TM. We have renamed Pardon's α, D_α, $\overline{\mathcal{M}}_\alpha$, λ_α, ϕ_α, e_α to a, Z_a, $\overline{M}_a$, Θ_a, ψ_a, ε_a, to avoid clashes with our notation. $\mathcal{M}_I$ has the Gromov topology, which for $I \neq \emptyset$ is by definition the topology of Hausdorff distance on the images $((\psi_a)_{a\in I}, u)(\Sigma) \subset \prod_{a\in I} \overline{\mathcal{C}}_a \times M$, together with the ε_a, [31, Definition 9.2.4, footnote 34]. As the ψ_a are injective, this image and the ε_a determine the tuple up to isomorphism, so this is a metric topology.

The group $\Gamma_I = \prod_{a\in I} \Gamma_a$ acts on $\mathcal{M}_I$ through the ψ_a and the ε_a. Set $E_I := \bigoplus_{a\in I} E_a$. The Kuranishi map $s_I : \mathcal{M}_I \to E_I$ is $s_I(u, \psi, \varepsilon) = \varepsilon$. For $I \subset I'$ we write $s_{I'\setminus I} := (\varepsilon_a)_{a\in I'\setminus I} : \mathcal{M}_{I'} \to E_{I'\setminus I}$. The forgetful map $s^{-1}_{I'\setminus I}(0) \to \mathcal{M}_I$ drops the ψ_a and the $\varepsilon_a = 0$ for $a \in I' \setminus I$. For $I = \emptyset$ its target is $\mathcal{M}_\emptyset = \mathcal{M}$. The zero set $s_I^{-1}(0) = \{\varepsilon = 0\}$ consists of tuples with $u \in \mathcal{M}$, and the forgetful map identifies $s_I^{-1}(0)/\Gamma_I$ with the open subset of $\mathcal{M}$ of curves transverse to the Z_a, $a \in I$, with exactly r_a intersection points, away from the nodes and from ∂Z_a, whose domains with these points added are isomorphic to fibers of the $\overline{\mathcal{C}}_a$: the footprint of $\mathcal{M}_I$. The regular locus $\mathcal{M}_I^{reg} \subset \mathcal{M}_I$ consists of the tuples with trivial automorphism group at which the linearization of (5.4) in (u, ε), for the fixed domain Σ, is onto. Triviality of the automorphism group is automatic for $I \neq \emptyset$, as the ψ_a are injective.

This is Pardon's atlas on $\mathcal{M}$, when $\mathcal{M}$ is compact. Here $\mathcal{M}_I$ is open and closed in Pardon's thickened space of all genus one stable maps in class A: convergence in the topology above implies convergence in the C^0-topology of Lemma 4.9, so the charge class is locally constant on that space.

For our N we set

$$N_I := \{(u, \psi, \varepsilon) \in \mathcal{M}_I : \varepsilon \neq 0 \text{ or } u \in N\},$$

discarding the honest solutions, those with $\varepsilon = 0$, outside N. We set $N_I^{reg} := N_I \cap \mathcal{M}_I^{reg}$. The openness of N in $\mathcal{M}$ is used to show that N_I is open in $\mathcal{M}_I$. Indeed, $\mathcal{Z}_{N,I} := \{\varepsilon = 0,\ u \in N\}$ is open in the closed set $s_I^{-1}(0)$, as the forgetful map $s_I^{-1}(0) \to \mathcal{M}$ is continuous. Continuity holds because, for $a \in I$, $(\psi_a, u)(\Sigma)$ is the graph of u over its domain stabilized by the points $u^{-1}(Z_a)$, as in the topology of [31, Section 9.1]. So $s_I^{-1}(0) \setminus \mathcal{Z}_{N,I}$ is closed in $\mathcal{M}_I$, and its complement N_I is open in $\mathcal{M}_I$. Also $s_I^{-1}(0) \cap N_I = \mathcal{Z}_{N,I}$ and $N_\emptyset = N$. For a compact N not open in $\mathcal{M}$, honest solutions outside N would accumulate at points of $\mathcal{Z}_{N,I}$, which would then have no neighborhood in N_I that is a manifold, and the openness and submersion axioms below would fail. The virtual fundamental class is extracted from the system $\{(N_I, \Gamma_I, E_I, s_I, N_I^{reg})\}_I$, with the forgetful maps, by homological algebra, [31, Sections 4, 5].

Pardon's verification, [31, Theorem 9.2.10], of the axioms of an implicit atlas, [31, Definition 3.1.1], for the system $\{\mathcal{M}_I\}_I$ uses a taming form only through the compactness of $\mathcal{M}$. The compatibility and

homeomorphism axioms, concerning the forgetful maps $s^{-1}_{I'\setminus I}(0) \to \mathcal{M}_I$, $I \subset I'$, are checked from the definitions, using Lemma 5.3 for $I = \emptyset$, to identify the topology of [31, Section 9.1] on the target $\mathcal{M}$ with the Gromov topology. The openness and submersion axioms are local. They say that $\mathcal{M}_I^{reg}$ is open in $\mathcal{M}_I$ and a topological manifold of dimension $2c_1(A) + \dim E_I$, and that $s_{I'\setminus I} : \mathcal{M}_{I'} \to E_{I'\setminus I}$ is a topological submersion over the preimage of $\mathcal{M}_I^{reg}$ under the forgetful map. They are proved by the implicit function theorem near points with smooth domain, and by gluing near points with nodal domain, [31, Proposition 9.2.6 and Appendix B].

The covering axiom, that every $u \in \mathcal{M}$ lies in the image under the forgetful map of $s_I^{-1}(0) \cap \mathcal{M}_I^{reg}$ for some I, follows from domain stabilization, [31, Lemma 9.2.8], and from the choice of Θ_a: one chooses it so that $\varepsilon_a \mapsto \Theta_a(\varepsilon_a)(\psi_a, u)$, followed by the projection to the cokernel of the linearized $\overline{\partial}_J$ operator at u, maps E_a onto this cokernel. Hausdorffness of the thickenings, also required, holds for any J: for $I \neq \emptyset$ the topology is metric, hence Hausdorff. For $I = \emptyset$ it is the Gromov topology of $\mathcal{M}$, Hausdorff in any genus for any almost complex structure, Siebert [39, Propositions 3.8, 3.11, Remark 3.12].

All these conditions except the covering axiom are conditions on open subsets and on local models, so they restrict from $\{\mathcal{M}_I\}_I$ to the system $\{N_I\}_I$ of open subsets. For the homeomorphism axiom note also that an element of $s^{-1}_{I'\setminus I}(0)$ and its image in $\mathcal{M}_I$ have the same u and the same ε_a, $a \in I$, the others being 0, so the first lies in $N_{I'}$ iff the second lies in N_I. The covering axiom restricts as well: a lift of $u \in N$ to $s_I^{-1}(0) \cap \mathcal{M}_I^{reg}$ lies in $\mathcal{Z}_{N,I}$, hence in N_I^{reg}. So in the construction of the atlas the taming form is used in one place: [31, Definition 3.1.1] asks that the moduli space carrying the atlas be compact, which for $\mathcal{M}$ is Gromov compactness. In our situation the moduli space carrying the atlas is N, and its compactness is a hypothesis. So an open compact N as above carries a natural implicit atlas, hence a virtual fundamental class in the sense of [31]. The atlas is oriented, as in [31]: the linearized operators are complex linear up to compact perturbation. This uses no taming form.

5.2. **The Gromov–Witten invariant and its dependence.** We may thus define, when $c_1(A) = 0$, so that the virtual dimension $2c_1(A)$ of N vanishes,

$$GW_1^k(N, J, A, \beta) := \int_{[N]^{vir}} 1 \in \mathbb{Q}, \tag{5.5}$$

the degree of the virtual fundamental class of N, [31, Section 5.1, Definition 9.3.1].

We now study how these counts depend on N, J. Given a Fréchet smooth family $\{J_t\}$, $t \in [0,1]$, of almost complex structures on M, a smooth family for short, we denote by $\overline{\mathcal{M}}_1^k(\{J_t\}, A, \beta)$, the ***cobordism moduli space***, the space of pairs (u,t), $u \in \overline{\mathcal{M}}_1^k(J_t, A, \beta)$. It is topologized as a subspace of $\mathcal{S}^k(A,\beta) \times [0,1]$, for $\mathcal{S}^k(A,\beta)$ the space of all smooth stable maps of charge class (A, β, k) with respect to α, defined in Section 5.3. The subspace topology of $\mathcal{S}^k(A,\beta)$ on J'-holomorphic curves is the Gromov topology for any J', see the text following property (c) there. Likewise, by (a), (b), (c), Lemma 4.9 and the sequential compactness for varying J_j in Theorem 5.2, the cobordism moduli space is metrizable, and its convergent sequences are the Gromov convergent ones, with $t_j \to t$. So its topology does not depend on the auxiliary metric g of Section 5.3.

Lemma 5.6. *Let $\{J_t\}$, $t \in [0,1]$, be a smooth family of almost complex structures on M, and α a closed integral 1-form on M with $[\alpha] \neq 0$. Suppose that $\widetilde{N}$ is an open compact subset of the cobordism moduli space $\overline{\mathcal{M}}_1^k(\{J_t\}, A, \beta)$, with $c_1(A) = 0$. Let*

$$N_i := \{u \,|\, (u,i) \in \widetilde{N}\}, \quad i = 0, 1,$$

the **slices** *of $\widetilde{N}$, open compact subsets of $\overline{\mathcal{M}}_1^k(J_i, A, \beta)$, see the proof; then*

$$GW_1^k(N_0, J_0, A, \beta) = GW_1^k(N_1, J_1, A, \beta).$$

In particular if $GW_1^k(N_0, J_0, A, \beta) \neq 0$, there is an element of $\overline{\mathcal{M}}_1^k(J_1, A, \beta)$, a J_1-holomorphic stable map of a nodal elliptic curve into M, of charge class (A, β, k).

Proof of Lemma 5.6. The set $N_i \times \{i\} = \widetilde{N} \cap (\mathcal{S}^k(A,\beta) \times \{i\})$ is closed in $\widetilde{N}$, hence compact, and open in $\overline{\mathcal{M}}_1^k(J_i, A, \beta) \times \{i\}$, as $\widetilde{N}$ is open. The subspace topology there is the Gromov topology, Section 5.3.

The construction of Section 5.1 goes through unchanged for the cobordism moduli space. Here Lemma 5.3 is used for J_{t_i}-holomorphic u_i, $t_i \to t_0$. Its proof applies with ω_g replaced by $g_{t_i}(J_{t_i}\cdot, \cdot)$, for a smooth family $\{g_t\}$ of J_t-invariant metrics, and with Theorem 5.2 for varying J_j. The equality $e(u_0) = 2\int u_0^* \omega_g$ there becomes an equality up to an error tending to 0, as $g_{t_i}(J_{t_i}\cdot, \cdot) \to g_{t_0}(J_{t_0}\cdot, \cdot)$. This suffices for the bound on the energies. With (c), the lemma then identifies the topology of the cobordism moduli space with that of [31, Section 9.1], as for a single J.

Exactly as in the proof that the Gromov–Witten invariants do not depend on J, [31, Lemma 9.3.2], we construct a natural implicit atlas on $\widetilde{N}$, with boundary in the sense of [31, Section 3.2]. Its boundary is $N_0^{op} \sqcup N_1$, *op* denoting the opposite orientation. And so we immediately get

$$GW_1^k(N_0, J_0, A, \beta) = GW_1^k(N_1, J_1, A, \beta).$$

□

Remark 5.7. Positivity of the charge, $k \geq 1$, rules out degenerations to constant curves. This is not essential. Gromov limits are stable maps of type $(1,0)$, and none of these is constant, as there are no stable curves of type $(1,0)$. Energy quantization for non-constant J-holomorphic curves of any genus on a closed manifold gives a second proof. This follows from the monotonicity theorem, Zinger [47, Proposition 3.12], as noted to me by Spencer Cattalani.

5.3. **The space of smooth stable maps.** Let $\mathcal{S}^k(A,\beta)$ denote the space of equivalence classes of all smooth, stable maps of nodal elliptic curves into M of charge class (A,β,k), with the topology defined by the quasi-distance functions $\{\rho_\epsilon\}_\epsilon$ of McDuff–Salamon [28, 2nd ed., pages 143–144] on the set of stable maps. These involve the C^0 distance, the domains and the energy e_g, for an auxiliary Riemannian metric g on M. McDuff–Salamon treat the genus 0 case. The higher genus case is likewise well known. Elements of $\mathcal{S}^k(A,\beta)$ are like elements of $\overline{\mathcal{M}}_1^k(J, A, \beta)$, but are not required to be J-holomorphic. Stable means as in Lemma 4.9.

We use only the following three properties of $\mathcal{S}^k(A,\beta)$. (a) Every element of $\mathcal{S}^k(A,\beta)$ has a neighborhood on which e_g is bounded. (b) Gromov convergence of J_j-holomorphic stable maps, $J_j \to J'$ in C^∞, for any almost complex structure J' on M, as in Theorem 5.2, implies convergence in $\mathcal{S}^k(A,\beta)$. (c) For any smooth family $\{J_t\}$, $t \in [0,1]$, of almost complex structures on M, the subspace of $\mathcal{S}^k(A,\beta) \times [0,1]$ of pairs (u,t) with u J_t-holomorphic is metrizable, so that its subsets are compact if and only if they are sequentially compact. For a constant family, the genus 0 analogue of (c) is the metrizability of the Gromov topology, [28, 2nd ed., Theorem 5.6.6]. The general case is likewise well known. We do not use metrizability of $\mathcal{S}^k(A,\beta)$ itself.

By (a), (b), (c), the sequential compactness in Theorem 5.2, and Lemmas 4.9 and 5.3, for any almost complex structure J' on M the subspace topology on $\overline{\mathcal{M}}_1^k(J', A, \beta) \subset \mathcal{S}^k(A,\beta)$ is the Gromov topology: a sequence in $\overline{\mathcal{M}}_1^k(J', A, \beta)$ converging in $\mathcal{S}^k(A,\beta)$ to an element of $\overline{\mathcal{M}}_1^k(J', A, \beta)$ has bounded energy by (a), hence a Gromov convergent subsequence. Its limit has charge class (A,β,k) by Lemma 4.9. So it is the given limit, by (b) and (c). This applies to every subsequence. Gromov convergence implies convergence in the Gromov topology, as noted before Lemma 5.3. So the sequence converges to that limit in the Gromov topology. Conversely, convergence in the Gromov topology implies Gromov convergence, by Lemma 5.3, hence convergence in $\mathcal{S}^k(A,\beta)$, by (b). The subspace topology is first countable, being metrizable by (c). The Gromov topology is first countable, as noted before Lemma 5.3. So the two topologies, having the same convergent sequences, coincide.

6. Elliptic curves in the lcs-fication of a contact manifold and the Fuller index

The main result of this section is Theorem 6.14. Its proof, and that of Proposition 7.1, rest on the following elementary lemma.

Lemma 6.1. *Let (λ, α, J) be a tamed first kind lcs pair on M. Then every non-constant closed connected (nodal) J-holomorphic curve $u : \Sigma \to M$ is a Reeb 2-curve.*

Proof of Lemma 6.1. Let $u : \Sigma \to M$ be as in the lemma. We may assume M connected, replacing it by its component containing $u(\Sigma)$. By Lemma 4.5, for each $z \in \widetilde{\Sigma}$ the image of $d\widetilde{u}(z)$ lies in $\mathcal{V}_\lambda = \mathcal{D}$. Being J-invariant, it is $\mathcal{D}$ wherever it is non-zero, $\mathcal{D}$ being 2-dimensional. So it is enough to show that $u^*[\alpha] \neq 0$ in $H^1(\Sigma, \mathbb{R})$. Let $\widetilde{M}$ denote the α-covering space of M, that is the space of equivalence classes of paths p starting at $x_0 \in M$, with a pair p_1, p_2 equivalent if $p_1(1) = p_2(1)$ and

$$\int_{[0,1]} p_1^*\alpha = \int_{[0,1]} p_2^*\alpha.$$

Then the lift of ω to $\widetilde{M}$ is

$$\widetilde{\omega} = \frac{1}{f} d(f\widetilde{\lambda}),$$

where $\widetilde{\lambda}$ is the lift of λ and $f = e^{-g}$. Here $g([p]) := \int_{[0,1]} p^*\alpha$, well defined by the equivalence relation. It is a primitive of the lift $\widetilde{\alpha}$ of α to $\widetilde{M}$, that is $\widetilde{\alpha} = dg$. In particular $f\widetilde{\omega} = d(f\widetilde{\lambda})$ is an exact symplectic form on $\widetilde{M}$. Let $\widetilde{J}$ be the lift of J. It is $f\widetilde{\omega}$-compatible, as J is ω-compatible and $f > 0$. So any closed connected $\widetilde{J}$-curve is constant by Stokes theorem. Now suppose $u^*[\alpha] = 0$. Then $\int_{u\circ c} \alpha = 0$ for every loop c in Σ. So u lifts to a $\widetilde{J}$-holomorphic map $v : \Sigma \to \widetilde{M}$. Since Σ is closed, it follows by the above that v is constant, so that u is constant, which is impossible. □

6.1. **Preliminaries on Reeb tori.** Throughout the rest of this section and the next, (C, λ) is a closed contact manifold. Let (λ, α) be the lcs-fication of (C, λ) on $M = C \times S^1$. For $\beta \in \pi_1^{\text{free}}(C)$ and $k \geq 1$ we set $A_\beta^k := k \cdot \beta \otimes [S^1] \in H_2(M, \mathbb{Z})$, for β also denoting the Hurewicz image of the class in $H_1(C, \mathbb{Z})$. We abbreviate A_β^1 by A_β. Let $\mathcal{O}(R^\lambda, \beta)$ be the orbit space as in the Introduction. Let J^λ on $C \times S^1$ be as in Example 4.6 and Section 4.2. We have a map:

$$\mathcal{P} : \mathcal{O}(R^\lambda, \beta) \to \overline{\mathcal{M}}_1^1(J^\lambda, A_\beta^1, \beta), \quad \mathcal{P}(o) = u_o, \tag{6.2}$$

for u_o the Reeb torus of o, Section 4.2.

We set up some terminology. Let $pr_C : C \times S^1 \to C$ and $pr_{S^1} : C \times S^1 \to S^1$ be the projections. We write ξ both for $\ker \lambda \subset TC$ and for its horizontal lift $\ker \lambda \cap \ker \alpha \subset TM$ of Example 4.6. The two are identified by dpr_C. For $o \in \mathcal{O}(R^\lambda, \beta)$ let $m = m(o)$ be its multiplicity, Appendix A, so that $o = m\, o_s$ for a simple orbit o_s. Here $m\, o_s$ is the m-fold iterate of o_s, defined there. Set

$$\mathcal{T}_o := \text{image}\, u_o = \text{image}\, o_s \times S^1,$$

an embedded torus, tangent to the distribution $\mathcal{D} = \text{span}(R^\lambda, \partial_\theta)$. For $0 \leq \ell < m$ define the ***sheared Reeb tori***

$$u_{o,\ell} : T^2 \to M, \quad u_{o,\ell}(s,t) = (o_s(ms + \ell t), t), \tag{6.3}$$

covering maps of degree m onto $\mathcal{T}_o$, of charge 1, class A_β and π-class β for the orientation (∂_s, ∂_t) of T^2, with $u_{o,0} = u_o$. Each representative of o is the m-fold iterate of a representative of o_s. Replacing o_s by $\theta \cdot o_s$ precomposes $u_{o,\ell}$ with the translation $(s,t) \mapsto (s + \theta/m, t)$. So the class of $u_{o,\ell}$ in the moduli spaces below depends only on (o, ℓ). In particular $\mathcal{P}$ is well defined. For J admissible the complex orientation of the domain of $u_{o,\ell}$ is (∂_s, ∂_t), as $J(R^\lambda, 0)$ has positive ∂_θ-component, see the proof of Proposition 6.4. So the fiber class of $u_{o,\ell}$ is that of $S^1 \times \{t\}$ oriented by ∂_s, as for u_o in Section 4.2.

Let $u : \Sigma \to C \times S^1$ be a smooth map from a closed oriented surface and t_0 a regular value of $pr_{S^1} \circ u$. Then each component S_0 of $(pr_{S^1} \circ u)^{-1}(t_0)$ is an embedded circle in Σ. We orient S_0 so that (w, v) is a positive basis of $T\Sigma$ for w a positive tangent vector of S_0 and v any vector with $d(pr_{S^1} \circ u)(v) > 0$. When $pr_{S^1} \circ u$ is a fibration of a torus, this is the orientation of the fiber class, fixed before Definition 4.7 by $[\gamma] \cdot [\rho] = 1$ and $\langle [\rho], u^*[\alpha] \rangle > 0$. We say that u ***traces*** o on S_0 if $pr_C \circ u|_{S_0}$ is, up to orientation preserving reparametrization of S_0, a representative of the string o.

The sheared Reeb torus $u_{o,\ell}$ traces o on every component of every fiber of $pr_{S^1} \circ u_{o,\ell}$, these fibers being the circles $S^1 \times \{t_0\}$. We can say more:

Proposition 6.4. *Let J be (λ, α)-admissible. Each $u_{o,\ell}$ is J-holomorphic for a unique complex structure on T^2, and every $u \in \overline{\mathcal{M}}_1^1(J, A_\beta, \beta)$ is the class of $u_{o,\ell}$ for a unique pair (o, ℓ), $o \in \mathcal{O}(R^\lambda, \beta)$, $0 \le \ell < m(o)$. Here o is the orbit string traced by u, in the sense above, on any component of any regular fiber of $pr_{S^1} \circ u$. In particular the map (6.2), with J in place of J^λ, is injective, with image*

$$\overline{\mathcal{M}}^{Reeb}(J, A_\beta, \beta) := \mathcal{P}(\mathcal{O}(R^\lambda, \beta)),$$

whose elements, the classes of the maps u_o, we also call **Reeb tori**.

In the particular case of J^λ, the domains of the Reeb tori are "rectangular", that is quotients of the complex plane by rectangular lattices. The domains of the sheared Reeb tori (6.3), $\ell \neq 0$, need not be rectangular: the lattice of $u_{o,\ell}$ is $\mathbb{Z} + \mathbb{Z}(\frac{\ell}{m} + \frac{i}{c})$, for c the period of o. Rectangularity can also fail for the generalized Reeb tori of Section 7.1, for the mapping torus of some strict contactomorphism.

Proof of Proposition 6.4. Since $\mathcal{T}_o$ is tangent to the J-invariant distribution $\mathcal{D}$, it is a J-holomorphic torus, so each covering map $u_{o,\ell} : T^2 \to \mathcal{T}_o$ is J-holomorphic for the unique pulled back complex structure. In particular $\mathcal{P}(o)$, the class of $u_o = u_{o,0}$, lies in $\overline{\mathcal{M}}_1^1(J, A_\beta, \beta)$. We now classify the elements of the moduli space. Suppose we have a curve $u \in \overline{\mathcal{M}}_1^1(J, A_\beta, \beta)$, represented by $u : \Sigma \to M$. As u has charge 1, it is non-constant. So u is a Reeb 2-curve, by Lemma 6.1.

We show that $\widetilde{\Sigma}$ has no spherical components. A non-constant one gives a J-holomorphic map $u' : \mathbb{CP}^1 \to M$. By Lemma 6.1 it is a Reeb 2-curve. This is impossible by the second property in Definition 1.12, as $H^1(\mathbb{CP}^1, \mathbb{R}) = 0$. So all spherical components are constant. As u is non-constant, $\widetilde{\Sigma}$ then has a genus one component. The constant spheres form trees attached to it at single nodes. A leaf of such a tree carries a single special point, contradicting stability. As $\widetilde{\Sigma}$ has no spherical components, Σ is a smooth torus by the description of nodal elliptic curves in Section 4.1. That is, u is normal.

We now show that u is a covering map onto $\mathcal{T}_o$ for some o. By the charge 1 condition $pr_{S^1} \circ u$ is surjective. By the Sard theorem we have a regular value $t_0 \in S^1$, so that $(pr_{S^1} \circ u)^{-1}(t_0)$ contains an embedded circle $S_0 \subset \Sigma$. Now $d(pr_{S^1} \circ u)$ is surjective onto $T_{t_0}S^1$ along $T\Sigma|_{S_0}$. So, by the first property in Definition 1.12, $o := pr_C \circ u|_{S_0}$ is an immersion. Moreover, again by the first property, o is tangent to $\ker d\lambda$. It follows that o is an unparametrized closed λ-Reeb orbit. We also write o for its orbit string, of a priori unknown class, and $\mathcal{T}_o := \operatorname{image} o_s \times S^1$ as above. That the class is β is shown below.

The map $pr_C \circ u|_{S_0}$ is positively oriented, that is u traces o on S_0. Indeed, let w be a positive tangent vector of S_0 and v as in the orientation convention, so that $jw = \epsilon v + \sigma w$ with $\epsilon > 0$. Then $du(w) = (aR^\lambda, 0)$ with $a \neq 0$, and $J(R^\lambda, 0) = (a'R^\lambda, b\,\partial_\theta)$ with $b = \omega((R^\lambda, 0), J(R^\lambda, 0)) > 0$, as J is admissible and $\omega((R^\lambda, 0), (0, \partial_\theta)) = \lambda(R^\lambda) = 1$. So $d(pr_{S^1} \circ u)(jw) = d(pr_{S^1})(Jdu(w)) = ab$. But $d(pr_{S^1} \circ u)(jw) = \epsilon\, d(pr_{S^1} \circ u)(v) > 0$. Hence $a > 0$.

Next, du vanishes at only finitely many points, the zeros of the differential of a non-constant J-holomorphic map being isolated, as for holomorphic maps, by the Carleman similarity principle [28, 2nd ed., Theorem 2.3.5, Lemma 2.4.1]. Elsewhere $du(T\Sigma) = \mathcal{D}$. The distribution $\mathcal{D} = \operatorname{span}(R^\lambda, \partial_\theta)$ is

integrable, with leaves image $\gamma \times S^1$ for γ a Reeb trajectory. Off the zeros of du, a finite set, u is an immersion of a connected surface tangent to $\mathcal{D}$. So it maps into a single leaf, the leaf image $o \times S^1 = \mathcal{T}_o$ through a point of o. This leaf is closed, so image $u \subset \mathcal{T}_o$.

As $\mathcal{T}_o$ is a J-holomorphic torus, $u : T^2 \to \mathcal{T}_o$ is a holomorphic map, hence a covering map, unbranched by the Riemann–Hurwitz formula as both tori have Euler characteristic 0. Coverings $T^2 \to \mathcal{T}_o$ are classified by the finite index subgroups of $\pi_1(\mathcal{T}_o) = \mathbb{Z}^2$, with generators the simple orbit o_s underlying o and the S^1 factor. The covering has charge one iff its subgroup surjects onto the second factor. So $pr_{S^1} \circ u$, a submersion of the closed T^2, is a fibration with connected fibers, as $(pr_{S^1} \circ u)_*$ is onto $\pi_1(S^1)$. Its fiber over t_0 is S_0, the preimage of image $o_s \times \{t_0\}$. Hence S_0 covers image $o_s \times \{t_0\}$ with degree the index m of the subgroup. So $o = m\, o_s$, and m is the multiplicity of o. Oriented as above, S_0 represents the fiber class of u, defined before Definition 4.7. So o has class the π-class β of u, that is $o \in \mathcal{O}(R^\lambda, \beta)$.

The index m subgroups surjecting onto the second factor are $\Lambda_\ell = \langle (m, 0), (\ell, 1) \rangle$, $0 \le \ell < m$, and Λ_ℓ is the subgroup of the map $u_{o,\ell}$ of (6.3). Two coverings with the same subgroup are isomorphic as coverings, and the isomorphism is a biholomorphism for the pulled back complex structures. Conversely, a biholomorphism ϕ with $u_{o,\ell'} \circ \phi = u_{o,\ell}$ is an isomorphism of coverings. So $\Lambda_{\ell'} = \Lambda_\ell$, as $\pi_1(\mathcal{T}_o)$ is abelian, and $\ell' = \ell$. Also o is determined by u, as the orbit string it traces. This proves the classification. In particular $\mathcal{P}$ is injective. By definition of Reeb tori, it is a bijection onto the set of Reeb tori. □

Lemma 6.5. *For a (λ, α)-admissible J, $\overline{\mathcal{M}}^{Reeb}(J, A_\beta, \beta)$ is open and closed in $\overline{\mathcal{M}}^1_1(J, A_\beta, \beta)$. The same holds for the subset of Reeb tori of a cobordism moduli space $\overline{\mathcal{M}}^1_1(\{J_t\}, A_\beta, \beta)$, for $\{J_t\}$ a smooth family of almost complex structures with each J_t admissible for the lcs-fication (λ_t, α) of a contact form λ_t on C, $\{\lambda_t\}$ a smooth family.*

Proof. By the proof of Proposition 6.4 (the subgroups Λ_ℓ) a curve u tracing o is a Reeb torus iff the subgroup $u_*\pi_1(T^2) \subset \pi_1(\mathcal{T}_o)$ contains the class of the S^1 factor. Suppose $u_i \to u$ in the Gromov topology, tracing o_i and o. By Lemma 5.3 the u_i Gromov converge to u. All domains are smooth tori, so after reparametrization $u_i \to u$ in C^∞. Hence the fibers of $pr_{S^1} \circ u_i$ converge to those of $pr_{S^1} \circ u$, and $o_i \to o$ in $\mathcal{O}(R^\lambda, \beta)$. For i large the nearest point projection $\varpi_i : \mathcal{T}_{o_i} \to \mathcal{T}_o$ is a covering map taking the class of the S^1 factor to the class of the S^1 factor. The maps $\varpi_i \circ u_i$ are C^0-close to u after reparametrization, hence homotopic to u as maps into $\mathcal{T}_o$. So $u_*\pi_1(T^2) = (\varpi_i \circ u_i)_*\pi_1(T^2)$. This contains the S^1 class iff $u_{i*}\pi_1(T^2)$ does, as ϖ_{i*} is injective and takes the S^1 class to the S^1 class. Thus the set of Reeb tori and its complement are both closed. The cobordism case is identical, with $\mathcal{T}_{o_i}$ the images of the curves at times $t_i \to t$. In the cobordism moduli space convergent sequences are Gromov convergent, Section 5.2. □

For a (λ, α)-admissible J, as in Proposition 6.4, we also have the map

$$\mathrm{Orb} : \overline{\mathcal{M}}^1_1(J, A_\beta, \beta) \to \mathcal{O}(R^\lambda, \beta), \quad u_{o,\ell} \mapsto o, \tag{6.6}$$

a left inverse of $\mathcal{P}$, whose fiber over o consists of the $m(o)$ curves (6.3). It is continuous, by the argument of Lemma 6.5. It is also proper. Indeed, let $p(o)$ be the period of o, Appendix A. As $u_{o,\ell}^*\omega = p(o)\, ds \wedge dt$, the curves over o have energy $2p(o)$ for the metric $\omega(\cdot, J\cdot)$. This is bounded over a compact set of orbit strings. So a sequence of curves over such a set has a Gromov convergent subsequence by Theorem 5.2, whose limit lies in $\overline{\mathcal{M}}^1_1(J, A_\beta, \beta)$ by Lemma 4.9. The limit lies over the same compact set, as Gromov convergence implies convergence in the Gromov topology, noted before Lemma 5.3, and Orb is continuous. Thus for an open compact $N \subset \mathcal{O}(R^\lambda, \beta)$ both $\mathrm{Orb}^{-1}(N)$ and $\mathcal{P}(N) = \mathrm{Orb}^{-1}(N) \cap \overline{\mathcal{M}}^{Reeb}(J, A_\beta, \beta)$ are open compact subsets of $\overline{\mathcal{M}}^1_1(J, A_\beta, \beta)$, the latter by Lemma 6.5.

Proposition 6.7. *Let (C,λ) be a closed contact manifold and o a period c closed R^λ-Reeb orbit. The kernel of the normal Cauchy–Riemann operator $D:\Omega^0(\mathcal{W})\to\Omega^{0,1}(\mathcal{W})$ of the Reeb torus u_o for J^λ, defined in the proof, is naturally identified with the 1-eigenspace of $F^\lambda_{c,*}$, the time c linearized return map $\xi(o(0))\to\xi(o(0))$ induced by the R^λ Reeb flow. In particular, if o is non-degenerate then u_o is a regular curve, that is the operator (6.8) below is surjective at it. So if λ is non-degenerate, meaning that all its closed Reeb orbits are, all the elements of $\overline{\mathcal{M}}^{Reeb}(J^\lambda,A_\beta,\beta)$ are regular.*

Proof. Abbreviate u_o by u, a J^λ-holomorphic map $u:(T^2,j)\to(M,J^\lambda)$, Section 4.2. As u is immersed, we get a splitting $u^*TM\simeq\mathcal{W}\oplus T(T^2)$, using the metric $\omega(\cdot,J^\lambda\cdot)$, where $\mathcal{W}\to T^2$ denotes the pull-back of the normal bundle to image u for this metric. It is identified with the pullback of $\xi=\mathcal{D}^\perp$, Example 4.6.

The full associated real linear Cauchy–Riemann operator takes the form:

$$D_u^{J^\lambda}:\Omega^0(\mathcal{W}\oplus T(T^2))\oplus T_j\mathfrak{S}\to\Omega^{0,1}(\mathcal{W}\oplus T(T^2)). \tag{6.8}$$

Here $\mathfrak{S}\subset\mathcal{J}(T^2)$, for $\mathcal{J}(T^2)$ the space of complex structures on T^2, is a Teichmüller slice through j, [45, Definition 3.2]: a 2-dimensional family of complex structures through j representing the Teichmüller space of the torus near j. The summand $T_j\mathfrak{S}$ enters (6.8) through $y\mapsto\frac{1}{2}J^\lambda\,du\circ y$. The operator $D_u^{J^\lambda}$ is Fredholm of index 2 (after standard Sobolev completions). Its restriction to $\Omega^0(\mathcal{W}\oplus T(T^2))$ preserves the splitting, that is the restricted operator splits as

$$D\oplus D':\Omega^0(\mathcal{W})\oplus\Omega^0(T(T^2))\to\Omega^{0,1}(\mathcal{W})\oplus\Omega^{0,1}(T(T^2)).$$

The splitting holds since $\omega(\cdot,J^\lambda\cdot)$ is the product metric $g_C+d\theta^2$. Here $g_C:=\lambda^2+g_J(\pi_\xi\cdot,\pi_\xi\cdot)$, for g_J as in Example 4.6, viewed as a metric on $\ker\lambda$ by S^1-invariance, and $\pi_\xi:TC\to\ker\lambda$ the projection along R^λ. Let ∇ be the Levi-Civita connection of $g_C+d\theta^2$. Then R^λ is a unit geodesic field, with $\nabla_\nu R^\lambda\in\xi$ for $\nu\in\xi$. So, by the formula for $D\zeta$ in terms of ∇ recalled below, the linearized operator takes sections of $\mathcal{W}$ to $\Omega^{0,1}(\mathcal{W})$. Sections of $T(T^2)$ go to $\Omega^{0,1}(T(T^2))$ for any immersed J-curve, as $u\circ\phi_\tau$ is J-holomorphic for ϕ_τ^*j, for ϕ_τ the flow of a vector field on T^2. Also the restricted Fredholm index 2 operator

$$\Omega^0(T(T^2))\oplus T_j\mathfrak{S}\to\Omega^{0,1}(T(T^2))$$

is surjective by classical Teichmüller theory, see also [45, Lemma 3.1, Definition 3.2] for a precise argument in this setting. It follows that $D_u^{J^\lambda}$ will be surjective if the restricted Fredholm index 0 operator, the ***normal Cauchy–Riemann operator*** of u,

$$D:\Omega^0(\mathcal{W})\to\Omega^{0,1}(\mathcal{W})$$

has no kernel.

The bundle $\mathcal{W}$ is symplectic, with the form $d\lambda$ on its fibers $\mathcal{W}_z=\xi(u(z))$. Together with J^λ this gives a Hermitian structure (g_λ,j_λ) on $\mathcal{W}$. We have a linear symplectic connection $\mathcal{A}$ on $\mathcal{W}$, which over the slices $S^1\times\{t\}\subset T^2$ is induced by the pullback by u of the linearized R^λ Reeb flow. Specifically the $\mathcal{A}$-transport map from the fiber $\mathcal{W}_{(s_0,t)}$ to the fiber $\mathcal{W}_{(s_1,t)}$ over the path $[s_0,s_1]\times\{t\}\subset T^2$ is given by

$$(u_*|_{\mathcal{W}_{(s_1,t)}})^{-1}\circ(F^\lambda_{c(s_1-s_0)})_*\circ u_*|_{\mathcal{W}_{(s_0,t)}},$$

where $F^\lambda_{c(s_1-s_0)}$ is the time $c\cdot(s_1-s_0)$ map for the R^λ Reeb flow. Here $u_*:\mathcal{W}\to TM$ denotes the natural map (it is the universal map in the pull-back diagram).

The connection $\mathcal{A}$ is defined to be trivial in the t direction, where trivial means that the parallel transport maps along the segments $\{s\}\times[t_0,t_1]$ are the identity under the identifications $\mathcal{W}_{(s,t)}=\xi(o(s))$. In particular the curvature $R_{\mathcal{A}}$, understood as a Lie algebra valued 2-form, of this connection vanishes. The connection $\mathcal{A}$ determines a real linear CR operator $D_{\mathcal{A}}$ on $\mathcal{W}$ in the standard way: take the complex anti-linear part of the vertical differential of a section. Explicitly,

$$D_{\mathcal{A}}:\Omega^0(\mathcal{W})\to\Omega^{0,1}(\mathcal{W})$$

is defined by

$$D_{\mathcal{A}}(\mu)(z) = \tfrac{1}{2}\left(\pi^{vert}(\mu(z)) \circ d\mu(z) + j_\lambda \circ \pi^{vert}(\mu(z)) \circ d\mu(z) \circ j\right),$$

where

$$\pi^{vert}(\mu(z)) : T_{\mu(z)}\mathcal{W} \to T^{vert}_{\mu(z)}\mathcal{W} \simeq \mathcal{W}_z$$

is the $\mathcal{A}$-projection, and where $T^{vert}_{\mu(z)}\mathcal{W}$ is the kernel of the projection $T_{\mu(z)}\mathcal{W} \to T_z T^2$.

Both D and $D_{\mathcal{A}}$ are real Cauchy–Riemann operators on $\mathcal{W}$. They have index 0 by the Riemann–Roch theorem, as $\mathcal{W}$ is pulled back from the circle $S^1 \times \{0\}$, so that $c_1(\mathcal{W}) = 0$.

In fact $D = D_{\mathcal{A}}$. Indeed, for this ∇ and $J := J^\lambda$, $D\zeta = \frac{1}{2}(\nabla\zeta + J\nabla\zeta \circ j) - \frac{1}{2}J(\nabla_\zeta J)\,du$ for $\zeta \in \Omega^0(\mathcal{W})$, [28, 2nd ed., Proposition 3.1.1]. Here $\nabla_{\partial_s}\zeta = c\nabla_{R^\lambda}\zeta = c(L_{R^\lambda}\zeta + \nabla_\zeta R^\lambda)$, as ∇ is torsion free. Also $\nabla_{\partial_t}\zeta = \partial_\theta\zeta$, and $(\nabla_\zeta J)R^\lambda = -J\nabla_\zeta R^\lambda$, as $JR^\lambda = \partial_\theta$ is parallel. So $D\zeta(\partial_s) = \frac{c}{2}(L_{R^\lambda}\zeta + J\partial_\theta\zeta) = D_{\mathcal{A}}\zeta(\partial_s)$, the $\mathcal{A}$-transport along ∂_s being the linearized Reeb flow. As $D\zeta$ and $D_{\mathcal{A}}\zeta$ are $(0,1)$-forms, they are determined by their values on ∂_s. So $D = D_{\mathcal{A}}$.

This is the standard computation of the linearized operator at a trivial cylinder over a Reeb orbit in a symplectization, of which u_o is the $\mathbb{Z}$-quotient. Indeed, $(C \times S^1, J^\lambda)$ is the quotient of $\mathbb{R} \times C$, with $\theta = -r$ as in Section 1.2 and the cylindrical almost complex structure $J\partial_r = R^\lambda$, $J = J^\lambda$ on ξ, by $r \mapsto r + 1$. There the linearized operator at the trivial cylinder $(s,t) \mapsto (cs, o(t))$, holomorphic for $j\partial_s = \partial_t$ as $J\partial_r = R^\lambda$, is $\partial_s + J\nabla^\xi_t$, with t the orbit parameter. That is, the roles of s and t are exchanged relative to u_o. Here ∇^ξ is the connection on ξ along o whose parallel transport is the linearized Reeb flow, see Wendl [43, Definition 3.22, Remark 3.24]. See also [29, Section 10.1] for a computation of this kind in much greater generality.

We have a differential 2-form Ω on the total space of $\mathcal{W}$ defined as follows. On the vertical spaces $T^{vert}\mathcal{W}$, $\Omega = (u_*)^*\omega$, for $\omega = d_\alpha\lambda$. On the $\mathcal{A}$-horizontal distribution Ω vanishes, and $\Omega(v,h) := 0$ for v vertical and h horizontal. The 2-form Ω is closed, which we may check explicitly by using that $R_{\mathcal{A}}$ vanishes to obtain local symplectic trivializations of $\mathcal{W}$ in which $\mathcal{A}$ is trivial. Clearly $\int_0 \Omega = 0$ over the 0-section, since it is an $\mathcal{A}$-flat section. Any section is homotopic to the 0-section and Ω is closed, so in particular if $\mu \in \ker D$ then $\int_\mu \Omega = 0$.

Let $\mu \in \ker D$. Its vertical differential is complex linear. It follows that the vertical differential vanishes. To see this note that $\Omega(v, j_\lambda v) > 0$, for $0 \neq v \in T^{vert}\mathcal{W}$. So if the vertical differential did not vanish we would have $\int_\mu \Omega > 0$. So μ is $\mathcal{A}$-flat. In particular the restriction of μ over all slices $S^1 \times \{t\}$ is identified with a period c orbit of the R^λ Reeb flow linearized at o. This orbit does not depend on t as $\mathcal{A}$ is trivial in the t variable. So the kernel of D is identified with the vector space of period c orbits of the R^λ Reeb flow linearized at o, as needed. □

Proposition 6.9. *Let λ be a contact form on a $(2n+1)$-fold C, and o a non-degenerate, period c, λ-Reeb orbit of class β. Then the orientation of $[u_o]$ induced by the determinant line bundle orientation of $\overline{\mathcal{M}}^1_1(J^\lambda, A_\beta, \beta)$ is $(-1)^{CZ(o)-n}$, for $CZ(o)$ the Conley–Zehnder index of Appendix A. By (A.2) this is*

$$\operatorname{sign}\operatorname{Det}(\mathrm{id}\,|_{\xi(o(0))} - F^\lambda_{c,*}|_{\xi(o(0))}).$$

The sign of a Reeb torus is thus the fixed point index of the return map of its orbit. This is the analogue of the computation of Ionel–Parker [21, Proposition 4.3, Lemmas 4.5, 4.6] for the symplectic mapping torus of a symplectomorphism, where the torus over a fixed point counts with its Lefschetz sign. There it is obtained by gluing, here by spectral flow.

Proof of Proposition 6.9. Abbreviate u_o by u. Let $\mathcal{W} \to T^2$ be the vector bundle associated to u as in the proof of Proposition 6.7. Fix a unitary trivialization τ of $\mathcal{W}$ induced by a unitary trivialization

of the contact distribution $(\xi, d\lambda, J)$ along o in the obvious sense: $\mathcal{W}$ is the pullback of ξ along the composition

$$T^2 \to S^1 \xrightarrow{o} C.$$

Let the symplectic connection $\mathcal{A}$ on $\mathcal{W}$ be defined as before. Then the push-forward connection $\mathcal{A}' := \tau_*\mathcal{A} = (\tau^{-1})^*\mathcal{A}$ on $T^2 \times \mathbb{R}^{2n}$, for $\mathbb{R}^{2n} = \mathbb{C}^n$ with its standard structures ω_{st}, $J_{\mathrm{st}} = i$, is a connection whose parallel transport paths $p : [0,1] \to Sp(2n, \mathbb{R})$, along the closed loops $S^1 \times \{t\}$, start at id and are independent of t. The parallel transport of $\mathcal{A}'$ along $\{s\} \times S^1$ is the identity, as $\mathcal{A}$ is trivial in the t direction by definition and τ is pulled back from the circle. That is, $\mathcal{A}'$ is trivial in the t variable. We shall call such a connection $\mathcal{A}'$ on $T^2 \times \mathbb{R}^{2n}$ ***induced by*** p.

By the non-degeneracy assumption on o, the map $p(1)$ has no 1-eigenvalues. Let $p'' : [0,1] \to Sp(2n, \mathbb{R})$ be a path from $p(1)$ to a unitary map $p''(1)$, with $p''(1)$ having no 1-eigenvalues, and s.t. p'' has only simple and regular crossings with the Maslov cycle, that is at each crossing time the 1-eigenspace is one-dimensional and the crossing form on it is non-zero. Let p' be the concatenation of p and p''. The path p' is homotopic relative end points to a unitary geodesic path h starting at id and having regular crossings, so $CZ(p') = CZ(h)$. At each interior crossing of h the crossing form has even signature, by unitarity. At time 0 the crossing form of h is $\Gamma(h,0) = \bigoplus_j \vartheta_j |z_j|^2$ on $\mathbb{C}^n$, for $h(t) = \bigoplus_j e^{i\vartheta_j t}$ in unitary coordinates z_j diagonalizing h, with all $\vartheta_j \neq 0$. So $\frac{1}{2} \operatorname{sign} \Gamma(h,0)$ is the number of positive ϑ_j minus the number of negative ones. This is congruent to n modulo 2. Hence

$$CZ(p') \equiv CZ(h) \equiv n \mod 2,$$

in the notation of [32]. Consequently,

$$CZ(p'') \equiv CZ(p) - n \mod 2, \tag{6.10}$$

by additivity of the Conley–Zehnder index.

Let us then define a free homotopy $\{p_r\}$ of p to p': p_r is the concatenation of p with $p''|_{[0,r]}$, all paths being reparametrized to be constant near their endpoints and to have domain $[0,1]$ at each moment r. This determines a homotopy $\{\mathcal{A}'_r\}$ of connections induced by $\{p_r\}$, with CR operators D_r. Here p_0 is a reparametrization of p with the same endpoints. Gradual reparametrization joins $\mathcal{A}'$ to $\mathcal{A}'_0$ through connections with monodromy $p(1)$. By the proof of Proposition 6.7 their CR operators are invertible. The operator of $\mathcal{A}'$ is D in the trivialization τ, as $D = D_{\mathcal{A}}$. So D and D_0 have the same sign. Again by the proof of Proposition 6.7, D_r is surjective except at finitely many times $r_i \in (0,1)$, $i \in \{1, \dots, \nu\}$, the crossing times of p'' with the Maslov cycle. The dimension of the kernel of D_{r_i} is the dimension of the 1-eigenspace of $p''(r_i)$, which is 1 by the assumption that the crossings of p'' are simple.

The operator D_1 need not be complex linear. To fix this we concatenate the homotopy $\{D_r\}$ with the homotopy $\{\widetilde{D}_r\}$ defined as follows. Let $\{\widetilde{\mathcal{A}}_r\}$ be a homotopy of $\mathcal{A}'_1$ to a unitary connection $\widetilde{\mathcal{A}}_1$, where the homotopy $\{\widetilde{\mathcal{A}}_r\}$ is through connections induced by paths $\{\widetilde{p}_r\}$, giving a path homotopy of $p' = \widetilde{p}_0$ to h. All paths are reparametrized to be constant near their endpoints, so that the induced connections are smooth. Then $\{\widetilde{D}_r\}$ is defined to be induced by $\{\widetilde{\mathcal{A}}_r\}$.

Let us denote by $\{D^\sharp_r\}$ the concatenation of $\{D_r\}$ with $\{\widetilde{D}_r\}$. By construction, in the second half of the homotopy $\{D^\sharp_r\}$, $D^\sharp_r$ is surjective. And $D^\sharp_1$ is induced by a unitary connection, since it is induced by the unitary path $\widetilde{p}_1$. Consequently, $D^\sharp_1$ is complex linear. By the above construction, for the homotopy $\{D^\sharp_r\}$, $D^\sharp_r$ is surjective except for ν times in $(0,1)$, where the kernel has dimension one.

Each of these crossings is transversal, that is the orientation of the determinant line changes sign there. Indeed, at the i-th of these times $D^\sharp$ equals D_{r_i}, whose kernel is the 1-eigenspace of the monodromy $p_{r_i}(1) = p''(r_i)$. The crossing form of the spectral flow of $\{D^\sharp_r\}$ is non-zero, Robbin–Salamon [33]. Indeed, restricted to the kernel, the flat sections, $\langle \mu, -J_{\mathrm{st}} \dot{D}^\sharp_r \mu(\partial_s) \rangle_{L^2} = \frac{1}{2} \int \omega_{\mathrm{st}}(\mu, \dot{a}_r \mu)$, for a_r the ds-components of the connection forms. This is a non-zero multiple of the crossing form of the monodromies at $\mu(0)$, which is non-zero by regularity. Also $CZ(p'') \equiv \nu \mod 2$, each simple

regular crossing of p'' contributing ± 1, while $p''(0)$ and $p''(1)$ have no 1-eigenvalue. In particular the sign of $[u]$ by the definition via the determinant line bundle is exactly

$$(-1)^\nu = (-1)^{CZ(p)-n} = (-1)^{CZ(o)-n},$$

by (6.10). Here $CZ(p) = CZ(o)$, as p is, up to reparametrization, the path $t \mapsto F^\lambda_{t,*}|_{\xi(o(0))}$, $0 \le t \le c$, in the trivialization τ, by the definition of $\mathcal{A}$. This is what was to be proved. □

The sheared Reeb tori (6.3) are handled by the same argument.

Lemma 6.11. *Let λ be a contact form on a $(2n+1)$-fold C, and let $o = m\, o_s$ be a λ-Reeb orbit of multiplicity m, with o_s simple of period c_s. Let $P := F^\lambda_{c_s,*}|_{\xi(o(0))}$ be the linearized return map of o_s. Let β be the class of o. For the sheared Reeb torus $u_{o,\ell} \in \overline{\mathcal{M}}^1_1(J^\lambda, A_\beta, \beta)$, $0 \le \ell < m$, set $d := \gcd(\ell, m)$, with $\gcd(0,m) = m$. Then the kernel of the normal Cauchy–Riemann operator $D : \Omega^0(\mathcal{W}) \to \Omega^{0,1}(\mathcal{W})$ of $u_{o,\ell}$, defined as in the proof of Proposition 6.7, is naturally identified with $\ker(P^d - \mathrm{id})$, the 1-eigenspace of the linearized return map of $d\, o_s$. Suppose that o is non-degenerate. Then so is $d\, o_s$, as $\ker(P^d - \mathrm{id}) \subset \ker(P^m - \mathrm{id})$. So $u_{o,\ell}$ is regular. Its orientation sign is*

$$(-1)^{CZ(d\, o_s)-n} = i(d\, o_s),$$

in the notation of (A.2).

Proof. We have $u_{o,\ell} = u_{o_s} \circ \kappa_{m,\ell}$, for $u_{o_s}(\sigma, t) = (o_s(\sigma), t)$ the Reeb torus of o_s and $\kappa_{m,\ell} : T^2 \to T^2$, $\kappa_{m,\ell}(s,t) = (ms + \ell t, t)$, a covering map of degree m. The map $\kappa_{m,\ell}$ is holomorphic for the complex structures on the two tori for which $u_{o,\ell}$ and u_{o_s} are J^λ-holomorphic, as u_{o_s} is an immersion. Hence $\mathcal{W} = \kappa^*_{m,\ell}\mathcal{W}_{o_s}$ and $D = \kappa^*_{m,\ell} D_{o_s}$, for $\mathcal{W}_{o_s}$, D_{o_s} the normal bundle and normal Cauchy–Riemann operator of u_{o_s}. Let $\mathcal{A}_{o_s}$ be the connection on $\mathcal{W}_{o_s}$ of the proof of Proposition 6.7: parallel transport by the linearized R^λ flow along the projection to C of paths in T^2. Its holonomy along the loop $S^1 \times \{t\}$ is P, and along $\{\sigma\} \times S^1$ it is the identity. Set $\mathcal{A} := \kappa^*_{m,\ell}\mathcal{A}_{o_s}$. The loops $s \mapsto (s,t)$ and $t \mapsto (s,t)$ in the domain of $u_{o,\ell}$ are mapped by $\kappa_{m,\ell}$ to loops in the classes $(m,0)$ and $(\ell,1)$ of $\pi_1(T^2) = \mathbb{Z}^2$. So the holonomies of $\mathcal{A}$ along them are P^m and P^ℓ. As $D_{o_s} = D_{\mathcal{A}_{o_s}}$ by the proof of Proposition 6.7, we get $D = \kappa^*_{m,\ell} D_{\mathcal{A}_{o_s}} = D_{\mathcal{A}}$, and the argument there, with the pulled back 2-form Ω, shows that every $\mu \in \ker D$ is $\mathcal{A}$-flat. So $\ker D$ is identified with the subspace of $\xi(o(0))$ fixed by both holonomies, which is $\ker(P^m - \mathrm{id}) \cap \ker(P^\ell - \mathrm{id}) = \ker(P^d - \mathrm{id})$, since d divides m and ℓ and is an integer combination of them.

For the sign we run the proof of Proposition 6.9 with p the path of the linearized R^λ flow along o_s over one period, so that $p(1) = P$, choosing the path p'' generic: $p''(1)$ has no eigenvalue which is a d-th root of unity, and the path $r \mapsto p''(r)^d$ has only regular crossings with the Maslov cycle. The kernels at these crossings need not be one-dimensional, as the eigenvalues $e^{\pm 2\pi i q/d}$ cross together.

Along the resulting homotopy of connections, the operator induced on $\kappa^*_{m,\ell}\mathcal{W}_{o_s}$ has kernel identified, by the above, with the 1-eigenspace of $p_r(1)^d$. Its crossing forms are positive multiples of those of the operator induced on the normal bundle of the Reeb torus $u_{d\, o_s} = u_{o_s} \circ \kappa_{d,0}$ of the d-fold cover, by the pullbacks under $\kappa_{d,0}$ of the same connections, whose kernel is again the 1-eigenspace of $p_r(1)^d$. Indeed $\Lambda_\ell \subset \langle (d,0), (0,1) \rangle$, so $\kappa_{m,\ell} = \kappa_{d,0} \circ \kappa'$ for a covering κ' of degree m/d. The kernel elements on $\kappa^*_{m,\ell}\mathcal{W}_{o_s}$ are the pullbacks under κ' of those on $\kappa^*_{d,0}\mathcal{W}_{o_s}$, both being the flat sections. And L^2 pairings of pulled back sections multiply by a positive constant. At a regular crossing with k-dimensional kernel the orientation of the determinant line changes by $(-1)^k$. Both homotopies end at complex linear operators. So the two spectral flows coincide, and the sign of $u_{o,\ell}$ is that of $u_{d\, o_s}$, which is $(-1)^{CZ(d\, o_s)-n}$ by Proposition 6.9. □

For an open compact $N \subset \mathcal{O}(R^\lambda, \beta)$ and $e \ge 1$ let

$$N^{(e)} := \{o' \in \textstyle\bigsqcup_\gamma \mathcal{O}(R^\lambda, \gamma) \,|\, e\, o' \in N\}$$

be the set of e-th roots of elements of N, where $e\,o'$ is the e-fold iterate of o'. For o' of class γ it has class $e\gamma$, the class of the e-fold iterate of a loop in class γ. Iteration $o' \mapsto e\,o'$ is continuous on the space of all orbit strings. It is proper by the Arzelà–Ascoli theorem, as $p(o') = p(e\,o')/e$ and C is closed. Both facts hold likewise for the orbit strings of a homotopy, Appendix A. So $N^{(e)}$ is open and compact and meets only finitely many classes γ. We write $i(N^{(e)}, R^\lambda)$ for the sum of the Fuller indices of its intersections with these $\mathcal{O}(R^\lambda, \gamma)$. Clearly $N^{(1)} = N$, and $N^{(e)} = \emptyset$ once e exceeds the multiplicity of every element of N. The multiplicity is bounded on N, as the period is continuous, hence bounded on N, while the periods of closed R^λ-orbits are bounded below by a positive constant, as C is closed and R^λ vanishes nowhere. Set

$$\hat{i}(N, R^\lambda, \beta) := \sum_{e\geq 1} \frac{\varphi(e)}{e}\, i(N^{(e)}, R^\lambda), \tag{6.12}$$

for φ Euler's totient function. We call $\hat{i}(N, R^\lambda, \beta)$ the ***inertia Fuller index*** of N. For N finite and consisting of non-degenerate orbits, write $o = m(o)\, o_s$ with o_s simple. Then $o \mapsto o/e := (m(o)/e)\, o_s$ is a bijection from $\{o \in N \,|\, e \mid m(o)\}$ onto $N^{(e)}$, with $m(o/e) = m(o)/e$. So for such N, (6.12) reads

$$\hat{i}(N, R^\lambda, \beta) = \sum_{o\in N} \frac{1}{m(o)} \sum_{d\mid m(o)} \varphi\left(\tfrac{m(o)}{d}\right) i(d\, o_s) = \sum_{o\in N} \frac{1}{m(o)} \sum_{g\in \mathbb{Z}/m(o)} i\left((m(o)/\operatorname{ord}(g))\, o_s\right), \tag{6.13}$$

the last sum being over the stabilizer $\mathbb{Z}/m(o) \subset S^1$ of o under reparametrization. So, for such N, $\hat{i}$ counts pairs (o, g), g in the stabilizer of o, each weighted by $1/m(o)$ times the fixed point index of the quotient orbit string $o/\langle g\rangle = (m(o)/\operatorname{ord}(g))\, o_s$. It is thus an analogue of a Lefschetz number on the inertia groupoid of the orbit space. It coincides with $i(N, R^\lambda, \beta)$ when all elements of N are simple.

Theorem 6.14. *Let (C, λ) be a closed contact manifold, $N \subset \mathcal{O}(R^\lambda, \beta)$ open and compact, and let $\mathcal{P}$,* Orb *be as in (6.2), (6.6), for $J = J^\lambda$. Then*

$$GW_1^1(\mathcal{P}(N), J^\lambda, A_\beta, \beta) = i(N, R^\lambda, \beta), \tag{6.15}$$

$$GW_1^1(\mathrm{Orb}^{-1}(N), J^\lambda, A_\beta, \beta) = \hat{i}(N, R^\lambda, \beta), \tag{6.16}$$

where i is the Fuller index of Appendix A, $\hat{i}$ is as in (6.12), and the left-hand sides are the counts (5.5).

Proof. Suppose first that N is finite and consists of non-degenerate orbits. For $o \in N$ write $o = m(o)\, o_s$, with o_s simple, Appendix A. By Propositions 6.7 and 6.9 the Reeb tori u_o, $o \in N$, are regular with sign $(-1)^{CZ(o)-n} = i(o)$, (A.2), where $\dim C = 2n+1$. By Lemma 6.11 the sheared Reeb tori $u_{o,\ell}$ are regular with sign $i(\gcd(\ell, m(o))\, o_s)$. Each of these curves has isotropy group of order $m(o)$. Indeed, the automorphisms of $u_{o,\ell} = u_{o_s} \circ \kappa_{m(o),\ell}$, in the notation of Lemma 6.11, are the deck transformations of the covering $\kappa_{m(o),\ell}$. This covering is normal of degree $m(o)$, as $\pi_1(T^2)$ is abelian. Hence, by the last paragraph of Appendix B,

$$GW_1^1(\mathcal{P}(N), J^\lambda, A_\beta, \beta) = \sum_{o\in N} \frac{(-1)^{CZ(o)-n}}{m(o)} = i(N, R^\lambda, \beta),$$

and

$$GW_1^1(\mathrm{Orb}^{-1}(N), J^\lambda, A_\beta, \beta) = \sum_{o\in N} \frac{1}{m(o)} \sum_{\ell\in \mathbb{Z}/m(o)} i\left(\gcd(\ell, m(o))\, o_s\right) = \hat{i}(N, R^\lambda, \beta),$$

by (6.13). Here $m(o)/\operatorname{ord}(\ell) = \gcd(\ell, m(o))$ for $\ell \in \mathbb{Z}/m(o)$. Thus, the theorem follows for N as above.

In general, take a sufficiently C^1-small smooth family $\{\lambda_t\}$ of contact forms, $\lambda_0 = \lambda$, with λ_1 non-degenerate. It exists as non-degenerate contact forms are C^∞-generic, see [17], whose argument is independent of the dimension. Take a smooth family $\{J^{\lambda_t}\}$ as in Example 4.6, with $J^{\lambda_0} = J^\lambda$. For this choose S^1-invariant $d\lambda_t$-compatible structures on $\xi_t := \ker\lambda_t \cap \ker\alpha$ smoothly in t, equal at $t = 0$ to the structure on ξ that defines J^λ. This is possible as the space of such structures is contractible.

By the proof of [37, Lemma 4.1] and the Arzelà–Ascoli theorem, as in Appendix A, there is an open compact subset $\widetilde{N}_{\mathcal{O}}$ of $S(\{R^{\lambda_t}\},\beta)/S^1$, in the notation of Appendix A, consisting of the class β orbit strings of the R^{λ_t} near N, with $\widetilde{N}_{\mathcal{O}} \cap (L_\beta C \times \{0\})/S^1 = N$. Set $N_1 := \widetilde{N}_{\mathcal{O}} \cap (L_\beta C \times \{1\})/S^1$, a finite set of non-degenerate orbits. By the invariance of the Fuller index, $i(N_1, R^{\lambda_1}, \beta) = i(N, R^\lambda, \beta)$. Likewise $i(N_1^{(e)}, R^{\lambda_1}) = i(N^{(e)}, R^\lambda)$ for every e, applying the invariance to the set of e-th roots of elements of $\widetilde{N}_{\mathcal{O}}$, open and compact as noted before (6.12), whose slices at 0 and 1 are $N^{(e)}$ and $N_1^{(e)}$. So also $\hat{i}(N_1, R^{\lambda_1}, \beta) = \hat{i}(N, R^\lambda, \beta)$.

On the other hand, the maps Orb of (6.6) for the J^{λ_t} assemble to a continuous proper map $\widetilde{\mathrm{Orb}} : \overline{\mathcal{M}}_1^1(\{J^{\lambda_t}\}, A_\beta, \beta) \to S(\{R^{\lambda_t}\},\beta)/S^1$, $(u,t) \mapsto (\mathrm{Orb}(u), t)$. It is continuous by the cobordism case of the argument of Lemma 6.5. It is proper. Indeed, the curves over (o,t) have energy $2p(o)$ for the metric $d_\alpha \lambda_t(\cdot, J^{\lambda_t}\cdot)$, and p is bounded on compact sets. So the argument given for (6.6) applies, with Theorem 5.2 for varying J_j. Hence $\widetilde{N} := \widetilde{\mathrm{Orb}}^{-1}(\widetilde{N}_{\mathcal{O}})$ is an open compact subset of $\overline{\mathcal{M}}_1^1(\{J^{\lambda_t}\}, A_\beta, \beta)$, with slices $\mathrm{Orb}^{-1}(N)$ at $t=0$ and $\mathrm{Orb}^{-1}(N_1)$ at $t=1$. Intersecting $\widetilde{N}$ with the open and closed subset of Reeb tori of Lemma 6.5, we get the same with $\mathcal{P}(N)$ and $\mathcal{P}(N_1)$ in place of $\mathrm{Orb}^{-1}(N)$ and $\mathrm{Orb}^{-1}(N_1)$. Then by Lemma 5.6, as $c_1(A_\beta) = 0$ by Section 4.1, the left-hand sides of (6.15) and (6.16) do not change when (N,λ) is replaced by (N_1, λ_1). So the theorem follows from the finite non-degenerate case. □

What about higher genus invariants of $C \times S^1$? Following the proof of Proposition 6.4, it is not hard to see that every non-constant J^λ-holomorphic map of a closed connected Riemann surface is a branched cover of a sheared Reeb torus. If one can show that these branched covers are regular when the underlying tori are regular, the calculation of invariants would be fairly automatic from this data. See [46], [44] where these kinds of regularity calculations are made.

7. Proofs of main theorems

7.1. **Mapping tori and Reeb 2-curves.** Let (C,λ) be a closed contact manifold, ϕ a strict contactomorphism of it, and $M = (M_{\phi,1}, \lambda_\phi, \alpha)$ the mapping torus of ϕ, as also appearing in Theorem 2.10. More specifically, $M = C \times \mathbb{R}/\sim$, where the equivalence relation $\sim$ is generated by $(x,\theta) \sim (\phi(x), \theta+1)$, with λ_ϕ and $\alpha = d\theta$ as in Example 2.9 for $c=1$; see also [3]. Then (λ_ϕ, α) is an integral first kind lcs pair on M. We write $\omega := d_\alpha \lambda_\phi$. Mapping tori enter the proofs of Theorems 1.14 and 1.8.

As the pair is first kind, $\mathcal{V}_{\lambda_\phi} = \mathcal{D}$, by the computation following Definition 2.6. In mapping torus coordinates at a point (x,θ), it is spanned by $X_{\lambda_\phi} = (0, -\partial_\theta)$ and $X_\alpha = (R^{\lambda_\theta}, 0)$, for R^{λ_θ} the λ_θ-Reeb vector field, where $\lambda_\theta = \lambda_\phi|_{C_\theta}$, C_θ the fiber over θ of the projection $M \to S^1$. In these coordinates $\lambda_\theta = \lambda$ and $R^{\lambda_\theta} = R^\lambda$; below we write R^λ.

Analogously to Example 4.6 there is a (λ_ϕ, α)-admissible almost complex structure J^{λ_ϕ} on M with $J^{\lambda_\phi}(R^\lambda, 0) = (0, \partial_\theta)$. Indeed, let $\{J_\theta\}_{\theta\in\mathbb{R}}$ be a family of $d\lambda$-compatible almost complex structures on the contact distribution ξ of C, smooth in θ and constant near the integers, with $J_{\theta+1} = \phi_* J_\theta$. It exists as $\phi_* J_0$ is again $d\lambda$-compatible and the space of such structures is contractible. Let J^{λ_ϕ} be the almost complex structure on $C \times \mathbb{R}$ equal to J_θ on $\xi \subset T(C \times \{\theta\})$, with $J^{\lambda_\phi}(R^\lambda, 0) = (0, \partial_\theta)$. It descends to M, as $J_{\theta+1} = \phi_* J_\theta$ and $\phi_* R^\lambda = R^\lambda$. It preserves $\mathcal{D}$ and $\xi = \mathcal{D}^\perp$. It is ω-compatible. Indeed, $\omega = d\lambda$ on ξ, and $\omega((R^\lambda, 0), (0, \partial_\theta)) = \lambda(R^\lambda) = 1$. Hence it is admissible.

We first construct certain Reeb 2-curves in M. By Proposition 7.1, every Reeb 2-curve in M maps each non-constant component onto the image of one of them. Let $o : S^1 \to C$ be a closed λ-Reeb orbit and suppose that image $\phi^k \circ o =$ image o, for some $k > 0$, so that

$$\forall t \in S^1 : \phi^k(o(t)) = o(t + \theta_0)$$

for some $\theta_0 \in [0,1)$, which is determined modulo $1/m(o)$, for $m(o)$ the multiplicity of o. So there are exactly $m(o)$ choices of θ_0. Fix one. Let $\widetilde{o} : S^1 \times [0,k] \to C \times \mathbb{R}$ be the map

$$\widetilde{o}(t,\tau) = (o(t + \theta_0 \cdot \frac{\tau}{k}), \tau).$$

Then $\widetilde{o}$ induces a map $T^2 \simeq S^1 \times ([0,k]/0 \sim k) \to M$, as $(o(t+\theta_0), k) \sim (\phi^{-k}(o(t+\theta_0)), 0) = (o(t), 0)$. We denote it by u^k_{o,θ_0} and call it a ***charge*** k ***generalized Reeb torus of*** o. The $m(o)$ choices of θ_0 give $m(o)$ distinct maps with the same image, denoted $\mathcal{T}_o$: an embedded torus, the leaf through image $o \times \{0\}$ of the distribution $\mathcal{D}$, see the proof of Proposition 7.1. When $\phi^k \circ o = o$ we may take $\theta_0 = 0$. The other choices are the analogues of the sheared Reeb tori (6.3), to which they reduce for $\phi = \mathrm{id}$, $k = 1$. Replacing o by $s \cdot o$ keeps θ_0 and precomposes u^k_{o,θ_0} with the translation $(t,\tau) \mapsto (t+s, \tau)$. So for an orbit string o, u^k_{o,θ_0} is well defined up to translation of the domain.

By construction, u^k_{o,θ_0} is a charge k Reeb 2-curve. Its image $\mathcal{T}_o$ is tangent to $\mathcal{D}$, hence J-holomorphic for every (λ_ϕ, α)-admissible J. So u^k_{o,θ_0} is J-holomorphic with respect to a uniquely determined complex structure on T^2, similarly to the case of Reeb tori of Section 4.2. However, for J^{λ_ϕ}, unlike the case of Reeb tori and J^λ, this complex structure need not be "rectangular" when $\theta_0 \neq 0$: its lattice is $\mathbb{Z} + \mathbb{Z}(\theta_0 + \frac{ik}{p(o)})$, for $p(o)$ the period of o.

We denote by A^k_o the class of u^k_{o,θ_0} in $H_2(M,\mathbb{Z})$. It does not depend on θ_0: the $m(o)$ maps are coverings of the embedded torus $\mathcal{T}_o$ with the same ω-area, k times the period of o, hence of the same degree. For $\phi = \mathrm{id}$ and $\beta = [o]$ we have $A^k_o = k \cdot \beta \otimes [S^1] = A^k_\beta$, the class of Section 6.1. As before we abbreviate A^1_β by A_β. We do not claim that A^k_o depends only on k and $[o]$. So A is a free parameter in Proposition 7.1.

Proposition 7.1. *Let $M = (M_{\phi,1}, \lambda_\phi, \alpha)$ be the mapping torus of a strict contactomorphism ϕ as above, and let J be a (λ_ϕ, α)-admissible almost complex structure on M. Then:*

(1) *Every Reeb 2-curve $u : \Sigma \to M$ maps, via $\widetilde{u}$, each component of $\widetilde{\Sigma}$ on which $\widetilde{u}$ is not constant onto a torus $\mathcal{T}_o = \operatorname{image} u^k_{o,\theta_0}$, for an orbit string o with $\operatorname{image}\phi^k \circ o = \operatorname{image} o$, $k \geq 1$. If Σ is a nodal elliptic curve and u has charge k, then the image of u is a single such torus, with $\operatorname{image}\phi^k \circ o = \operatorname{image} o$ for this k.*

(2) *Let $\beta \in \pi_1^{\mathrm{free}}(C)$ and $A \in H_2(M,\mathbb{Z})$. Let $\iota : C \to M$, $x \mapsto [x,0]$, be the inclusion of the fiber over 0, and $pr_{S^1} : M \to S^1$ the projection. Every element of $\overline{\mathcal{M}}^1_1(J, A, \iota_*\beta)$ is represented by a J-holomorphic covering map $u : T^2 \to \mathcal{T}_o$ of charge 1. Here o is the orbit string traced by u on any component of any regular fiber of $pr_{S^1} \circ u$, in the sense of Section 6.1, applied in the coordinates $C \times \{\theta\}$, $\theta \in [0,1)$. All these components trace the same o, which is thus determined by u. It satisfies $\operatorname{image}\phi \circ o = \operatorname{image} o$. Its class $\beta' \in \pi_1^{\mathrm{free}}(C)$ satisfies $\iota_*\beta' = \iota_*\beta$. For $o \in \mathcal{O}(R^\lambda, \beta')$ with $\operatorname{image}\phi \circ o = \operatorname{image} o$ and $\iota_*\beta' = \iota_*\beta$, the charge 1 elliptic J-curves tracing o are exactly the $m(o)$ generalized Reeb tori u^1_{o,θ_0}, one for each choice of θ_0. They all have class A^1_o. So they lie in $\overline{\mathcal{M}}^1_1(J, A, \iota_*\beta)$ iff $A = A^1_o$.*

(3) *As u^k_{o,θ_0} is immersed, the real linear CR operator (6.8) at it is defined as there, with $\mathcal{W}$ the pullback of ξ. Its Fredholm index is 2. So, subtracting the dimension of the translations, the expected dimension of $\overline{\mathcal{M}}^k_1(J, A^k_o, \iota_*[o])$ is 0, $\iota_*[o]$ being the π-class of u^k_{o,θ_0}.*

(4) *For β and A as in part 2, the map*

$$\mathrm{Orb} : \overline{\mathcal{M}}^1_1(J, A, \iota_*\beta) \to \bigsqcup_{\beta'} \mathcal{O}(R^\lambda, \beta'), \quad u \mapsto o,$$

the union over the $\beta' \in \pi_1^{\mathrm{free}}(C)$ with $\iota_*\beta' = \iota_*\beta$, with o as in part 2, is continuous and proper. Its fiber over o is empty unless $\operatorname{image}\phi \circ o = \operatorname{image} o$ and $A^1_o = A$, and then has exactly $m(o)$ elements. In particular Orb *is injective over simple orbit strings.*

Proof. The proof of part 1 is a variant of the proof of Proposition 6.4. Here $\mathcal{D} = \operatorname{span}(R^\lambda, \partial_\theta)$ is integrable. Its leaves are the surfaces swept out by a Reeb trajectory under the flow of ∂_θ. Such a leaf is compact iff the trajectory is closed and ϕ^k preserves its image for some $k \geq 1$, that is iff the leaf is of the form $\mathcal{T}_o$. Indeed, a leaf closed in M is embedded, and pr_{S^1} restricts to it as a submersion. So its fibers are compact, and their components are closed Reeb orbits in the fibers C_θ, finitely many in each. The ∂_θ-holonomy ϕ^{-1} permutes the components of the fiber over 0 cyclically, as the leaf is connected. Let Σ_i be a component of $\widetilde{\Sigma}$ on which $\widetilde{u}$ is not constant. By the first condition of Definition 1.12 the rank of $d\widetilde{u}$ is 0 or 2 at each point of Σ_i. So on a component W of $\{d\widetilde{u} \neq 0\} \cap \Sigma_i$ the map $\widetilde{u}$ is an immersion into a single leaf L, and $\widetilde{u}(W)$ is open in L. The frontier F of $\widetilde{u}(W)$ in L lies in $\widetilde{u}(\overline{W} \setminus W) \subset \widetilde{u}(\{d\widetilde{u} = 0\})$, which has Hausdorff dimension 0 by the Sard–Federer theorem [9, 3.4.3], as $\widetilde{u}$ is C^∞. A closed set of dimension 0 does not disconnect the surface L. So $L \setminus F$ is connected. As $\widetilde{u}(W)$ is open and closed in $L \setminus F$, $\widetilde{u}(W) = L \setminus F$. The closure $\overline{L}$ of L in M lies in the compact set $\widetilde{u}(\overline{W})$, and $\overline{L} \setminus L \subset \widetilde{u}(\overline{W} \setminus W)$ has Hausdorff dimension 0. As the closure of a leaf is a union of leaves, each of dimension 2, $\overline{L} \setminus L = \emptyset$. So $L = \widetilde{u}(\overline{W})$ is compact, hence a torus $\mathcal{T}_{o_W}$.

There are finitely many such W. Indeed, fix Riemannian metrics on M and Σ_i. As $\widetilde{u}(W) = \mathcal{T}_{o_W} \setminus F$, with F of Hausdorff dimension 0, the area of $\widetilde{u}(W)$ is that of $\mathcal{T}_{o_W}$. It is at most $\int_W |d\widetilde{u}|^2$. These integrals sum to at most $\int_{\Sigma_i} |d\widetilde{u}|^2 < \infty$. The area of a torus $\mathcal{T}_o$ is at least $K^{-1} \int_{\mathcal{T}_o} \omega$, for K a bound on $|\omega|$. As $\omega = \lambda_\phi \wedge \alpha$ on $\mathcal{D}$, this integral is the integral over S^1 of the λ-length of the fiber of $pr_{S^1}|_{\mathcal{T}_o}$, a union of closed Reeb orbits. So it is at least the minimal period of a closed R^λ-orbit. This period T_0 is positive, as C is closed and R^λ vanishes nowhere. So $\int_W |d\widetilde{u}|^2 \geq K^{-1} T_0$ for each W, and there are at most $K T_0^{-1} \int_{\Sigma_i} |d\widetilde{u}|^2$ components W. On the interior of $\{d\widetilde{u} = 0\}$ the map $\widetilde{u}$ is locally constant. Each component of this interior has a frontier point in the closure of some W. The other points of Σ_i lie in the closure of $\{d\widetilde{u} \neq 0\}$, the union of the finitely many $\overline{W}$. So $\widetilde{u}(\Sigma_i)$ is contained in the finite union of the tori $\mathcal{T}_{o_W}$. Distinct leaves are disjoint, so the sets $\widetilde{u}^{-1}(\mathcal{T}_{o_W}) \cap \Sigma_i$ are closed, pairwise disjoint or equal, and cover the connected Σ_i. So they are all equal, and $\widetilde{u}(\Sigma_i)$ is a single torus $\mathcal{T}_o$.

The same argument on the connected Σ, the constant components of $\widetilde{\Sigma}$ mapping to points of the tori of the non-constant ones, shows that a charge k curve u has image a single $\mathcal{T}_o$. Then image $\phi^k \circ o =$ image o. Indeed, let $k' \geq 1$ be least with image $\phi^{k'} \circ o =$ image o. The fibers of $pr_{S^1}|_{\mathcal{T}_o}$ have k' components, the images of $\phi^{-j} \circ o$, $0 \leq j < k'$. So $(pr_{S^1}|_{\mathcal{T}_o})_* \pi_1(\mathcal{T}_o) = k'\mathbb{Z}$, that is the periods of α on $\mathcal{T}_o$ lie in $k'\mathbb{Z}$. Hence the charge k of u is a multiple of k', and ϕ^k preserves image o.

For part 2, let $u \in \overline{\mathcal{M}}_1^1(J, A, \iota_*\beta)$. As u has charge 1, it is non-constant. So it is a Reeb 2-curve by Lemma 6.1, and its domain is a smooth torus, as in the proof of Proposition 6.4. By part 1 u has image $\mathcal{T}_o$, which is an embedded J-holomorphic torus as it is tangent to $\mathcal{D}$. So u is a J-holomorphic map $T^2 \to \mathcal{T}_o$, a covering map by the Riemann–Hurwitz formula.

As u has charge 1, part 1 gives image $\phi \circ o =$ image o. So $\mathcal{T}_o \cap C_\theta =$ image o in the coordinates $C \times \{\theta\}$. So $pr_{S^1}|_{\mathcal{T}_o}$ is a fibration with connected fibers. Its composite $pr_{S^1} \circ u$ with the covering u is a fibration as well. Its fibers are connected, as u has charge 1, so that $(pr_{S^1} \circ u)_*$ is onto $\pi_1(S^1)$. Each fiber covers image o with some degree d, which does not depend on θ. The fibers are positively oriented by the orientation argument in the proof of Proposition 6.4, which applies verbatim. So every fiber traces the d-fold iterate of the simple orbit. Replace o by this iterate. This does not change $\mathcal{T}_o$, which depends only on image o. Oriented as in Section 6.1, a fiber represents the fiber class of u. So $\iota_*[o]$ is the π-class $\iota_*\beta$ of u, as C_θ is isotopic to C_0 in M.

As in the proof of Proposition 6.4, the charge 1 coverings of $\mathcal{T}_o$ whose fibers trace o are classified by the index $m(o)$ subgroups of $\pi_1(\mathcal{T}_o) \simeq \mathbb{Z}^2$ surjecting onto π_1 of the base circle of $\mathcal{T}_o \to S^1$, of which there are $m(o)$. The $m(o)$ maps u^1_{o,θ_0} realize them, their subgroups being pairwise distinct. All of these have class A^1_o, by the definition of A^1_o.

We prove part 3. First $c_1(A^k_o) = 0$. As u^k_{o,θ_0} covers $\mathcal{T}_o$, it suffices that $\langle c_1(TM, J), [\mathcal{T}_o] \rangle = 0$. Now $TM|_{\mathcal{T}_o} = T\mathcal{T}_o \oplus \xi|_{\mathcal{T}_o}$ as complex vector bundles, J preserving $\mathcal{D} = T\mathcal{T}_o$ and $\mathcal{D}^\perp = \xi$. Also

$\langle c_1(T\mathcal{T}_o), [\mathcal{T}_o]\rangle = \chi(T^2) = 0$. The symplectic bundle $(\xi|_{\mathcal{T}_o}, d\lambda)$ has a flat symplectic connection: parallel transport along ∂_θ is the identity in the coordinates $C \times \mathbb{R}$, and along R^λ it is the linearized Reeb flow, cf. the proof of Proposition 6.7. These commute as $[\partial_\theta, R^\lambda] = 0$ on $C \times \mathbb{R}$, so the connection is flat. It descends to M as $\phi_* R^\lambda = R^\lambda$.

A flat symplectic bundle over T^2 with commuting holonomies $\Psi_1, \Psi_2 \in Sp(2n, \mathbb{R})$ has $c_1 = 0$. Indeed, cut $T^2 = (\mathbb{R}/\mathbb{Z})^2 \ni (x, y)$ along $x = 0$ into a cylinder, with Ψ_1, Ψ_2 the holonomies along the x- and y-circles. For b a path from id to Ψ_2, trivializing the bundle over the cylinder by b exhibits the bundle as obtained by clutching with the loop $\gamma(y) = b(y)^{-1}\Psi_1\, b(y)$, closed as Ψ_1 and Ψ_2 commute. So $\langle c_1, [T^2]\rangle$ is, up to sign, the Maslov index of γ. This vanishes since the map $\rho : Sp(2n, \mathbb{R}) \to S^1$ of [27, 2nd ed., Theorem 2.29], see also Gutt [14, Proposition 2.3, (2)], is conjugation invariant, hence constant on γ. By the Riemann–Roch theorem the index of (6.8) is then $(n+1)\chi(T^2) + 2c_1(A_o^k) + 2 = 2$, where $\dim C = 2n + 1$ and the summand 2 is $\dim T_j\mathfrak{S}$.

For part 4, the fiber of Orb over o consists of the charge 1 curves of class A tracing o, which by part 2 are the $m(o)$ generalized Reeb tori u^1_{o,θ_0} if $\operatorname{image}\phi \circ o = \operatorname{image} o$ and $A = A^1_o$, and there are none otherwise. The map Orb is continuous, by the argument of Lemma 6.5. It is proper. Indeed, the curves over o have energy $2p(o)$ for the metric $\omega(\cdot, J\cdot)$, and p is bounded on compact sets of orbit strings. So the argument given for (6.6) applies. □

Lemma 7.2. *Let (C, λ) be a closed contact manifold and λ' a 1-form on $M = C \times S^1$, C^∞-close to the pull-back λ_0 of λ, with $\ker d\lambda'$ of rank 2 everywhere, for instance the first form of a first kind lcs pair. Let λ'_0 be the restriction of λ' to $C \times \{0\} = C$, a contact form C^∞-close to λ. Then there is a strict contactomorphism ψ of (C, λ'_0), C^∞-close to the identity, and a diffeomorphism $M_{\psi,1} \to M$ pulling λ' back to $(\lambda'_0)_\psi$, Example 2.9. Under it the compact leaves of $\ker d\lambda'$ are the tori $\mathcal{T}_o$ over the Reeb strings o of λ'_0 with $\operatorname{image}\psi^k \circ o = \operatorname{image} o$ for some $k \geq 1$.*

Proof. As $\ker d\lambda'$ is C^∞-close to $\ker d\lambda_0 = \operatorname{span}(R^\lambda, \partial_\theta)$, there is a unique section Z of $\ker d\lambda'$ with $\lambda'(Z) = 0$ and $d\theta(Z) = 1$, and it is C^∞-close to ∂_θ. Then $L_Z\lambda' = d(\lambda'(Z)) + \iota_Z d\lambda' = 0$. So the flow ψ_s of Z preserves λ' and maps $C \times \{\theta\}$ to $C \times \{\theta + s\}$. Set $\psi := (\psi_1|_{C\times\{0\}})^{-1}$. Then $\psi^*\lambda'_0 = \lambda'_0$, and ψ is C^∞-close to the identity. The map $C \times \mathbb{R} \to M$, $(x, s) \mapsto \psi_s(x)$, identifies (x, θ) with $(\psi(x), \theta + 1)$, so it descends to a diffeomorphism $M_{\psi,1} \to M$. It pulls λ' back to $(\lambda'_0)_\psi$, as $\psi_s^*\lambda' = \lambda'$ and $\lambda'(Z) = 0$. The last claim is the description of the compact leaves of $\ker d\lambda_\phi$ in the proof of part 1 of Proposition 7.1, applied to (C, λ'_0) and $\phi = \psi$. □

So, under the diffeomorphism of Lemma 7.2, a first kind lcs pair (λ', α') near the lcs-fication is $((\lambda'_0)_\psi, \alpha')$ on $M_{\psi,1}$, for ψ a strict contactomorphism near the identity and α' a closed 1-form near $d\theta$. For $\dim C \geq 3$ and C connected, by Theorem 2.10, $[\alpha']$ decides whether the pair is Liouville isomorphic to a mapping torus. The compact leaves of $\ker d\lambda'$ do not depend on α'. For every such α' the canonical distribution of (λ', α') is $\ker d\lambda'$, by the computation following Definition 2.6. So the inclusions of these leaves are Reeb 2-curves, by Example 1.13.

Proposition 7.3. *Let o be a non-degenerate closed Reeb orbit of a closed contact manifold (C, λ). Then for every first kind lcs pair (λ', α') on $C \times S^1$ sufficiently C^∞-close to the lcs-fication, the foliation $\ker d\lambda'$ has a compact leaf T, an embedded torus C^∞-close to $\operatorname{image} o \times S^1$. For any complex structure on T, its inclusion is a normal embedded Reeb 2-curve of (λ', α').*

Proof. Replacing o by its underlying simple orbit, which is also non-degenerate, we may assume o simple. Let ψ, ψ_s and λ'_0 be as in Lemma 7.2 and its proof. Let V be a thin tubular neighborhood of $\operatorname{image} o$. As o is non-degenerate, the implicit function theorem applies to the return map on a disk transverse to o. So λ'_0, being C^∞-close to λ, has, up to reparametrization, exactly one closed Reeb orbit o' homotopic to o in V. Moreover o' is C^∞-close to o. As ψ is strict and C^∞-close to the identity, $\psi \circ o'$ is a closed Reeb orbit of λ'_0 in V homotopic to o in V. So $\operatorname{image}\psi \circ o' = \operatorname{image} o'$, and the

image T of $\mathcal{T}_{o'}$ in M is a compact leaf of $\ker d\lambda'$, by Lemma 7.2. It is an embedded torus, C^∞-close to image $o \times S^1$, as o' is C^∞-close to o and ψ_s to $(x,\theta) \mapsto (x, \theta + s)$. So the inclusion of T is a Reeb 2-curve, as noted before the proposition. □

In particular, if (C,λ) has a non-degenerate closed Reeb orbit, then every first kind lcs pair near its lcs-fication has a Reeb 2-curve. So, near the lcs-fication of such (C,λ), the first assertion of Theorem 1.14 has no analogue.

Proof of Theorem 1.14. Write $M = M_{\phi,1}$ and $C = T^2 \times (\mathbb{R}/2\pi\mathbb{Z}) \ni (x,\sigma)$, with $\lambda = \cos\sigma\, dx_1 + \sin\sigma\, dx_2$. Then $R^\lambda = \cos\sigma\, \partial_{x_1} + \sin\sigma\, \partial_{x_2}$, the geodesic field. The translation ϕ preserves dx_1, dx_2. So α_ϵ is a well defined closed 1-form on M.

We show that $(\lambda_\phi, \alpha_\epsilon)$ is a first kind Reeb pair. By Example 2.9, $\ker d\lambda_\phi = \operatorname{span}(R^\lambda, \partial_\theta)$. So $d\lambda_\phi \wedge d\lambda_\phi = 0$, and $\omega_\epsilon := d_{\alpha_\epsilon}\lambda_\phi$ satisfies $\omega_\epsilon \wedge \omega_\epsilon = -2\,\alpha_\epsilon \wedge \lambda_\phi \wedge d\lambda_\phi$. The 3-form $\lambda_\phi \wedge d\lambda_\phi$ is annihilated by ∂_θ and restricts to a volume form on each fiber C_θ, while $\alpha_\epsilon(\partial_\theta) = 1$. So $\omega_\epsilon \wedge \omega_\epsilon$ vanishes nowhere, and $(\lambda_\phi, \alpha_\epsilon)$ is an lcs pair. It is first kind, as $\mathcal{V}_{\lambda_\phi} = \ker d\lambda_\phi$ is a 2-distribution. By the computation following Definition 2.6, $\lambda_\phi(X_{\alpha_\epsilon}) = 1$, so it is a Reeb pair, and $\mathcal{D} = \mathcal{V}_{\lambda_\phi} = \operatorname{span}(R^\lambda, \partial_\theta)$ for every ϵ. Explicitly, for $v \in \mathcal{V}_{\lambda_\phi}$ we have $\iota_v\omega_\epsilon = \lambda_\phi(v)\alpha_\epsilon - \alpha_\epsilon(v)\lambda_\phi$. Hence

$$X_{\alpha_\epsilon} = R^\lambda - c_\epsilon\, \partial_\theta, \quad c_\epsilon := \alpha_\epsilon(R^\lambda) = \epsilon_1 \cos\sigma + \epsilon_2 \sin\sigma, \qquad X_{\lambda_\phi} = -\partial_\theta. \tag{7.4}$$

Let u be a Reeb 2-curve of $(\lambda_\phi, \alpha_\epsilon)$. By the second condition of Definition 1.12, $\widetilde{u}$ is non-constant on some component of $\widetilde{\Sigma}$. Otherwise u is locally constant, and $u^*[\alpha_\epsilon] = 0$. The proof of the first sentence of part 1 of Proposition 7.1 uses the pair only through $\mathcal{D}$ and through the restriction of the lcs form to the tori $\mathcal{T}_o$. It uses that u is a Reeb 2-curve only through the first condition of Definition 1.12, which depends only on $\mathcal{D}$. Here $\mathcal{D} = \operatorname{span}(R^\lambda, \partial_\theta)$ for every ϵ, as shown above. The lcs forms of $(\lambda_\phi, \alpha_\epsilon)$ and (λ_ϕ, α_0) differ by $-(\epsilon_1 dx_1 + \epsilon_2 dx_2) \wedge \lambda_\phi$, which vanishes on $\mathcal{D}$. So that proof applies: $\widetilde{u}$ maps some component of $\widetilde{\Sigma}$ onto a torus $\mathcal{T}_o$, with image $\phi^k \circ o =$ image o for some $k \geq 1$. The image of o projects to the image of a closed geodesic of T^2. Its preimage in $\mathbb{R}^2$ is $x_0 + \mathbb{R}m + \mathbb{Z}^2$, for some $x_0 \in \mathbb{R}^2$ and $0 \neq m \in \mathbb{Z}^2$. The translation by $k(a,b)$ preserves this set iff $k(a,b) = tm + \ell$ for some $t \in \mathbb{R}$ and $\ell \in \mathbb{Z}^2$. Eliminating t gives $k(m_2 a - m_1 b) = m_2\ell_1 - m_1\ell_2$, a non-trivial rational relation between $1, a, b$. This is impossible. So there are no Reeb 2-curves.

Now suppose that $1, \epsilon_1, \epsilon_2$ are linearly independent over $\mathbb{Q}(a,b)$. Suppose that X_{α_ϵ} has a closed orbit of period $T > 0$ through the image of $(x,\sigma,\theta) \in C \times \mathbb{R}$. By (7.4) the flow of X_{α_ϵ} on $C \times \mathbb{R}$ fixes σ and takes (x,σ,θ) at time T to $(x + Te_\sigma, \sigma, \theta - Tc_\epsilon(\sigma))$, for $e_\sigma := (\cos\sigma, \sin\sigma)$. As $(x,\sigma,\theta) \sim (x + n(a,b), \sigma, \theta + n)$, $n \in \mathbb{Z}$, are the identifications defining M, the orbit closes up iff $Te_\sigma = n(a,b) + q$ and $-Tc_\epsilon(\sigma) = n$ for some $n \in \mathbb{Z}$ and $q \in \mathbb{Z}^2$. Then $n = -T\,\epsilon \cdot e_\sigma = -\epsilon \cdot (n(a,b) + q)$, that is

$$n + (na + q_1)\epsilon_1 + (nb + q_2)\epsilon_2 = 0.$$

The coefficients lie in $\mathbb{Q}(a,b)$. So $n = 0$, $na + q_1 = 0$ and $nb + q_2 = 0$. Hence $q = 0$ and $Te_\sigma = 0$, contradicting $T > 0$.

Finally, the loops $t \mapsto (x_1 + t, x_2, \sigma, \theta)$ and $t \mapsto (x_1, x_2 + t, \sigma, \theta)$, $t \in [0,1]$, in a fiber have α_ϵ-periods ϵ_1 and ϵ_2. If $c\alpha_\epsilon$ were integral for some $c \neq 0$, then ϵ_1, ϵ_2 would be linearly dependent over $\mathbb{Q}$. This contradicts the linear independence of $1, \epsilon_1, \epsilon_2$ over $\mathbb{Q}(a,b) \supset \mathbb{Q}$. So α_ϵ is scale irrational. □

7.2. **Proofs of the remaining theorems.**

Proof of Theorem 1.8. Let (C,λ) and ϕ be as in the hypothesis. Let $\{\phi_t\}$, $t \in [0,1]$, be a smooth family of strict contactomorphisms with $\phi_0 = \mathrm{id}$, $\phi_1 = \phi$. This gives a smooth fibration $\widehat{M} := C \times \mathbb{R} \times [0,1]/((x,\theta,t) \sim (\phi_t(x), \theta+1, t)) \to [0,1]$, with fiber $M_{\phi_t,1}$ over $t \in [0,1]$. This fiber carries the first kind lcs pair $(\lambda_{\phi_t}, \alpha)$ of Section 7.1. Let $tr : \widehat{M} \to (C \times S^1) \times [0,1]$ be given by $tr_t[(x,\theta)] := (\phi^{-1}_{t\mu(\theta)}(x), \theta)$,

$\theta \in [0,1]$, for $\mu : [0,1] \to [0,1]$ smooth, 0 near 0 and 1 near 1. It is well defined, as $(x,0) \sim (\phi_t(x),1)$ and both map to $(x,0)$. It is a smooth trivialization covering the identity of $S^1 \times [0,1]$. Each tr_t is the identity on the fiber over $\theta = 0$, and tr_0 is the identity of $C \times S^1$. Pushing forward the above family of lcs pairs by the bundle map tr, we get a smooth family $\{(\lambda_t, \alpha)\}$, $t \in [0,1]$, of first kind integral lcs pairs on $C \times S^1$, with $(\lambda_0, \alpha) = (\lambda, \alpha)$ the lcs-fication of λ.

Fix a smooth family $\{J^{\lambda_t}\}$ of almost complex structures on $C \times S^1$ with each J^{λ_t} admissible with respect to (λ_t, α), and with $J^{\lambda_0} = J^\lambda$ the almost complex structure of Example 4.6. It exists as the admissible structures for each pair form a contractible space, see the paragraph following Definition 4.3. (Here λ_t is not a contact form, and J^{λ_t} is only notation.) Let $N \subset \mathcal{O}(R^\lambda, \beta)$ be an open compact set satisfying $i(N, R^\lambda, \beta) \neq 0$.

We first reduce to the case $\hat{i}(N, R^\lambda, \beta) \neq 0$, for $\hat{i}$ as in (6.12). Below we count $\mathrm{Orb}^{-1}(N)$, not $\mathcal{P}(N)$. By (6.16) this count is $\hat{i}(N, R^\lambda, \beta)$. As noted before (6.12), $N^{(e)} = \emptyset$ for e large. Also $i(N^{(1)}, R^\lambda) = i(N, R^\lambda, \beta) \neq 0$. So there is a largest $e_0 \geq 1$ with $i(N^{(e_0)}, R^\lambda) \neq 0$. Set $N' := N^{(e_0)}$, an open compact set of orbit strings meeting finitely many classes γ, each with $e_0\gamma = \beta$. For $e \geq 2$ the set $(N')^{(e)} := \{o'' \,|\, e\, o'' \in N'\} = \bigsqcup_\gamma (N' \cap \mathcal{O}(R^\lambda, \gamma))^{(e)}$ is $N^{(e_0 e)}$, so it has vanishing Fuller index. So (6.12) gives $\sum_\gamma \hat{i}(N' \cap \mathcal{O}(R^\lambda, \gamma), R^\lambda, \gamma) = i(N', R^\lambda) \neq 0$. Hence $\hat{i}(N' \cap \mathcal{O}(R^\lambda, \gamma), R^\lambda, \gamma) \neq 0$ for some γ. If $o' \in N'$ is a fixed Reeb string of ϕ then so is its e_0-fold iterate $e_0\, o' \in N$, which is in class β. Replacing (N, β) by $(N' \cap \mathcal{O}(R^\lambda, \gamma), \gamma)$ we may therefore assume that $\hat{i}(N, R^\lambda, \beta) \neq 0$.

For each t, part 4 of Proposition 7.1, applied to $M_{\phi_t,1}$ with the class $tr_{t*}^{-1} A_\beta$ and transported by tr_t, gives a continuous proper map $\mathrm{Orb} : \overline{\mathcal{M}}_1^1(J^{\lambda_t}, A_\beta, \beta) \to \mathcal{O}(R^\lambda, \beta)$, the traced string having class β in C, as tr_t is the identity on the fiber over $\theta = 0$ and the fiber inclusion $C \to C \times S^1$ is injective on free homotopy classes. These induce a map

$$\widetilde{\mathrm{Orb}} : \widetilde{\mathcal{M}} = \overline{\mathcal{M}}_1^1(\{J^{\lambda_t}\}, A_\beta, \beta) \to \mathcal{O}(R^\lambda, \beta) \times [0,1],$$

defined by $\widetilde{\mathrm{Orb}}(u,t) = (\mathrm{Orb}(u), t)$. It is continuous by the argument of Lemma 6.5, the fibers of $pr_{S^1} \circ u_i$ converging to those of $pr_{S^1} \circ u$ when $(u_i, t_i) \to (u,t)$. It is proper. Indeed, the curves over (o,t) have energy $2p(o)$ for the metric $d_\alpha \lambda_t(\cdot, J^{\lambda_t}\cdot)$, and p is bounded on compact sets. So a sequence of curves over a compact set K of pairs (o,t) has a Gromov convergent subsequence by Theorem 5.2 for varying J_j, whose limit lies in $\widetilde{\mathcal{M}}$ by Lemma 4.9. The limit lies over K, as Gromov convergence implies convergence in $\widetilde{\mathcal{M}}$, Section 5.2, and $\widetilde{\mathrm{Orb}}$ is continuous.

Set

$$\widetilde{N} = \widetilde{\mathrm{Orb}}^{-1}(N \times [0,1]).$$

As N is open and compact in $\mathcal{O}(R^\lambda, \beta)$ and $\widetilde{\mathrm{Orb}}$ is continuous and proper, $\widetilde{N}$ is an open and compact subset of $\widetilde{\mathcal{M}}$. Set $N_i := \{u \,|\, (u,i) \in \widetilde{N}\}$ for $i = 0, 1$, so that by construction $N_0 = \mathrm{Orb}^{-1}(N)$, for Orb as in (6.6) for the lcs-fication (λ_0, α). We use $\mathrm{Orb}^{-1}(N)$ rather than the set of Reeb tori $\mathcal{P}(N)$. Indeed, for $t > 0$ the pair (λ_t, α) need not be an lcs-fication. So Lemma 6.5 does not apply to the family $\{J^{\lambda_t}\}$. Hence we do not know that $\mathcal{P}(N) = \{u \,|\, (u,0) \in \widetilde{\mathcal{R}}\}$ for some open and closed $\widetilde{\mathcal{R}} \subset \widetilde{\mathcal{M}}$.

By (6.16) of Theorem 6.14, as $N_0 = \mathrm{Orb}^{-1}(N)$,

$$GW_1^1(N_0, J^{\lambda_0}, A_\beta, \beta) = \hat{i}(N, R^\lambda, \beta) \neq 0.$$

Then applying Lemma 5.6, as $c_1(A_\beta) = 0$ by Section 4.1, we get that

$$GW_1^1(N_1, J^{\lambda_1}, A_\beta, \beta) \neq 0.$$

So $N_1 \neq \emptyset$. By construction of $\widetilde{N}$ and Proposition 7.1, an element u of N_1 is the transport by tr_1 of a generalized Reeb torus u^1_{o,θ_0} of $M_{\phi,1}$, with $o = \mathrm{Orb}(u) \in N$. In particular image $\phi \circ o =$ image o. So, for a representative o, $\phi \circ o = \theta_0 \cdot o$ for some θ_0, as in Section 7.1. Hence $\phi \circ o = o$ in $\mathcal{O}(R^\lambda, \beta)$. So $o \in N$ is a class β fixed Reeb string of ϕ, Definition 1.1. For the original (N, β) take the e_0-fold iterate, as noted above. □

Proof of Theorem 1.7. Suppose that λ is Morse–Bott. Let $N \subset \mathcal{O}(R^\lambda, \beta)$ be a connected component with $\chi^{orb}(N) \neq 0$ and with O_N orientable, for O_N the union of the images of its orbits. By Appendix A, N is open and compact, and $i(N, R^\lambda, \beta) = \pm\chi^{orb}(N) \neq 0$ by (A.4). If λ' is sufficiently C^1-close to λ then $R^{\lambda'}$ is C^0-close to R^λ. So by Lemma A.1 there is an open compact $N' \subset \mathcal{O}(R^{\lambda'}, \beta)$ with

$$i(N', R^{\lambda'}, \beta) = i(N, R^\lambda, \beta) \neq 0.$$

Then the result follows by Theorem 1.8. □

Proof of Theorem 1.2. The form λ_{st} is Morse–Bott, see [4]. The simple Hopf fibers form a connected component $N \simeq \mathbb{CP}^n$ of $\mathcal{O}(R^{\lambda_{st}}, \beta)$, for β the trivial class, with $O_N = S^{2n+1}$ in the notation of Appendix A, orientable. Its orbits are simple, so $m \equiv 1$ on N and $\chi^{orb}(N) = \chi(\mathbb{CP}^n) = n + 1 \neq 0$. Apply Theorem 1.7. □

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A. Fuller index

Let X be a complete vector field without zeros on a smooth manifold M. Set

$$S(X, \beta) = \{o \in L_\beta M \,|\, \exists p \in (0, \infty),\ \ o : \mathbb{R}/\mathbb{Z} \to M \text{ is a periodic orbit of } pX\},$$

where $L_\beta M$ denotes the free homotopy class β component of the free loop space

$$LM = \{o : S^1 \to M \,|\, o \text{ is smooth}\},$$

with the compact open topology. Here $S^1 = \mathbb{R}/\mathbb{Z}$. The above p is uniquely determined and we denote it by $p(o)$, the ***period*** of o.

There is a natural S^1 reparametrization action on $S(X, \beta)$: $\theta \cdot o$ is the loop $(\theta \cdot o)(\tau) = o(\theta + \tau)$. The elements of $\mathcal{O}(X, \beta) := S(X, \beta)/S^1$ will be called ***orbit strings***. Slightly abusing notation we just write o for the equivalence class of o.

An orbit string is ***simple*** if its parametrization by $\mathbb{R}/\mathbb{Z}$ is injective. Every orbit string o covers a unique simple one, o_s. The ***multiplicity*** $m(o)$ of an orbit string is the ratio $m(o) := p(o)/p(o_s) \in \mathbb{Z}_{\geq 1}$. We write $d\,o$ for the ***d-fold iterate*** of o, $d \geq 1$, $(d\,o)(\tau) := o(d\tau)$, of period $d\,p(o)$. So $o = m(o)\,o_s$.

We want a kind of fixed point index of an open compact subset $N \subset \mathcal{O}(X, \beta)$, which counts orbit strings o with certain weights. Suppose first that $N \subset \mathcal{O}(X, \beta)$ is finite. Each $o \in N$ is then isolated in $\mathcal{O}(X, \beta)$, as N is open and $\mathcal{O}(X, \beta)$ is Hausdorff. To such an (N, X, β) Fuller [13] associates an index:

$$i(N, X, \beta) = \sum_{o \in N} \frac{1}{m(o)} i(o),$$

where $i(o)$ is the fixed point index at $o(0)$ of the Poincaré return map, with return time near $p(o)$, of a local hypersurface of section in M through $o(0)$ transverse to the image of o.

In this appendix a homotopy $\{X_t\}$, $t \in [0, 1]$, of vector fields is continuous in t for the C^1 topology, and each X_t is complete without zeros. For a homotopy $\{X_t\}$ set

$$S(\{X_t\}, \beta) = \{(o, t) \in L_\beta M \times [0, 1] \,|\, o \in S(X_t, \beta)\}.$$

Fuller does not fix the class β, but $S(\{X_t\}, \beta)/S^1$ is open and closed in the space of all orbit strings of the homotopy, the class being locally constant. For $\widetilde{N}$ an open compact subset of $S(\{X_t\}, \beta)/S^1$ and $s \in [0, 1]$, the ***slice*** of $\widetilde{N}$ at $t = s$ is $\widetilde{N} \cap (L_\beta M \times \{s\})/S^1$, identified with a subset of $\mathcal{O}(X_s, \beta)$. It is compact, being closed in $\widetilde{N}$, and open in $\mathcal{O}(X_s, \beta)$, as $\widetilde{N}$ is open.

Fuller shows the following invariance property. Let $\{X_t\}$ be a homotopy with $X_0 = X$, and $\widetilde{N}$ an open compact subset of $S(\{X_t\},\beta)/S^1$ with slice N at $t=0$. If its slice N_1 at $t=1$ is also finite, then

$$i(N,X,\beta) = i(N_1,X_1,\beta).$$

For a general open compact $N \subset \mathcal{O}(X,\beta)$ the index is defined by perturbation. There is a C^1-small homotopy $\{X_t\}$, $X_0 = X$, supported near the compact union K of the images of the orbits in N, and an open compact $\widetilde{N}$ as above, consisting of orbit strings near N, with N_1 finite. Indeed, $\widetilde{N}$ exists by the proof of [37, Lemma 4.1] and by the Arzelà–Ascoli theorem. These use only that N is open and compact in $\mathcal{O}(X,\beta)$, and that each X_t is sufficiently C^0-close to X. For a generic such X_1 the class β orbits near K with period in a bounded range are ***non-degenerate***, that is their linearized return maps have no eigenvalue 1 (Kupka–Smale). Hence they are isolated, so that the compact N_1 is finite. We set $i(N,X,\beta) := i(N_1,X_1,\beta)$.

This is independent of the choices. Indeed, let $(\{X'_t\},\widetilde{N}')$ be another choice, with end slice N'_1. Concatenate the reverse of $\{X_t\}$ with $\{X'_t\}$. For this homotopy the union of $\widetilde{N}$ and $\widetilde{N}'$, glued along N, is open and compact. Its end slices N_1 and N'_1 are finite. So $i(N_1,X_1,\beta) = i(N'_1,X'_1,\beta)$, by the invariance property for finite end slices. The invariance property then holds for general N, N_1, see Fuller [13].

Lemma A.1. *Let M be closed, and let $N \subset \mathcal{O}(X,\beta)$ be open and compact with $i(N,X,\beta) \neq 0$. Then every vector field X' sufficiently C^0-close to X has a closed orbit of class β. More precisely, for $X_t := (1-t)X + tX'$ there is an open compact $\widetilde{N} \subset S(\{X_t\},\beta)/S^1$ with slice N at $t=0$. Its slice N' at $t=1$ is open and compact in $\mathcal{O}(X',\beta)$, and $i(N',X',\beta) = i(N,X,\beta) \neq 0$. So $N' \neq \emptyset$.*

Proof. As M is closed and X vanishes nowhere, each X_t is complete and vanishes nowhere once X' is C^0-close to X. Also $t \mapsto X_t$ is affine, so it is continuous for the C^1 topology. So $\{X_t\}$ is a homotopy as above. The set $\widetilde{N}$ exists by the proof of [37, Lemma 4.1] and the Arzelà–Ascoli theorem, as above. Its slice N' at $t=1$ is open and compact, as noted before the invariance property. The equality is the invariance property. □

Let now $X = R^\lambda$ be the Reeb vector field of a closed contact manifold (C^{2n+1},λ), and F^λ_t its time t flow map. Set $\xi = \ker\lambda$. For a closed R^λ-orbit o set $x = o(0)$. The linearized return map of o is conjugate to $F^\lambda_{p(o),*}|_{\xi(x)}$, as $F^\lambda_{p(o),*}$ fixes R^λ and preserves ξ. So o is non-degenerate iff 1 is not an eigenvalue of $F^\lambda_{p(o),*}$ on $\xi(x)$. In this case:

$$i(o) = \operatorname{sign}\operatorname{Det}(\mathrm{id}\,|_{\xi(x)} - F^\lambda_{p(o),*}|_{\xi(x)}) = (-1)^{CZ(o)-n}, \tag{A.2}$$

where $CZ(o)$ is the Conley–Zehnder index of the path $t \mapsto F^\lambda_{t,*}|_{\xi(x)}$, $0 \le t \le p(o)$, in a trivialization of ξ along o. The parity of $CZ(o)$ does not depend on the trivialization, see [32]. The second equality is the determinant property of the Conley–Zehnder index, Salamon [34], see also Gutt [14, Proposition 2.6, (7)].

The Morse–Bott case. The ***action spectrum*** of λ is the image of the period map $p : S(R^\lambda) \to (0,\infty)$, for $S(R^\lambda) := \bigcup_\beta S(R^\lambda,\beta)$. Note that $p(o) = \int_{S^1} o^*\lambda$. We say that the action spectrum is ***discrete*** if it is a discrete subset of $\mathbb{R}$. Recall [4] that a contact form λ on a closed manifold C^{2n+1} is ***Morse–Bott*** if its action spectrum is discrete, and if for every a in the action spectrum

$$P_a := \{x \in C \mid F^\lambda_a(x) = x\}$$

is a closed submanifold of C with $T_xP_a = \ker(F^\lambda_{a,*} - \mathrm{id})_x$ and with $\operatorname{rank} d\lambda|_{P_a}$ locally constant.

Suppose that λ is Morse–Bott and let $N \subset \mathcal{O}(R^\lambda,\beta)$ be a connected component. As the period map is continuous with discrete image, all $o \in N$ have the same period a. Let $O_N \subset P_a$ be the union of the images of the orbits in N. The circle $\mathbb{R}/a\mathbb{Z}$, identified with S^1 by $t \mapsto t/a$, acts on O_N by the flow

of R^λ, and $N = O_N/S^1$. So O_N is connected, as N and S^1 are. It is a connected component of P_a, as N is a connected component of $\mathcal{O}(R^\lambda, \beta)$ and the free homotopy class of the period a orbit through $x \in P_a$ is locally constant in x. This action is locally free, the isotropy group of $x \in O_N$ being cyclic of order $m(x)$, the multiplicity of the element of N through x. Thus N is a closed orbifold, in general non-effective, of dimension $\dim O_N - 1$, whose local group at $q \in N$ has order $m(q)$, the multiplicity of q. We assume that O_N is orientable. This holds for instance when $O_N = C$.

The ***orbifold Euler characteristic*** of N is

$$\chi^{orb}(N) := \sum_\sigma \frac{(-1)^{\dim \sigma}}{m(\sigma)}.$$

Here σ runs over the open cells of a cell decomposition of N with m constant on each cell, equal to $m(\sigma)$ on σ. Such a decomposition exists by Satake [36], applied to N with the effective orbifold structure given at the end of this appendix. That structure has the same underlying space, and its local group orders m/m_0, for $m_0 := \min m$, have the same level sets as m. The sum is $\sum_k \chi_c(m^{-1}(k))/k$, over the values k of m, for χ_c the Euler characteristic with compact supports, so it does not depend on the decomposition. When m is constant on N, N is a closed manifold and $\chi^{orb}(N) = \chi(N)/m$. In general $\chi^{orb}(N)$ and $\chi(N)$ need not vanish simultaneously: for a 2-sphere with four cone points of order 2, $\chi^{orb} = 0$ while $\chi = 2$.

We now show that N is open in $\mathcal{O}(R^\lambda, \beta)$, and compute $i(N, R^\lambda, \beta)$, following Bourgeois [4, Section 2.2], cf. [37, Section 2.1.1]. Choose $\delta > 0$ so that a is the only point of the action spectrum in $[a-\delta, a+\delta]$. Let U be a small tubular neighborhood of O_N with $\overline{U} \cap P_a = O_N$, so that the closed R^λ-orbits with period in $[a-\delta, a+\delta]$ meeting $\overline{U}$ are exactly the elements of N. Let $\bar f : N \to \mathbb{R}$ be a Morse function on the orbifold N, meaning that its pullback to O_N is Morse–Bott with critical set a finite union of S^1-orbits. The critical points of $\bar f$ are the images of these orbits. For such a critical point q, $\operatorname{ind}_{\bar f}(q)$ denotes the Morse–Bott index of the pullback along the orbit over q. Such $\bar f$ exist, see [4, Section 2.2]. Extend the pullback of $\bar f$ to a smooth function f on C, supported in U and constant near O_N along the fibers of a tubular projection $U \to O_N$, so that in particular $df(R^\lambda) = 0$ on O_N, and set

$$\lambda_\mu := (1 + \mu f)\lambda, \quad \mu \geq 0 \text{ small}.$$

For a critical point q of $\bar f$ let $o_q \in L_\beta C$ be a representative of q, a period a orbit of R^λ. Along the image of o_q we have $df = 0$, as f is constant along the fibers of the tubular projection near O_N, and its restriction to O_N is critical there. So $R^{\lambda_\mu} = (1+\mu f)^{-1} R^\lambda$ along the image of o_q. Consequently o_q is a closed R^{λ_μ}-orbit of period $(1 + \mu \bar f(q))a$ with the same image, hence with the same multiplicity $m(o_q) = m(q)$ and in the same free homotopy class β.

For μ small the closed R^{λ_μ}-orbits contained in U with period in $(a-\delta, a+\delta)$ lie in any given neighborhood of O_N. Indeed, by Arzelà–Ascoli, as $\mu \to 0$ a subsequence of such orbits converges to a closed R^λ-orbit with period in $[a-\delta, a+\delta]$, hence a. Its image lies in $\overline{U} \cap P_a = O_N$. By the proofs of [4, Lemmas 2.3, 2.4], which are local near O_N and use only that f is constant along the fibers near O_N and that $\bar f$ is Morse, for $\mu > 0$ sufficiently small these orbits are exactly the o_q. By the same proofs, the orbits o_q are non-degenerate as R^{λ_μ}-orbits, and the Conley–Zehnder index $CZ(o_q)$ of o_q as an R^{λ_μ}-orbit is

$$CZ(o_q) = CZ(N) - \tfrac{1}{2}\dim N + \operatorname{ind}_{\bar f}(q), \tag{A.3}$$

where $CZ(N)$ is the generalized Conley–Zehnder index [32] of the orbits in N, which is constant on N. (As usual $CZ(N)$ depends on the choice of a trivialization of the contact distribution along the orbits, but $CZ(N) - \frac{1}{2}\dim N$ is an integer whose parity does not.)

By the same argument, after shrinking μ if necessary, no closed $R^{\lambda_{t\mu}}$-orbit, $t \in [0,1]$, with period in $[a-\delta, a+\delta]$ meets ∂U or has period $a \pm \delta$. Indeed, a limit orbit has period a. If it meets ∂U, it meets $\overline{U} \cap P_a = O_N$, so it lies in $O_N \subset U$, which is disjoint from ∂U. Consequently the pairs (o, t), $t \in [0,1]$, with o a class β orbit string of $R^{\lambda_{t\mu}}$ with image in U and period in $(a-\delta, a+\delta)$, form an open compact

subset $\widetilde{N}$ of $S(\{R^{\lambda_{t\mu}}\},\beta)/S^1$. Its slice at $t=0$ is N. So N is open and compact in $\mathcal{O}(R^\lambda,\beta)$, as slices are. By the above, its slice at $t=1$ is $\{o_q\}_{q\in\operatorname{Crit}\bar f}$. So by the invariance of the Fuller index, (A.2) and (A.3):

$$i(N,R^\lambda,\beta)=\sum_{q\in\operatorname{Crit}\bar f}\frac{(-1)^{CZ(o_q)-n}}{m(o_q)}=(-1)^{CZ(N)-\frac12\dim N-n}\sum_{q\in\operatorname{Crit}\bar f}\frac{(-1)^{\operatorname{ind}_{\bar f}(q)}}{m(q)}=\pm\chi^{orb}(N). \tag{A.4}$$

The last equality is the Morse theoretic computation of the orbifold Euler characteristic, that is the Poincaré–Hopf theorem for orbifolds [36] applied to the gradient of $\bar f$. When m is constant on N it is just the classical Morse count of $\chi(N)$, divided by m.

Satake's theorem is for effective orbifolds whose singular set has codimension at least two. As S^1 is connected it preserves the orientation of O_N. So the isotropy group of $x\in O_N$ acts on the normal space $T_xO_N/\mathbb{R}R^\lambda$ by orientation preserving linear maps of finite order, whose fixed subspaces have even codimension. Hence the second condition holds once the first does. Also N is orientable, the normal spaces being oriented by the orientation of O_N and R^λ, compatibly with the local groups. We reduce to the effective case.

A finite subgroup of S^1 fixing an open subset of O_N pointwise fixes all of O_N: the fixed point set of a diffeomorphism of finite order is a closed submanifold, whose full dimensional part is then open and closed, and O_N is connected. Recall that $m_0=\min m$. The set $\{m=m_0\}\subset O_N$ is open, as by the slice theorem the isotropy group of a point near $x\in O_N$ is a subgroup of that of x. Each point of $\{m=m_0\}$ has isotropy group $\mathbb{Z}/m_0$, the only subgroup of S^1 of order m_0. So $\mathbb{Z}/m_0$ fixes the open set $\{m=m_0\}$ pointwise, hence acts trivially on O_N. So m_0 divides every $m(q)$. The group $S^1/(\mathbb{Z}/m_0)$ acts on O_N with isotropy groups of orders m/m_0. These act effectively on the normal spaces. Indeed, by the slice theorem a neighborhood of the orbit of x is equivariantly $S^1\times_{\Gamma_x}D$, for Γ_x the isotropy group of x and D a disk in $T_xO_N/\mathbb{R}R^\lambda$. As S^1 is abelian, an element of Γ_x acting trivially on D fixes this neighborhood. Hence it fixes all of O_N. So it lies in the isotropy group $\mathbb{Z}/m_0$ of a point of $\{m=m_0\}$. Thus N, with the orbifold structure given by this action, is effective. Its orbifold Euler characteristic is $m_0\chi^{orb}(N)$, and its Morse count for $\bar f$ is m_0 times $\sum_q(-1)^{\operatorname{ind}_{\bar f}(q)}/m(q)$. So the last equality of (A.4) for this structure implies it for N. In particular $i(N,R^\lambda,\beta)\neq0$ if and only if $\chi^{orb}(N)\neq0$.

B. Remark on multiplicity

This is a small note on how the virtual fundamental class technology deals with curves having non-trivial isotropy groups. We need this for the proof of Theorem 6.14.

Given a closed oriented effective orbifold $\mathfrak{X}$ with an oriented orbibundle E over $\mathfrak{X}$, Fukaya–Ono [12] construct the rational homology Euler class of E, using multi-sections. The effectivity assumption is not necessary for the construction of the Euler class, but it is convenient for other technical reasons.

McDuff–Wehrheim use a different approach to the virtual fundamental class, see for example McDuff [24], [26]. It emphasizes branched manifolds, and does not have the effectivity assumption. A similar use of branched manifolds appears in [7]. In the case of a non-effective orbibundle $E\to\mathfrak{X}$, McDuff [25] constructs using multi-sections, a homological Euler class $e(E)$, which extends the construction of [12]. McDuff shows that this class $e(E)$ is Poincaré dual to the usual cohomological Euler class of E. We shall assume the following black box property of the virtual fundamental class technology.

Axiom B.1. *Suppose that an open compact subset of a moduli space of stable maps is cleanly cut out. This means that it is represented by a (possibly non-effective) orbifold $\mathfrak{X}$ whose tangent space at each $[u]$ is $\ker D_u$ modulo the infinitesimal automorphisms of the domain. Here D_u is the linearized CR operator at u, coupled with the variations of the domain, cf. (6.8). Then the cokernels of the D_u form an orbibundle E over $\mathfrak{X}$, the obstruction bundle. The axiom is that the virtual fundamental class $[\mathfrak{X}]^{vir}$ coincides with $e(E)$, for the orientation of E relative to $\mathfrak{X}$ given by the determinant lines of the D_u.*

As $e(E)$ extends the Fukaya–Ono class, $[\mathfrak{X}]^{vir}$ is the Fukaya–Ono class when $\mathfrak{X}$ is effective. When $[\mathfrak{X}]^{vir}$ is in degree 0, the corresponding Gromov–Witten invariant is $\int_{[\mathfrak{X}]^{vir}} 1$. Given this axiom it does not matter to us which virtual moduli cycle technique we use. It is satisfied automatically by the construction of McDuff–Wehrheim (at the moment in genus 0). It can be shown to be satisfied in the approach of Pardon [31], cf. [31, Section 5.3] for the case of a manifold with obstruction bundle. And it is satisfied by the construction of Fukaya–Oh–Ohta–Ono [10]. This was communicated to me by Kaoru Ono. When $\mathfrak{X}$ is 0-dimensional, the case needed in this paper, the axiom follows immediately from the construction in [12]. Let (V, E_V, Γ, s) be a Kuranishi neighborhood of $\mathfrak{X}$ at one of its points. Replace it by $(V \times W, E_V \oplus W, \Gamma, s \oplus \mathrm{id}_W)$, with Γ acting diagonally, for W a faithful Γ-representation. This neighborhood is effective, with zero set $s^{-1}(0) \times \{0\}$.

As a special case most relevant to us here, let $\mathfrak{X}$ be an open compact subset of a moduli space of elliptic curves in an almost complex manifold (M, J), as in (5.5). Suppose that $\mathfrak{X}$ is regular, with expected dimension 0. Then its underlying space is a finite collection of oriented points. However, some curves are multiply covered and may have non-trivial isotropy groups. So we must treat $\mathfrak{X}$ as a possibly non-effective 0-dimensional oriented orbifold. The contribution of each curve $[u]$ to the Gromov–Witten invariant $\int_{[\mathfrak{X}]^{vir}} 1$ is $\frac{\epsilon([u])}{|\Gamma([u])|}$, where $\epsilon([u]) = \pm 1$ is the orientation of the point $[u]$ and $|\Gamma([u])|$ is the order of the isotropy group $\Gamma([u])$ of $[u]$. In the McDuff–Wehrheim setup the analogous genus 0 statement is explained in [24, Section 5]. In the setup of Fukaya–Ono [12] the same follows by taking the Kuranishi neighborhoods $(V \times W, E_V \oplus W, \Gamma, s \oplus \mathrm{id}_W)$ above at the isolated points of $\mathfrak{X}$. In the setup of Pardon it follows from the description of the virtual class by a single chart, [31, Section 2.3.3], as $1/|\Gamma|$ times the signed count of the Γ-set of solutions, for Γ the covering group of that chart. The solutions over $[u]$ form one Γ-orbit, with stabilizer of order $|\Gamma([u])|$. So they contribute $\epsilon([u])/|\Gamma([u])|$.

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